Measure Theory and Probability

Lecture notes, Chapters 1–5

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Chapter 1. Measure Spaces, Measurable Functions and Integration

1.1 Systems of sets and measures

Motivation. A probability or a length cannot be assigned consistently to *every* subset of $\mathbb{R}$; one first fixes a family of "measurable" sets that is closed under the countable operations a limit argument needs. This section introduces those families (σ-algebras, Dynkin systems, rings) and the tool that makes them useful: a measure is determined by its values on any $\cap$-stable generator.

Definition 1.1 (σ-algebra)

Let Ω be a set and $\mathcal{P}(\Omega) := \{A \mid A \subset \Omega\}$ its power set. A family $\mathcal{A} \subset \mathcal{P}(\Omega)$ is a **σ-algebra** (or σ-field) on Ω if

(i) $\Omega \in \mathcal{A}$;

(ii) $A \in \mathcal{A} \Rightarrow A^c := \Omega \setminus A \in \mathcal{A}$;

(iii) $A_i \in \mathcal{A},\ i \in \mathbb{N} \Rightarrow \bigcup_{i\in\mathbb{N}} A_i \in \mathcal{A}$.

The prefix "σ" always signals closure under *countable* operations, as in the summation sign Σ of a series.

Example 1.2

(i) $\mathcal{P}(\Omega)$ is a σ-algebra; it is the natural choice for finite or countable Ω and in general far too large.

(ii) If $\mathcal{A}_i$, $i \in I$ (I arbitrary, possibly uncountable), are σ-algebras on Ω, then $\bigcap_{i\in I} \mathcal{A}_i = \{A \subset \Omega \mid A \in \mathcal{A}_i\ \forall i\}$ is a σ-algebra.

(iii) (*Generated σ-algebra*) For $\emptyset \neq \mathcal{A}_0 \subset \mathcal{P}(\Omega)$,

$$\sigma(\mathcal{A}_0) := \bigcap_{\mathcal{B}\ \sigma\text{-algebra},\ \mathcal{A}_0\subset\mathcal{B}} \mathcal{B}$$

is a σ-algebra by (ii) and is the smallest σ-algebra containing $\mathcal{A}_0$.

(iv) (*Borel sets*) If Ω is a topological space and $\mathcal{A}_0$ the family of open sets, $\mathcal{B}(\Omega) := \sigma(\mathcal{A}_0)$ is the **Borel σ-algebra**. Open sets, closed sets and countable unions of closed sets are Borel; nevertheless $\mathcal{B}(\mathbb{R}) \neq \mathcal{P}(\mathbb{R})$.

(v) (*Trace*) For a σ-algebra $\mathcal{A}$ on Ω and $\Omega' \subset \Omega$, $\Omega' \cap \mathcal{A} := \{A \cap \Omega' \mid A \in \mathcal{A}\}$ is a σ-algebra on Ω'. If $\Omega' \in \mathcal{A}$, then $\Omega' \cap \mathcal{A} = \{A \in \mathcal{A} \mid A \subset \Omega'\}$.

Definition 1.3 (Measure)

Let $\Omega \neq \emptyset$ and $\mathcal{A}$ a σ-algebra on Ω. A map $\mu : \mathcal{A} \to [0,\infty]$ is a **measure** on $(\Omega, \mathcal{A})$ if

(i) $\mu(\emptyset) = 0$;

(ii) (σ-additivity) $\mu(\dot{\bigcup}_{i\in\mathbb{N}} A_i) = \sum_{i=1}^{\infty} \mu(A_i)$ for pairwise disjoint $A_i \in \mathcal{A}$.

$(\Omega, \mathcal{A}, \mu)$ is a **measure space**, $(\Omega, \mathcal{A})$ a **measurable space**, and the elements of $\mathcal{A}$ are **measurable sets**. If moreover $\mu(\Omega) = 1$, μ is a **probability measure**; we then write $P := \mu$, call $(\Omega, \mathcal{A}, P)$ a **probability space** and its measurable sets **events**.

Notation. $\dot{\bigcup}_{i\in\mathbb{N}} A_i$ (or $A_1 \dot{\cup} A_2$) denotes a union whose members are pairwise disjoint.

Example 1.4

(i) (*Dirac measure*) For $x \in \mathbb{R}^d$, $\delta_x(A) := 1$ if $x \in A$ and 0 otherwise defines a probability measure on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$.

(ii) (*Restriction*) If $(\Omega, \mathcal{A}, \mu)$ is a measure space and $\Omega' \in \mathcal{A}$, then $(\Omega', \mathcal{A} \cap \Omega', \mu)$ is again a measure space.

(iii) (*Counting measure*) On $(\mathbb{N}_0, \mathcal{P}(\mathbb{N}_0))$, $\mu(A) := |A| \in [0, \infty]$ (number of elements) is a measure.

Dynkin systems (λ-systems) are weaker than σ-algebras but much easier to verify; they are the standard tool for uniqueness arguments.

Definition 1.5 (Dynkin system)

$\mathcal{D} \subset \mathcal{P}(\Omega)$ is a **Dynkin system** if (i) $\Omega \in \mathcal{D}$;
(ii) $A \in \mathcal{D} \Rightarrow A^c \in \mathcal{D}$;
(iii) $A_i \in \mathcal{D}$, $i \in \mathbb{N}$, pairwise disjoint $\Rightarrow \dot{\bigcup}_i A_i \in \mathcal{D}$.

The only difference from Definition 1.1 is that (iii) is required for disjoint sequences only.

Example 1.6

(i) Every σ-algebra is a Dynkin system.

(ii) If P_1, P_2 are probability measures on $(\Omega, \mathcal{A})$, then $\mathcal{D} := \{A \in \mathcal{A} \mid P_1(A) = P_2(A)\}$ is a Dynkin system: $\Omega \in \mathcal{D}$ because both total masses are 1, complements because $P_i(A^c) = 1 - P_i(A)$, and disjoint unions by σ-additivity. (It is in general *not* a σ-algebra.)

Remark 1.7

(i) If $\mathcal{D}$ is a Dynkin system and $A, B \in \mathcal{D}$ with $A \subset B$, then $B \setminus A = (B^c \dot{\cup} A)^c \in \mathcal{D}$.

(ii) A Dynkin system that is $\cap$-stable (closed under finite intersections) is a σ-algebra.

(a) For $A, B \in \mathcal{D}$: $A \cup B = A \dot{\cup} (B \setminus (A \cap B)) \in \mathcal{D}$, by (i) and $\cap$-stability.

(b) For $A_i \in \mathcal{D}$: $\bigcup_i A_i = \dot{\bigcup}_i \left(A_i \cap \left(\bigcup_{n<i} A_n\right)^c\right)$, and each piece lies in $\mathcal{D}$ by (a), $\cap$-stability and closure under complements.

Theorem 1.8 (π–λ theorem)

Let $\mathcal{A}_0 \subset \mathcal{P}(\Omega)$ be $\cap$-stable. Then $\sigma(\mathcal{A}_0) = \mathcal{D}(\mathcal{A}_0)$, where

$$\mathcal{D}(\mathcal{A}_0) := \bigcap_{\mathcal{D} \text{ Dynkin system, } \mathcal{A}_0 \subset \mathcal{D}} \mathcal{D}$$

is the Dynkin system generated by $\mathcal{A}_0$.

Proof.

Step 1. $\sigma(\mathcal{A}_0)$ is a Dynkin system containing $\mathcal{A}_0$, so $\mathcal{D}(\mathcal{A}_0) \subset \sigma(\mathcal{A}_0)$.

Step 2. For the converse it suffices, by Remark 1.7(ii), to show that $\mathcal{D}(\mathcal{A}_0)$ is $\cap$-stable. For

$D \in \mathcal{D}(\mathcal{A}_0)$ put

$$\mathcal{D}_D := \{Q \subset \Omega \mid Q \cap D \in \mathcal{D}(\mathcal{A}_0)\}.$$

This is a Dynkin system: $\Omega \cap D = D$; if $Q \cap D \in \mathcal{D}(\mathcal{A}_0)$ then $Q^c \cap D = D \setminus (Q \cap D) \in \mathcal{D}(\mathcal{A}_0)$ by Remark 1.7(i); and disjoint unions pass through $\cap D$.

Step 3. If $E \in \mathcal{A}_0$, then $\mathcal{A}_0 \subset \mathcal{D}_E$ because $\mathcal{A}_0$ is $\cap$-stable; hence $\mathcal{D}(\mathcal{A}_0) \subset \mathcal{D}_E$, i.e. $E \cap D \in \mathcal{D}(\mathcal{A}_0)$ for all $D \in \mathcal{D}(\mathcal{A}_0)$.

Step 4. Step 3 says $\mathcal{A}_0 \subset \mathcal{D}_D$ for every $D \in \mathcal{D}(\mathcal{A}_0)$, hence $\mathcal{D}(\mathcal{A}_0) \subset \mathcal{D}_D$, which is exactly the $\cap$-stability of $\mathcal{D}(\mathcal{A}_0)$. ■

Theorem 1.9 (Uniqueness of measures)

Let μ_1, μ_2 be measures on $(\Omega, \mathcal{A})$ and $\mathcal{A}_0 \subset \mathcal{A}$ a $\cap$-stable family with $\sigma(\mathcal{A}_0) = \mathcal{A}$ and $\mu_1 = \mu_2$ on $\mathcal{A}_0$. Suppose there are $\Omega_n \in \mathcal{A}_0$ with $\Omega_n \nearrow \Omega$ (i.e. $\Omega_n \subset \Omega_{n+1}$, $\bigcup_n \Omega_n = \Omega$) and $\mu_1(\Omega_n) = \mu_2(\Omega_n) < \infty$. Then $\mu_1 = \mu_2$. If μ_1, μ_2 are probability measures, the sets Ω_n are not needed.

Proof.

Step 1 (probability measures). By Example 1.6(ii), $\mathcal{D} := \{A \in \mathcal{A} \mid \mu_1(A) = \mu_2(A)\}$ is a Dynkin system containing $\mathcal{A}_0$. By Theorem 1.8, $\mathcal{A} = \sigma(\mathcal{A}_0) = \mathcal{D}(\mathcal{A}_0) \subset \mathcal{D}$, so $\mu_1 = \mu_2$.

Step 2 (localisation). In general, fix $E \in \mathcal{A}_0$ with $\mu_1(E) = \mu_2(E) < \infty$. Then $\mathcal{D}_E := \{A \in \mathcal{A} \mid \mu_1(A \cap E) = \mu_2(A \cap E)\}$ is a Dynkin system (the finiteness of $\mu_i(E)$ makes $\mu_i((A^c) \cap E) = \mu_i(E) - \mu_i(A \cap E)$ legitimate) and contains $\mathcal{A}_0$ by $\cap$-stability. Theorem 1.8 gives $\mathcal{D}_E = \mathcal{A}$.

Step 3. Apply Step 2 to $E = \Omega_n$ and use continuity from below (Theorem 1.15(ii)): $\mu_1(A) = \lim_n \mu_1(A \cap \Omega_n) = \lim_n \mu_2(A \cap \Omega_n) = \mu_2(A)$ for every $A \in \mathcal{A}$. ■

Measures are usually first defined on a small, explicit family of sets and then extended; rings are the natural such families.

Definition 1.10 (Rings, measures on rings, σ-finiteness)

(i) $\mathcal{R} \subset \mathcal{P}(\Omega)$ is a **ring** if $A, B \in \mathcal{R} \Rightarrow A \setminus B \in \mathcal{R}$ and $A \cup B \in \mathcal{R}$. Then $\emptyset \in \mathcal{R}$ and $A \cap B = A \setminus (A \setminus B) \in \mathcal{R}$, so rings are $\cap$-stable. A ring is a **σ-ring** if it is closed under countable unions; then also $\bigcap_n A_n = A_1 \setminus \bigcup_n (A_1 \setminus A_n) \in \mathcal{R}$. A ring containing Ω is an **algebra**, and a σ-ring containing Ω is a σ-algebra.

(ii) A **measure on a ring** $\mathcal{R}$ is defined as in Definition 1.3, with σ-additivity required only for disjoint $A_i \in \mathcal{R}$ whose union lies in $\mathcal{R}$.

(iii) A measure μ on a ring $\mathcal{R}$ is **σ-finite** if there are $\Omega_n \in \mathcal{R}$ with $\Omega_n \nearrow \Omega$ and $\mu(\Omega_n) < \infty$.

Theorem 1.11 (Carathéodory's extension theorem)

Let μ_0 be a σ-finite measure on a ring $\mathcal{R}$. Then there is a unique σ-finite measure μ on $\sigma(\mathcal{R})$ with $\mu = \mu_0$ on $\mathcal{R}$.

Proof [Supplement].

Step 1 (uniqueness). $\mathcal{R}$ is $\cap$-stable, and the sets $\Omega_n \in \mathcal{R}$ from σ-finiteness satisfy $\mu_1(\Omega_n) = \mu_2(\Omega_n) = \mu_0(\Omega_n) < \infty$ for any two extensions; Theorem 1.9 gives $\mu_1 = \mu_2$.

Step 2 (outer measure). For $Q \subset \Omega$ put

$$\mu^*(Q) := \inf\Big\{ \sum_{n=1}^{\infty} \mu_0(A_n) \;\Big|\; A_n \in \mathcal{R},\ Q \subset \bigcup_n A_n \Big\}.$$

Since $\Omega_n \nearrow \Omega$, covers exist. Clearly $\mu^*(\emptyset) = 0$ and μ^* is monotone. It is σ-subadditive: given $Q = \bigcup_k Q_k$ with $\mu^*(Q_k) < \infty$ and $\varepsilon > 0$, choose covers $(A_{kn})_n$ of Q_k with $\sum_n \mu_0(A_{kn}) \le \mu^*(Q_k) + \varepsilon 2^{-k}$; then $(A_{kn})_{k,n}$ covers Q and $\mu^*(Q) \le \sum_k \mu^*(Q_k) + \varepsilon$.

Step 3 ($\mu^* = \mu_0$ on $\mathcal{R}$). For $A \in \mathcal{R}$, $\mu^*(A) \le \mu_0(A)$ (cover by A itself). Conversely let $A \subset \bigcup_n A_n$ with $A_n \in \mathcal{R}$. Put $B_n := A \cap A_n \setminus \bigcup_{k<n} A_k \in \mathcal{R}$; these are disjoint with union $A \in \mathcal{R}$, so σ-additivity and monotonicity of μ_0 on $\mathcal{R}$ give $\mu_0(A) = \sum_n \mu_0(B_n) \le \sum_n \mu_0(A_n)$. Hence $\mu_0(A) \le \mu^*(A)$.

Step 4 (Carathéodory-measurable sets). Let $\mathcal{M}^* := \{A \subset \Omega \mid \mu^*(Q) = \mu^*(Q \cap A) + \mu^*(Q \setminus A)\ \forall Q \subset \Omega\}$. By subadditivity only "$\ge$" must be checked, and only for $\mu^*(Q) < \infty$. $\mathcal{M}^*$ contains Ω and is closed under complements (the defining identity is symmetric in A, A^c). If $A, B \in \mathcal{M}^*$, then for every Q

$$\mu^*(Q) = \mu^*(Q \cap A) + \mu^*(Q \cap A^c \cap B) + \mu^*(Q \cap A^c \cap B^c) \ge \mu^*(Q \cap (A \cup B)) + \mu^*(Q \setminus (A \cup B)),$$

using $Q \cap (A \cup B) = (Q \cap A) \cup (Q \cap A^c \cap B)$ and subadditivity; so $\mathcal{M}^*$ is an algebra. For disjoint $A_1, \dots, A_n \in \mathcal{M}^*$ induction gives $\mu^*(Q \cap \bigcup_{i \le n} A_i) = \sum_{i \le n} \mu^*(Q \cap A_i)$. Now let $A_i \in \mathcal{M}^*$ be disjoint, $A := \bigcup_i A_i$, $S_n := \bigcup_{i \le n} A_i \in \mathcal{M}^*$. Then

$$\mu^*(Q) = \mu^*(Q \cap S_n) + \mu^*(Q \setminus S_n) \ge \sum_{i \le n} \mu^*(Q \cap A_i) + \mu^*(Q \setminus A),$$

and letting $n \to \infty$ and using σ-subadditivity, $\mu^*(Q) \ge \sum_i \mu^*(Q \cap A_i) + \mu^*(Q \setminus A) \ge \mu^*(Q \cap A) + \mu^*(Q \setminus A)$. Thus $A \in \mathcal{M}^*$; since an algebra closed under disjoint countable unions is closed under all countable unions (disjointify as in Remark 1.7(ii)(b)), $\mathcal{M}^*$ is a σ-algebra. Taking $Q = A$ in the last display shows that μ^* is σ-additive on $\mathcal{M}^*$.

Step 5 ($\mathcal{R} \subset \mathcal{M}^*$). Let $A \in \mathcal{R}$, $Q \subset \Omega$ with $\mu^*(Q) < \infty$, and let $(A_n) \subset \mathcal{R}$ cover Q with $\sum \mu_0(A_n) \le \mu^*(Q) + \varepsilon$. Then $(A_n \cap A)$ covers $Q \cap A$, $(A_n \setminus A)$ covers $Q \setminus A$, and $\mu_0(A_n) = \mu_0(A_n \cap A) + \mu_0(A_n \setminus A)$, so $\mu^*(Q \cap A) + \mu^*(Q \setminus A) \le \mu^*(Q) + \varepsilon$.

Step 6. By Steps 4–5, $\sigma(\mathcal{R}) \subset \mathcal{M}^*$, and $\mu := \mu^*|_{\sigma(\mathcal{R})}$ is a measure extending μ_0 (Step 3). It is σ-finite because $\mu(\Omega_n) = \mu_0(\Omega_n) < \infty$. ∎

Remark 1.12

In the lectures the extension theorem is not used for Lebesgue measure. Instead, Section 1.2 constructs it directly, following Rudin's *Principles of Mathematical Analysis*, Ch. 11. The idea is the same outer-measure argument as above, but the abstract ring is replaced by the concrete ring of elementary sets in $\mathbb{R}^d$, so that compactness and open/closed approximation can be used. Rudin does not first prove σ-additivity on elementary sets: he introduces the outer measure at once, defines measurability by approximation with elementary sets, and obtains σ-additivity only on the resulting σ-ring. Much of the literature starts from the ring of half-open intervals instead; elementary sets give a shorter route here.

1.2 Construction of the Lebesgue measure

Motivation. We want a measure λ on $\mathbb{R}^d$ that gives a box its volume. Volume is obviously additive on finite unions of boxes; the work is to extend it to a σ-additive set function on a σ-algebra containing all open sets. The route is: outer measure → a pseudo-metric on sets → the sets that can be approximated by elementary sets.

Definition 1.13 (Additive and σ-additive set functions)

Let $\varphi : \mathcal{R} \to [0,\infty]$ on a ring $\mathcal{R}$ with $\varphi(\emptyset) = 0$. φ is **additive** if

$$A, B \in \mathcal{R},\ A \cap B = \emptyset \implies \varphi(A \cup B) = \varphi(A) + \varphi(B), \tag{1.1}$$

and **σ-additive** (a measure on $\mathcal{R}$) if $\varphi(\dot{\bigcup}_i A_i) = \sum_i \varphi(A_i)$ whenever the $A_i \in \mathcal{R}$ are pairwise disjoint and $\bigcup_i A_i \in \mathcal{R}$.

Remark 1.14

If φ is additive on $\mathcal{R}$, then by induction it is finitely additive, $\varphi(A_1 \dot{\cup} \cdots \dot{\cup} A_n) = \sum_i \varphi(A_i)$, and for all $A, B \in \mathcal{R}$:

$$\varphi(A \cup B) + \varphi(A \cap B) = \varphi(A) + \varphi(B), \tag{1.2}$$

$$A \subset B \Rightarrow \varphi(A) \le \varphi(B) \quad \text{(monotonicity)}, \tag{1.3}$$

$$A \subset B,\ \varphi(A) < \infty \Rightarrow \varphi(B \setminus A) = \varphi(B) - \varphi(A),$$

$$\varphi\Big(\bigcup_{i\le n} A_i\Big) \le \sum_{i\le n} \varphi(A_i) \quad \text{(finite subadditivity)}.$$

For (1.2) write $A\cup B = (A\setminus B)\dot{\cup}(A\cap B)\dot{\cup}(B\setminus A)$ and $A = (A\setminus B)\dot{\cup}(A\cap B)$, $B = (B\setminus A)\dot{\cup}(A\cap B)$. Monotonicity is $\varphi(B) = \varphi(A) + \varphi(B\setminus A)$, and subadditivity follows from (1.2) and induction.

Notation. $A_n \nearrow A$ means $A_n \subset A_{n+1}$ and $A = \bigcup_n A_n$; $A_n \searrow A$ means $A_n \supset A_{n+1}$ and $A = \bigcap_n A_n$.

Theorem 1.15 (Continuity of measures)

Let φ be additive on a ring $\mathcal{R}$ and consider

(i) φ is σ-additive on $\mathcal{R}$;

(ii) (continuity from below) $A_n \in \mathcal{R}$, $A_n \nearrow A \in \mathcal{R} \Rightarrow \varphi(A) = \lim_n \varphi(A_n)$;

(iii) (continuity from above) $A_n \in \mathcal{R}$, $A_n \searrow A \in \mathcal{R}$, $\varphi(A_{n_0}) < \infty$ for some $n_0 \Rightarrow \varphi(A) = \lim_n \varphi(A_n)$.

Then (i) $\Leftrightarrow$ (ii) $\Rightarrow$ (iii).

Proof.

Step (i)$\Rightarrow$(ii). Put $B_1 := A_1$, $B_n := A_n \setminus A_{n-1}$. The $B_n \in \mathcal{R}$ are disjoint, $A_n = B_1 \dot{\cup} \cdots \dot{\cup} B_n$ and $\bigcup_n B_n = A$, so $\lim_n \varphi(A_n) = \lim_n \sum_{i\le n} \varphi(B_i) = \sum_i \varphi(B_i) = \varphi(A)$.

Step (ii)$\Rightarrow$(i). For disjoint $A_n \in \mathcal{R}$ with $A := \bigcup_n A_n \in \mathcal{R}$, $B_N := \bigcup_{n\le N} A_n \nearrow A$, so $\varphi(A) = \lim_N \varphi(B_N) = \lim_N \sum_{n\le N} \varphi(A_n)$.

Step (ii)$\Rightarrow$(iii) [Supplement]. For $n \ge n_0$ the sets $C_n := A_{n_0} \setminus A_n \in \mathcal{R}$ increase to $A_{n_0} \setminus A \in \mathcal{R}$. All quantities are finite because $\varphi(A_n), \varphi(A) \le \varphi(A_{n_0}) < \infty$, so by Remark 1.14 and (ii)

$$\varphi(A_{n_0}) - \varphi(A) = \varphi(A_{n_0} \setminus A) = \lim_n \varphi(C_n) = \varphi(A_{n_0}) - \lim_n \varphi(A_n).$$

■

Definition 1.16 (Intervals and elementary sets)

For $a, b \in \mathbb{R}^d$ with $a_i \le b_i$, an **interval** (bounded box) is any set of points x with $a_i \lhd_i x_i \lhd_i' b_i$, $\lhd_i, \lhd_i' \in \{\le, <\}$. It is open (closed) if all inequalities are strict (non-strict); $a_i = b_i$ is allowed,

so an interval may be empty or a point. An **elementary set** is a finite union of intervals; $\mathcal{E}$ denotes the family of elementary sets. For an interval I put $\lambda(I) := \prod_{i=1}^{d}(b_i - a_i)$, and for $A = I_1 \dot\cup \cdots \dot\cup I_n$ with disjoint intervals

$$\lambda(A) := \sum_{i=1}^{n} \lambda(I_i). \tag{1.4}$$

For $d = 1, 2, 3$, λ is length, area and volume. The following facts are elementary. $\mathcal{E}$ is a ring but not a σ-ring: for $d = 1$, $\mathbb{Q} \cap [0, 1]$ is a countable union of one-point intervals but not a finite union of intervals. Every $A \in \mathcal{E}$ is a finite *disjoint* union of intervals: the endpoints of the intervals in all coordinates define a grid whose cells are intervals, and A is a union of cells. And $\lambda(A)$ in (1.4) does not depend on the decomposition: two decompositions have a common refinement into grid cells, and λ is additive on the cells of a grid inside one box. Hence λ is a finite, additive, nonnegative set function on $\mathcal{E}$.

Definition 1.17 (Regularity)

An additive $\varphi : \mathcal{E} \to [0, \infty]$ is **regular** if for every $A \in \mathcal{E}$ and $\varepsilon > 0$ there are $F, G \in \mathcal{E}$, F closed, G open, with $F \subset A \subset G$ and $\varphi(G) - \varepsilon \le \varphi(A) \le \varphi(F) + \varepsilon$.

Example 1.18

λ is regular and finite on $\mathcal{E}$: an interval can be shrunk to a closed one and enlarged to an open one, changing the product of side lengths by less than ε, and an elementary set is a finite disjoint union of intervals.

From now on μ denotes a finite, regular, additive set function on $\mathcal{E}$ (for example λ). The goal is to extend it to a σ-additive set function on a σ-ring containing $\mathcal{E}$.

Definition 1.19 (Outer measure)

For $E \subset \mathbb{R}^d$,

$$\mu^*(E) := \inf\Big\{ \sum_{n=1}^{\infty} \mu(A_n) \;\Big|\; A_n \in \mathcal{E} \text{ open},\ E \subset \bigcup_n A_n \Big\}.$$

Clearly $\mu^*(E) \ge 0$, $\mu^*(\emptyset) = 0$, and $E_1 \subset E_2 \Rightarrow \mu^*(E_1) \le \mu^*(E_2)$.

Theorem 1.20

(i) $\mu^*(A) = \mu(A)$ for $A \in \mathcal{E}$, so μ^* extends μ to $\mathcal{P}(\mathbb{R}^d)$.

(ii) (σ-subadditivity) If $E = \bigcup_n E_n$ with arbitrary $E_n \subset \mathbb{R}^d$, then

$$\mu^*(E) \le \sum_{n=1}^{\infty} \mu^*(E_n). \tag{1.5}$$

Proof. (i) Let $A \in \mathcal{E}$, $\varepsilon > 0$. By regularity there is an open $G \in \mathcal{E}$ with $G \supset A$ and $\mu(G) \le \mu(A) + \varepsilon$; G alone is an admissible cover, so $\mu^*(A) \le \mu(A)$.

Conversely, take open $A_n \in \mathcal{E}$ covering A with $\sum_n \mu(A_n) \le \mu^*(A) + \varepsilon$, and a closed $F \in \mathcal{E}$, $F \subset A$, with $\mu(A) \le \mu(F) + \varepsilon$. F is closed and bounded, hence compact, so $F \subset A_1 \cup \cdots \cup A_N$ for some N. By monotonicity and finite subadditivity,

$$\mu(A) \le \mu(F) + \varepsilon \le \mu(A_1 \cup \cdots \cup A_N) + \varepsilon \le \sum_{k \le N} \mu(A_k) + \varepsilon \le \mu^*(A) + 2\varepsilon.$$

(ii) Assume $\mu^*(E_n) < \infty$ for all n (otherwise there is nothing to show). For $\varepsilon > 0$ choose open covers $(A_{nk})_k \subset \mathcal{E}$ of E_n with $\sum_k \mu(A_{nk}) \le \mu^*(E_n) + 2^{-n}\varepsilon$. Then $(A_{nk})_{n,k}$ is an open cover of E, so $\mu^*(E) \le \sum_n \sum_k \mu(A_{nk}) \le \sum_n \mu^*(E_n) + \varepsilon$. ■

Definition 1.21 (Approximation by elementary sets)

For $A, B \subset \mathbb{R}^d$ let

$$S(A,B) := (A \setminus B) \cup (B \setminus A) \quad \text{(symmetric difference)}, \qquad d(A,B) := \mu^*(S(A,B)) \in [0,\infty].$$

Write $A_n \to A$ if $d(A, A_n) \to 0$, and define

$$A \in \mathcal{M}_F(\mu) \text{ (finitely } \mu\text{-measurable)} :\Longleftrightarrow \exists A_n \in \mathcal{E} : \ A_n \to A,$$
$$A \in \mathcal{M}(\mu) \text{ } (\mu\text{-measurable)} :\Longleftrightarrow \ A = \bigcup_n A_n \text{ with } A_n \in \mathcal{M}_F(\mu).$$

Obviously $\mathcal{E} \subset \mathcal{M}_F(\mu) \subset \mathcal{M}(\mu)$. Every $A \in \mathcal{M}_F(\mu)$ has $\mu^*(A) < \infty$: for n large, $A \subset S(A, A_n) \cup A_n$, so $\mu^*(A) \le d(A, A_n) + \mu(A_n) < \infty$.

Theorem 1.22

$\mathcal{M}(\mu)$ is a σ-ring and μ^* is σ-additive on $\mathcal{M}(\mu)$.

The proof needs two lemmas.

Lemma 1.23

For $A, B, C \subset \mathbb{R}^d$:

(1) $S(A,B) = S(B,A)$;

(2) $S(A,B) \subset S(A,C) \cup S(C,B)$;

(3) $S(A,B) = S(A^c, B^c)$;

(4) for A_1, A_2, B_1, B_2, each of $S(A_1 \cup A_2, B_1 \cup B_2)$, $S(A_1 \cap A_2, B_1 \cap B_2)$, $S(A_1 \setminus A_2, B_1 \setminus B_2)$ is contained in $S(A_1, B_1) \cup S(A_2, B_2)$.

Consequently, by monotonicity and subadditivity of μ^*: (i) $d(A,B) = d(B,A)$;
(ii) $d(A,B) \le d(A,C) + d(C,B)$;
(iii) $A = B \Rightarrow d(A,B) = 0$ (not conversely);
(iv) $d(A_1 \star A_2, B_1 \star B_2) \le d(A_1, B_1) + d(A_2, B_2)$ for $\star \in \{\cup, \cap, \setminus\}$.

Proof. (1) is clear.

(2) If $x \in A \setminus B$, then either $x \in C$ (so $x \in C \setminus B$) or $x \notin C$ (so $x \in A \setminus C$); hence $A \setminus B \subset (A \setminus C) \cup (C \setminus B)$, and symmetrically $B \setminus A \subset (B \setminus C) \cup (C \setminus A)$. The union of the right-hand sides is $S(A,C) \cup S(C,B)$.

(3) $S(A,B) = (A \cap B^c) \cup (B \cap A^c) = (B^c \setminus A^c) \cup (A^c \setminus B^c) = S(A^c, B^c)$.

(4) Since $(A_1 \cup A_2) \setminus (B_1 \cup B_2) \subset (A_1 \setminus B_1) \cup (A_2 \setminus B_2)$ and symmetrically,

$$S(A_1 \cup A_2, B_1 \cup B_2) \subset S(A_1, B_1) \cup S(A_2, B_2). \tag{1.6}$$

For intersections use (3) and (1.6): $S(A_1 \cap A_2, B_1 \cap B_2) = S(A_1^c \cup A_2^c, B_1^c \cup B_2^c) \subset S(A_1^c, B_1^c) \cup S(A_2^c, B_2^c) = S(A_1, B_1) \cup S(A_2, B_2)$. For differences write $A_1 \setminus A_2 = A_1 \cap A_2^c$ and use the intersection case together with (3). ■

Lemma 1.24

If $\mu^*(A) < \infty$ or $\mu^*(B) < \infty$, then

$$|\mu^*(A) - \mu^*(B)| \le d(A, B). \tag{1.7}$$

Proof. Say $\mu^*(B) \le \mu^*(A)$, so $\mu^*(B) < \infty$. Then by Lemma 1.23(ii),

$$\mu^*(A) = d(A, \emptyset) \le d(A, B) + d(B, \emptyset) = d(A, B) + \mu^*(B).$$

The case $\mu^*(A) \le \mu^*(B)$ is symmetric. ∎

Remark 1.25

(i) For $\mu = \lambda$, $d = 1$, a countable $A \subset \mathbb{R}$ and $B = \emptyset$: $d(A, B) = \lambda^*(A) \le \sum_{p \in A} \lambda([p, p]) = 0$, although $A \ne B$. So d is only a pseudo-metric; A and B differ by a λ^*-null set.

(ii) On $\mathcal{F} := \{A \subset \mathbb{R}^d \mid \mu^*(A) < \infty\} \supset \mathcal{M}_F(\mu)$ identify A and B when $d(A, B) = 0$. Then $(\mathcal{F}, d)$ is a metric space and $\mathcal{M}_F(\mu)$ is the closure $\overline{\mathcal{E}}$ of $\mathcal{E}$ in it. [Supplement] *Reason:* d is finite on $\mathcal{F}$ by subadditivity and is symmetric with the triangle inequality by Lemma 1.23. $\mathcal{M}_F(\mu) \subset \overline{\mathcal{E}}$ is the definition. For the reverse inclusion, a limit of points of $\overline{\mathcal{E}}$ is a limit of elementary sets by a diagonal choice and the triangle inequality.

Proof of Theorem 1.22.

Step 1 ($\mathcal{M}_F(\mu)$ is a ring). Let $A, B \in \mathcal{M}_F(\mu)$ with elementary $A_n \to A$, $B_n \to B$. By Lemmas 1.23(iv) and 1.24,

$$\begin{aligned} &d(A_n \cup B_n, A \cup B) + d(A_n \cap B_n, A \cap B) + d(A_n \setminus B_n, A \setminus B) + |\mu^*(A_n) - \mu^*(A)| + |\mu^*(B_n) - \mu^*(B)| \\ &\quad + |\mu^*(A_n \cup B_n) - \mu^*(A \cup B)| + |\mu^*(A_n \cap B_n) - \mu^*(A \cap B)| \le 6\big(d(A_n, A) + d(B_n, B)\big) \to 0. \end{aligned}$$

Since $A_n \cup B_n$, $A_n \setminus B_n \in \mathcal{E}$, we get $A \cup B$, $A \setminus B \in \mathcal{M}_F(\mu)$, and also

$$\mu^*(A_n) \to \mu^*(A), \quad \mu^*(A_n \cup B_n) \to \mu^*(A \cup B), \quad \mu^*(A_n \cap B_n) \to \mu^*(A \cap B). \tag{1.8}$$

Step 2 (μ^* is additive on $\mathcal{M}_F(\mu)$). By (1.2) for $\mu = \mu^*$ on $\mathcal{E}$, $\mu^*(A_n) + \mu^*(B_n) = \mu^*(A_n \cup B_n) + \mu^*(A_n \cap B_n)$. Letting $n \to \infty$ with (1.8): $\mu^*(A) + \mu^*(B) = \mu^*(A \cup B) + \mu^*(A \cap B)$, and $\mu^*(A \cap B) = 0$ if $A \cap B = \emptyset$.

Step 3 (disjoint representation). Every $A \in \mathcal{M}(\mu)$ is a disjoint union of sets from $\mathcal{M}_F(\mu)$: if $A = \bigcup_n A'_n$ with $A'_n \in \mathcal{M}_F(\mu)$, put $B'_n := A'_1 \cup \cdots \cup A'_n$, $A_1 := B'_1$, $A_n := B'_n \setminus B'_{n-1}$ (all in the ring $\mathcal{M}_F(\mu)$). For such a disjoint representation, σ-subadditivity gives $\mu^*(A) \le \sum_n \mu^*(A_n)$, while additivity and monotonicity give $\sum_{k \le n} \mu^*(A_k) = \mu^*(A_1 \dot\cup \cdots \dot\cup A_n) \le \mu^*(A)$. Hence

$$\mu^*(A) = \sum_{n=1}^{\infty} \mu^*(A_n). \tag{1.9}$$

Step 4 (finite measure implies $\mathcal{M}_F$). Let $A \in \mathcal{M}(\mu)$ with $\mu^*(A) < \infty$, written as in Step 3, and $B_n := A_1 \cup \cdots \cup A_n \in \mathcal{M}_F(\mu)$. Then $d(A, B_n) = \mu^*\big(\bigcup_{k>n} A_k\big) = \sum_{k>n} \mu^*(A_k) \to 0$ by (1.9) (Step 3 applied to $\bigcup_{k>n} A_k$). Choose elementary B_{n,k_n} with $d(B_{n,k_n}, B_n) < 1/n$; by the triangle inequality $B_{n,k_n} \to A$. Thus

$$A \in \mathcal{M}(\mu),\ \mu^*(A) < \infty \implies A \in \mathcal{M}_F(\mu). \tag{1.10}$$

Step 5 (σ-additivity on $\mathcal{M}(\mu)$). Let $A_n \in \mathcal{M}(\mu)$ be disjoint with union A. If $\mu^*(A) = \infty$, σ-subadditivity forces $\sum_n \mu^*(A_n) = \infty$. If $\mu^*(A) < \infty$, every A_n has finite outer measure, so $A_n \in \mathcal{M}_F(\mu)$ by (1.10), and (1.9) gives $\mu^*(A) = \sum_n \mu^*(A_n)$.

Step 6 ($\mathcal{M}(\mu)$ is a σ-ring). If $A_n = \bigcup_k A_{nk}$ with $A_{nk} \in \mathcal{M}_F(\mu)$, then $\bigcup_n A_n = \bigcup_{(n,k)\in\mathbb{N}^2} A_{nk} \in \mathcal{M}(\mu)$. For differences let $A = \bigcup_n A_n$, $B = \bigcup_l B_l$ with $A_n, B_l \in \mathcal{M}_F(\mu)$. Then $A_n \cap B = \bigcup_l (A_n \cap B_l) \in \mathcal{M}(\mu)$ has $\mu^*(A_n \cap B) \le \mu^*(A_n) < \infty$, so $A_n \cap B \in \mathcal{M}_F(\mu)$ by (1.10), and $A_n \setminus B = A_n \setminus (A_n \cap B) \in \mathcal{M}_F(\mu)$. Hence $A \setminus B = \bigcup_n (A_n \setminus B) \in \mathcal{M}(\mu)$. ■

Definition 1.26 (Lebesgue measure)

Set $\mu(A) := \mu^*(A)$ for $A \in \mathcal{M}(\mu)$. This extends μ from $\mathcal{E}$ to a σ-additive set function on the σ-ring $\mathcal{M}(\mu)$, i.e. a measure in the sense of Definition 1.10(ii). For $\mu = \lambda$ it is the **Lebesgue measure** on $\mathbb{R}^d$.

Remark 1.27

(i) $\mathcal{M}(\mu)$ contains the open sets, hence $\mathbb{R}^d$, so it is a σ-algebra and contains the closed sets too. Indeed, every open ball is a countable union of open intervals, and every open set is a countable union of balls with rational centres and radii; hence every open set is a countable union of open intervals, which lie in $\mathcal{E} \subset \mathcal{M}(\mu)$.

(ii) (*Regularity*) For $A \in \mathcal{M}(\mu)$ and $\varepsilon > 0$ there are F closed and G open with $F \subset A \subset G$, $\mu(G \setminus A) < \varepsilon$ and $\mu(A \setminus F) < \varepsilon$. The set G exists by Proposition 1.28. For F, apply Proposition 1.28 to A^c: there is an open $\tilde{G} \supset A^c$ with $\mu(\tilde{G} \setminus A^c) < \varepsilon$, and $F := \tilde{G}^c \subset A$ is closed with $A \setminus F = A \cap \tilde{G} = \tilde{G} \setminus A^c$.

(iii) With $G_n \supset A \supset F_n$ from (ii) for $\varepsilon = 1/n$, monotonicity gives $\mu(\bigcap_n G_n \setminus A) = 0 = \mu(A \setminus \bigcup_n F_n)$. Thus $A = B \dot\cup (A \setminus B)$ with $B := \bigcup_n F_n \in \mathcal{B}(\mathbb{R}^d)$ and $A \setminus B$ a μ-null set. Every μ-measurable set is a Borel set plus a null set, and always $\mathcal{B}(\mathbb{R}^d) \subset \mathcal{M}(\mu)$.

(iv) The μ-null sets form a σ-ring, and every $N \subset \mathbb{R}^d$ with $\mu^*(N) = 0$ is a μ-null set. [Supplement] *Reason:* countable unions of null sets are null by σ-subadditivity and differences by monotonicity; if $\mu^*(N) = 0$ then $d(N, \emptyset) = 0$ with $\emptyset \in \mathcal{E}$, so $N \in \mathcal{M}_F(\mu)$.

(v) For λ every countable set is a null set, but there are uncountable null sets, e.g. the Cantor set.

Proposition 1.28

μ is regular on $\mathcal{M}(\mu)$: for $A \in \mathcal{M}(\mu)$ and $\varepsilon > 0$ there is an open $G \supset A$ with $\mu(G \setminus A) < \varepsilon$ (and hence Remark 1.27(ii) holds).

Proof.

Step 1 ($\mu(A) < \infty$). Then $A \in \mathcal{M}_F(\mu)$ by (1.10), and there are open $O_n \in \mathcal{E}$ with $\bigcup_n O_n \supset A$ and $\sum_n \mu(O_n) < \mu(A) + \varepsilon$. $G := \bigcup_n O_n$ is open and $\mu(G) \le \sum_n \mu(O_n) < \mu(A) + \varepsilon$. Since $\mu(A) < \infty$, additivity gives $\mu(G \setminus A) = \mu(G) - \mu(A) < \varepsilon$.

Step 2 ($\mu(A) = \infty$) [Supplement]. Write $A = \bigcup_n A_n$ with $A_n \in \mathcal{M}_F(\mu)$ (so $\mu(A_n) < \infty$). By Step 1 choose open $G_n \supset A_n$ with $\mu(G_n \setminus A_n) < \varepsilon 2^{-n}$. Then $G := \bigcup_n G_n \supset A$ is open and $G \setminus A \subset \bigcup_n (G_n \setminus A_n)$, so $\mu(G \setminus A) < \varepsilon$. ■

1.3 Measure spaces and measurable functions

Motivation. Integration theory is no simpler on $\mathbb{R}^d$ than on an abstract set, so from now on $(X, \mathcal{M}, \mu)$ is an arbitrary measure space. Before integrating we need the functions whose level sets

are measurable, and we need to know that this class is stable under the operations used in analysis: limits, sums, products, continuous transformations.

Example 1.29

(i) $(\mathbb{R}^d, \mathcal{M}(\lambda), \lambda)$ is the **Lebesgue measure space**.
(ii) $\lambda|_{\mathcal{B}(\mathbb{R}^d)}$ is the **Lebesgue–Borel measure**, and $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \lambda)$ the Lebesgue–Borel measure space; in practice it is also simply called Lebesgue measure.

Definition 1.30 (Measurable function)

An extended real-valued $f : X \to \overline{\mathbb{R}} = [-\infty, \infty]$ is **measurable** if $\{f > \alpha\} := \{x \in X \mid f(x) \in (\alpha, \infty]\} \in \mathcal{M}$ for every $\alpha \in \mathbb{R}$.

This depends only on f and $\mathcal{M}$: for $(\mathbb{R}^d, \mathcal{M}(\lambda))$ one says *Lebesgue measurable*, for $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ *Borel measurable*.

Example 1.31

(i) Borel measurable functions are Lebesgue measurable. The converse fails: $\mathcal{B}(\mathbb{R}^d)$ has the cardinality $\mathfrak{c}$ of the continuum while $\mathcal{M}(\lambda)$ has cardinality $2^{\mathfrak{c}}$ (Dudley, *Real Analysis and Probability*, Prop. 3.4.1), so there is a Lebesgue measurable, non-Borel set L, and $\mathbb{1}_L$ is Lebesgue but not Borel measurable. (ii) A continuous $f : \mathbb{R}^d \to \mathbb{R}$ is Borel measurable, since $\{f > \alpha\} = f^{-1}((\alpha, \infty))$ is open.

Remark 1.32

For measurable spaces $(X, \mathcal{M})$, $(\tilde{X}, \tilde{\mathcal{M}})$, a map $T : X \to \tilde{X}$ is $\mathcal{M}/\tilde{\mathcal{M}}$**-measurable** if $T^{-1}(\tilde{A}) \in \mathcal{M}$ for all $\tilde{A} \in \tilde{\mathcal{M}}$. If $\tilde{\mathcal{M}} = \sigma(\tilde{\mathcal{M}}_0)$, it suffices that $T^{-1}(\tilde{A}) \in \mathcal{M}$ for all $\tilde{A} \in \tilde{\mathcal{M}}_0$. [Supplement] *Reason:* $\{\tilde{A} \subset \tilde{X} \mid T^{-1}(\tilde{A}) \in \mathcal{M}\}$ is a σ-algebra, since preimages commute with complements and unions, and it contains $\tilde{\mathcal{M}}_0$, hence $\sigma(\tilde{\mathcal{M}}_0)$. In particular, with $\mathcal{B}(\overline{\mathbb{R}}) := \{A \subset \overline{\mathbb{R}} \mid A \cap \mathbb{R} \in \mathcal{B}(\mathbb{R})\} = \sigma(\{(\alpha, \infty] \mid \alpha \in \mathbb{R}\})$, Definition 1.30 is exactly $\mathcal{M}/\mathcal{B}(\overline{\mathbb{R}})$-measurability.

Theorem 1.33

For $f : X \to \overline{\mathbb{R}}$ the following are equivalent: (i) $\{f > \alpha\} \in \mathcal{M}\ \forall \alpha \in \mathbb{R}$;
(ii) $\{f \geq \alpha\} \in \mathcal{M}\ \forall \alpha$;
(iii) $\{f < \alpha\} \in \mathcal{M}\ \forall \alpha$;
(iv) $\{f \leq \alpha\} \in \mathcal{M}\ \forall \alpha$.

Proof. For $\alpha \in \mathbb{R}$:

(i)⇒(ii): $\{f \geq \alpha\} = \bigcap_n \{f > \alpha - \frac{1}{n}\}$.

(ii)⇒(iii): $\{f < \alpha\} = X \setminus \{f \geq \alpha\}$.

(iii)⇒(iv): $\{f \leq \alpha\} = \bigcap_n \{f < \alpha + \frac{1}{n}\}$.

(iv)⇒(i): $\{f > \alpha\} = X \setminus \{f \leq \alpha\}$. ∎

Theorem 1.34

f measurable $\Rightarrow |f|$ measurable.

Proof. $\{|f| < \alpha\} = \{f < \alpha\} \cap \{f > -\alpha\} \in \mathcal{M}$; apply Theorem 1.33. ■

Theorem 1.35

If f_n, $n \in \mathbb{N}$, are measurable, so are $\inf_n f_n$, $\sup_n f_n$, $\liminf_n f_n$ and $\limsup_n f_n$.

Proof. $\{\sup_n f_n > \alpha\} = \bigcup_n \{f_n > \alpha\}$ and $\{\inf_n f_n < \alpha\} = \bigcup_n \{f_n < \alpha\}$. Then $\limsup_n f_n = \inf_n \sup_{m \geq n} f_m$ and $\liminf_n f_n = \sup_n \inf_{m \geq n} f_m$ are measurable by two applications. ■

Corollary 1.36

(i) f, g measurable $\Rightarrow$ $\max(f, g)$, $\min(f, g)$ measurable; in particular $f^+ := \max(f, 0)$ and $f^- := -\min(f, 0)$ are measurable. (ii) A pointwise limit of measurable functions is measurable.

Proof. (i) is Theorem 1.35 for the sequence $f, g, g, \ldots$.

(ii) If $f_n \to f$ pointwise, then $f = \limsup_n f_n$. ■

Theorem 1.37

Let $f, g : X \to \mathbb{R}$ be measurable and $F : \mathbb{R}^2 \to \mathbb{R}$ continuous. Then $h(x) := F(f(x), g(x))$ is measurable; in particular $f + g$ and fg are measurable.

Proof. Fix α. $G_\alpha := \{(u, v) \mid F(u, v) > \alpha\}$ is open in $\mathbb{R}^2$, hence (Remark 1.27(i)) a countable union of open rectangles $I_n = (\alpha_n, \beta_n) \times (\gamma_n, \delta_n)$. Each $\{(f, g) \in I_n\} = \{\alpha_n < f < \beta_n\} \cap \{\gamma_n < g < \delta_n\} \in \mathcal{M}$, so $\{h > \alpha\} = \bigcup_n \{(f, g) \in I_n\} \in \mathcal{M}$. ■

Remark 1.38

Compositions of Lebesgue measurable functions need not be Lebesgue measurable. There is a continuous increasing $f : [0, 1] \to [0, 1]$ and a Lebesgue measurable L with $f^{-1}(L)$ not Lebesgue measurable (Dudley, Prop. 4.2.1); then $g = \mathbb{1}_L$ gives $(g \circ f)^{-1}(\{1\}) = f^{-1}(L) \notin \mathcal{M}(\lambda)$. Compositions of Borel measurable functions are Borel measurable, and more generally $g \circ f$ is Lebesgue measurable if g is Borel and f Lebesgue measurable. Moreover, every Lebesgue measurable f agrees a.e. (Definition 1.49) with a Borel measurable function.

Definition 1.39 (Simple functions)

$f : X \to \mathbb{R}$ is **simple** if $\mathrm{im}(f) = \{f(x) \mid x \in X\}$ is finite. If $\mathrm{im}(f) = \{\alpha_1, \ldots, \alpha_n\}$ with distinct α_i and $E_i := \{f = \alpha_i\}$, then

$$f = \sum_{i=1}^{n} \alpha_i \mathbb{1}_{E_i}, \qquad \mathbb{1}_E(x) := \begin{cases} 1 & x \in E \\ 0 & x \notin E \end{cases} \tag{1.11}$$

($\mathbb{1}_E$ is the **indicator** or characteristic function of E). Since $\{\mathbb{1}_E \leq \alpha\}$ is $\emptyset$, E^c or X according as $\alpha < 0$, $0 \leq \alpha < 1$, $\alpha \geq 1$, the function $\mathbb{1}_E$ is measurable iff $E \in \mathcal{M}$; and f in (1.11) is measurable iff $E_1, \ldots, E_n \in \mathcal{M}$.

Theorem 1.40

For every $f : X \to \overline{\mathbb{R}}$ there are simple f_n with $f_n(x) \to f(x)$ for all x. If f is measurable the f_n can be chosen measurable, and if $f \geq 0$ they can be chosen pointwise increasing.

Proof. For $f \geq 0$ let

$$f_n := \sum_{i=1}^{n2^n} \frac{i-1}{2^n} \mathbb{1}_{\{f \in [\frac{i-1}{2^n}, \frac{i}{2^n})\}} + n \mathbb{1}_{\{f \geq n\}}. \tag{1.12}$$

Step 1 (increasing). Passing from n to $n+1$ halves each dyadic cell and the value on each half is at least the old one; on $\{f \geq n\}$ the new value is $\geq n$.

Step 2 (convergence). If $f(x) < \infty$, then for $n > f(x)$ we have $0 \leq f(x) - f_n(x) < 2^{-n}$. If $f(x) = \infty$, then $f_n(x) = n \to \infty$.

Step 3. The sets in (1.12) are measurable when f is. For general f apply this to f^+ and f^- and take $f_n^+ - f_n^-$. ∎

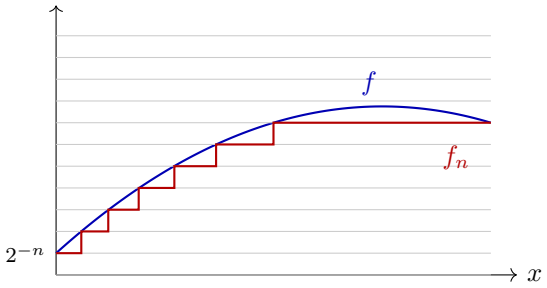


Figure. *The approximation (1.12): f_n rounds f down to the grid $2^{-n}\mathbb{Z}$ (and caps it at n). Refining the grid can only raise the step function.*

Note 1.41

If f is bounded, the sequence (1.12) for f^+ (resp. f^-) converges *uniformly*, because the error is $< 2^{-n}$ once $n > \sup |f|$. So a bounded (measurable) f is a uniform limit of (measurable) simple functions.

1.4 Integration

Motivation. The Lebesgue integral is defined by approximating f from below by simple functions, whose integral is an obvious finite sum. Its strength lies in the three limit theorems (monotone convergence, Fatou, dominated convergence), which let us exchange limits and integrals under conditions that are easy to check.

Throughout, $(X, \mathcal{M}, \mu)$ is a measure space.

Definition 1.42 (Integral of simple functions)

For $s = \sum_{i=1}^{n} \alpha_i \mathbb{1}_{E_i}$ with $\alpha_i > 0$, pairwise disjoint $E_i \in \mathcal{M}$, and $E \in \mathcal{M}$, put

$$I_E(s) := \sum_{i=1}^{n} \alpha_i \mu(E \cap E_i). \tag{1.13}$$

Lemma 1.43

If $s, \tilde{s}$ are measurable simple functions as in Definition 1.42 with $0 \le s \le \tilde{s}$, then $I_E(s) \le I_E(\tilde{s})$. In particular $I_E(s)$ does not depend on the representation of s.

Proof [Supplement]. Let $s = \sum_i \alpha_i \mathbb{1}_{E_i}$, $\tilde{s} = \sum_j \beta_j \mathbb{1}_{F_j}$. Where $s > 0$ also $\tilde{s} > 0$, so $E_i \subset \bigcup_j F_j$, and on a nonempty $E_i \cap F_j$ we have $\alpha_i \le \beta_j$. Hence, by finite additivity of μ,

$$I_E(s) = \sum_{i,j} \alpha_i \mu(E \cap E_i \cap F_j) \le \sum_{i,j} \beta_j \mu(E \cap E_i \cap F_j) \le \sum_j \beta_j \mu(E \cap F_j) = I_E(\tilde{s}).$$

For two representations of the same s, apply this in both directions. ∎

For measurable $f \ge 0$ define

$$\int_E f \, d\mu := \sup\{I_E(s) \mid s \text{ simple, measurable, } 0 \le s \le f\}. \tag{1.14}$$

Remark 1.44

$\int_E f \, d\mu \in [0, \infty]$, and $I_E(s) = I_E(s\mathbb{1}_E)$, so

$$\int_E f \, d\mu = \sup\{I_E(s) \mid s \text{ simple, measurable, } 0 \le s \le f\mathbb{1}_E\}. \tag{1.15}$$

Hence for measurable $g \ge 0$:

$$f = g \text{ on } E \Rightarrow \int_E f \, d\mu = \int_E g \, d\mu, \qquad f \le g \text{ on } E \Rightarrow \int_E f \, d\mu \le \int_E g \, d\mu, \tag{1.16}$$

and by Lemma 1.43, $\int_E s \, d\mu = I_E(s)$ for every measurable simple $s \ge 0$.

Definition 1.45 (Integral and integrability)

For measurable f with $\min\left(\int_E f^+ d\mu, \int_E f^- d\mu\right) < \infty$ put

$$\int_E f \, d\mu := \int_E f^+ d\mu - \int_E f^- d\mu \in [-\infty, \infty]. \tag{1.17}$$

If both integrals are finite, f is **(Lebesgue) integrable** on E w.r.t. μ, written $f \in \mathcal{L}(E, \mu)$ ($f \in \mathcal{L}(E)$ if $\mu = \lambda$).

Remark 1.46

For $E \in \mathcal{M}$:

(i) f measurable and bounded on E, $\mu(E) < \infty \Rightarrow f \in \mathcal{L}(E, \mu)$.

(ii) f measurable, $\alpha \le f \le \beta$ on E, $\mu(E) < \infty \Rightarrow \alpha\mu(E) \le \int_E f \, d\mu \le \beta\mu(E)$.

(iii) $f, g \in \mathcal{L}(E, \mu)$, $f \le g$ on $E \Rightarrow \int_E f \, d\mu \le \int_E g \, d\mu$.

(iv) $f \in \mathcal{L}(E, \mu) \Rightarrow \alpha f \in \mathcal{L}(E, \mu)$ and $\int_E \alpha f \, d\mu = \alpha \int_E f \, d\mu$ for $\alpha \in \mathbb{R}$.

(v) $\mu(E) = 0$, f measurable $\Rightarrow \int_E f \, d\mu = 0$.

(vi) $f \in \mathcal{L}(E, \mu)$, $A \subset E$, $A \in \mathcal{M} \Rightarrow f \in \mathcal{L}(A, \mu)$.

[Supplement] *Proofs.*
(ii) If s is simple with $0 \le s \le c\mathbb{1}_E$, then $I_E(s) \le c\mu(E)$, and the constant $c\mathbb{1}_E$ is itself admissible; hence $\int_E c \, d\mu = c\mu(E)$ for $c \ge 0$. Now $f^+ \le \beta^+$, $f^- \ge \beta^-$, $f^+ \ge \alpha^+$ and $f^- \le \alpha^-$

on E. By (1.16), $\int_E f \le \beta^+\mu(E) - \beta^-\mu(E) = \beta\mu(E)$ and similarly $\int_E f \ge \alpha\mu(E)$.
(i) follows from the bounds in (ii).
(iii) $f^+ \le g^+$ and $f^- \ge g^-$; use (1.16).
(iv) For $\alpha > 0$, $s \mapsto \alpha s$ is a bijection between simple functions below h and below αh, with $I_E(\alpha s) = \alpha I_E(s)$; so $\int_E \alpha h = \alpha \int_E h$ for $h \ge 0$, and $(\alpha f)^\pm = \alpha f^\pm$. For $\alpha < 0$ use $(\alpha f)^\pm = |\alpha| f^\mp$; $\alpha = 0$ is trivial.
(v) Every $I_E(s) = \sum \alpha_i \mu(E \cap E_i) = 0$.
(vi) $I_A(s) \le I_E(s)$ for $A \subset E$, so $\int_A f^\pm \le \int_E f^\pm < \infty$.

Theorem 1.47

(i) If $f \ge 0$ is measurable, then $\varphi(A) := \int_A f \, d\mu$, $A \in \mathcal{M}$, is σ-additive, i.e. a measure on $(X, \mathcal{M})$. (ii) The same holds for $f \in \mathcal{L}(X, \mu)$, except that φ is then a difference of two finite measures.

Proof. (ii) follows from (i) applied to f^+ and f^-. For (i) let $A = \dot\bigcup_i A_i$, $A_i \in \mathcal{M}$; we show $\varphi(A) = \sum_i \varphi(A_i)$.

Step 1 (simple f). For $f = \mathbb{1}_B$: $\varphi(A) = \mu(A \cap B) = \sum_i \mu(A_i \cap B) = \sum_i \varphi(A_i)$. For $f = \sum_j \beta_j \mathbb{1}_{B_j}$, $\beta_j \ge 0$, sum these identities (series of nonnegative terms may be rearranged).

Step 2 ($\le$). For simple measurable $0 \le s \le f$, Step 1 and (1.16) give $\int_A s \, d\mu = \sum_i \int_{A_i} s \, d\mu \le \sum_i \varphi(A_i)$. Taking the supremum over s: $\varphi(A) \le \sum_i \varphi(A_i)$.

Step 3 ($\ge$). φ is monotone in the set ($I_A(s) \le I_B(s)$ for $A \subset B$). So if some $\varphi(A_n) = \infty$, then $\varphi(A) = \infty$ and we are done. Otherwise, given $\varepsilon > 0$ choose simple $0 \le s_1, s_2 \le f$ with $\int_{A_k} s_k \ge \varphi(A_k) - \varepsilon$; then $s := \mathbb{1}_{A_1} s_1 + \mathbb{1}_{A_2} s_2 \le f$ and, by Step 1,

$$\varphi(A_1 \cup A_2) \ge \int_{A_1 \cup A_2} s \, d\mu = \int_{A_1} s_1 \, d\mu + \int_{A_2} s_2 \, d\mu \ge \varphi(A_1) + \varphi(A_2) - 2\varepsilon.$$

By induction $\varphi(A_1 \cup \cdots \cup A_n) \ge \sum_{i \le n} \varphi(A_i)$, and monotonicity gives $\varphi(A) \ge \sum_{i \le n} \varphi(A_i)$ for every n. ∎

Corollary 1.48

Let $A, E \in \mathcal{M}$, $A \subset E$, $\mu(E \setminus A) = 0$, and let $\int_E f \, d\mu$ be defined (e.g. $f \ge 0$ measurable or $f \in \mathcal{L}(E, \mu)$). Then $\int_E f \, d\mu = \int_A f \, d\mu$.

Proof. For $g = f^\pm$: $\int_E g = \int_A g + \int_{E \setminus A} g = \int_A g$ by Theorem 1.47 and Remark 1.46(v). ∎

Definition 1.49 (Almost everywhere)

A property P of points of $E \in \mathcal{M}$ holds **μ-almost everywhere** (μ-a.e.) on E if there is $N \in \mathcal{M}$ with $\mu(N) = 0$ such that P holds on $E \setminus N$. For Lebesgue measure one just says "a.e.".

Example 1.50

For measurable f, g: $f = g$ μ-a.e. on $E \iff \mu(\{f \ne g\} \cap E) = 0$. Here $\{f \ne g\} \in \mathcal{M}$ because $\{f < g\} = \bigcup_{q \in \mathbb{Q}} \{f < q\} \cap \{q < g\}$ and similarly for $\{f > g\}$. [Supplement]

Remark 1.51

"$f \sim g$ on $E :\iff f = g$ μ-a.e. on E" is an equivalence relation on measurable functions; for transitivity note that the union of two null sets is null. If $f \sim g$ on E and $\int_E f\,d\mu$ is defined, take a null set N with $f = g$ on $E \setminus N$. Then Corollary 1.48 and (1.16) give

$$\int_E f\,d\mu = \int_{E\setminus N} f\,d\mu = \int_{E\setminus N} g\,d\mu = \int_E g\,d\mu.$$

Null sets are negligible for integration.

Theorem 1.52

Let $E \in \mathcal{M}$.

(i) If f is measurable, $g \in \mathcal{L}(E,\mu)$ and $|f| \le g$ μ-a.e. on E, then $f \in \mathcal{L}(E,\mu)$.

(ii) $|f| \in \mathcal{L}(E,\mu)$ (i.e. $\int_E |f|\,d\mu < \infty$) $\iff$ $f \in \mathcal{L}(E,\mu)$.

(iii) $f \in \mathcal{L}(E,\mu) \Rightarrow |f| < \infty$ μ-a.e. on E.

(iv) $f \in \mathcal{L}(E,\mu) \Rightarrow |\int_E f\,d\mu| \le \int_E |f|\,d\mu$.

Proof. (i) With a null set N such that $|f| \le g$ on $E \setminus N$: $\int_E |f| = \int_{E\setminus N} |f| \le \int_{E\setminus N} g = \int_E g < \infty$. Since $f^{\pm} \le |f|$, (1.16) gives $f^{\pm} \in \mathcal{L}(E,\mu)$.

(ii) "$\Rightarrow$" by (i). "$\Leftarrow$": with $A := E \cap \{f \ge 0\}$, $B := E \cap \{f < 0\}$, $\int_E |f| = \int_A f^+ + \int_B f^- < \infty$ by Theorem 1.47 and Remark 1.46(vi).

(iii) For every n, $0 \le n\mathbb{1}_{\{|f|=\infty\}} \le |f|$, so $n\,\mu(\{|f| = \infty\} \cap E) \le \int_E |f| < \infty$; let $n \to \infty$.

(iv) By Remark 1.46(iii),(iv): $\pm \int_E f = \int_E (\pm f) \le \int_E |f|$. ∎

Theorem 1.53 (Monotone convergence theorem, Beppo Levi)

Let $0 \le f_1 \le f_2 \le \dots$ be measurable and $f := \lim_n f_n$. Then $\int_E f\,d\mu = \lim_n \int_E f_n\,d\mu$ for all $E \in \mathcal{M}$.

Proof. By (1.16), $\int_E f_n$ increases to some $\alpha \le \int_E f$. For the reverse inequality fix a simple measurable $0 \le s \le f$ and $c \in (0,1)$, and let $E_n := E \cap \{f_n \ge cs\}$. Then $E_n \nearrow E$ (if $s(x) > 0$ then $f_n(x) \to f(x) \ge s(x) > cs(x)$), and

$$\int_E f_n\,d\mu \ge \int_{E_n} f_n\,d\mu \ge c\int_{E_n} s\,d\mu.$$

Since $A \mapsto \int_A s\,d\mu$ is a measure (Theorem 1.47), continuity from below gives $\alpha \ge c\int_E s\,d\mu$. Let $c \uparrow 1$ and take the supremum over s. ∎

Theorem 1.54 (Linearity)

Let $f_1, f_2 \in \mathcal{L}(E,\mu)$ with $f_1 + f_2$ well defined (no $\infty - \infty$). Then:

(i) $f_1 + f_2$ is measurable;

(ii) if f is measurable and $f = f_1 + f_2$ μ-a.e. on E, then $f \in \mathcal{L}(E,\mu)$ and

$$\int_E f\,d\mu = \int_E f_1\,d\mu + \int_E f_2\,d\mu. \tag{1.18}$$

(1.18) also holds if $f_1, f_2 \ge 0$ are merely measurable and f is measurable with $f = f_1 + f_2$ μ-a.e. on E.

Remark 1.55

The measurability of f in Theorem 1.54(ii) is automatic if $(X, \mathcal{M}, \mu)$ is **complete**, i.e. every subset of a null set is measurable. In a complete space, g measurable and $h = g$ μ-a.e. imply h measurable. [Supplement] *Reason:* if $h = g$ off the null set N, then $\{h > \alpha\} = (\{g > \alpha\} \setminus N) \cup (\{h > \alpha\} \cap N)$, and the second set is a subset of N.

Proof of Theorem 1.54. (i) $f_1 + f_2 = f_1^+ - f_1^- + f_2^+ - f_2^-$ is the pointwise limit of the corresponding combinations of the simple approximations from Theorem 1.40, hence measurable by Corollary 1.36(ii).

(ii) By Remark 1.51 it suffices to treat $f = f_1 + f_2$.

Step 1 ($f_1, f_2 \geq 0$). For simple functions (1.18) is immediate from (1.13). In general take simple $s_n^i \nearrow f_i$ (Theorem 1.40); then $s_n^1 + s_n^2 \nearrow f_1 + f_2$ and the MCT gives $\int_E (f_1 + f_2) = \lim_n \int_E (s_n^1 + s_n^2) = \int_E f_1 + \int_E f_2$.

Step 2 (general). By Theorem 1.52(iii) and Remark 1.51 we may assume f_1, f_2 finite. Then $|f_1 + f_2| \leq |f_1| + |f_2|$, which is integrable by Step 1, so $f \in \mathcal{L}(E, \mu)$ by Theorem 1.52(i). From $f^+ - f^- = f_1^+ - f_1^- + f_2^+ - f_2^-$ we get $f^+ + f_1^- + f_2^- = f_1^+ + f_2^+ + f^-$. Step 1 gives equal integrals of both sides, all finite, and rearranging yields (1.18). ∎

Theorem 1.56

For measurable $f_k \geq 0$ and $E \in \mathcal{M}$: $\int_E \sum_{k=1}^{\infty} f_k \, d\mu = \sum_{k=1}^{\infty} \int_E f_k \, d\mu$.

Proof. $\sum_{k \leq n} f_k \nearrow \sum_k f_k$; apply the MCT and Theorem 1.54 to the partial sums. ∎

Theorem 1.57 (Fatou's lemma)

For measurable $f_n \geq 0$ and $E \in \mathcal{M}$: $\int_E \liminf_n f_n \, d\mu \leq \liminf_n \int_E f_n \, d\mu$.

Proof. $g_n := \inf_{k \geq n} f_k$ is measurable, $g_n \nearrow \liminf_n f_n$, and $g_n \leq f_k$ for $k \geq n$. By the MCT and (1.16),

$$\int_E \liminf_n f_n \, d\mu = \lim_n \int_E g_n \, d\mu \leq \liminf_n \inf_{k \geq n} \int_E f_k \, d\mu = \liminf_n \int_E f_n \, d\mu.$$

∎

Theorem 1.58 (Dominated convergence theorem)

Let f_n be measurable with $f_n(x) \to f(x)$ for all $x \in E$, and suppose there is $g \in \mathcal{L}(E, \mu)$ (an *integrable majorant*) with

$$|f_n(x)| \leq g(x) \qquad \forall n \geq 1, \ x \in E. \tag{1.19}$$

Then $f \in \mathcal{L}(E, \mu)$ and $\int_E f \, d\mu = \lim_n \int_E f_n \, d\mu$.

Remark 1.59

Pointwise convergence in the MCT and DCT may be weakened to μ-a.e. convergence, provided the a.e. limit is measurable. That is automatic in complete spaces (Remark 1.55) but not in general. Likewise (1.19) need only hold μ-a.e., and Fatou's lemma holds for $f_n \geq 0$ μ-a.e. [Supplement] *Reason:* let N be the (countable) union of the exceptional null sets. Replacing every function by its product with $\mathbb{1}_{N^c}$ changes no integral (Corollary 1.48), and the hypotheses then hold everywhere.

Proof of Theorem 1.58. $f, f_n \in \mathcal{L}(E,\mu)$ by Theorem 1.52(i), since $|f| \le g$ as well.

Step 1. $f_n + g \ge 0$ and $f + g = \liminf_n (f_n + g)$. Fatou and Theorem 1.54 give $\int_E f + \int_E g \le \liminf_n \int_E f_n + \int_E g$, i.e. $\int_E f \le \liminf_n \int_E f_n$.

Step 2. $g - f_n \ge 0$ and $g - f = \liminf_n (g - f_n)$, so $\int_E g - \int_E f \le \int_E g + \liminf_n \int_E (-f_n)$, i.e. $\int_E f \ge \limsup_n \int_E f_n$.

Step 3. Hence $\liminf_n \int_E f_n = \limsup_n \int_E f_n = \int_E f$. ∎

1.4.1 Comparison of the Lebesgue and Riemann integrals

Motivation. The Lebesgue integral should not change the value of any integral computed in calculus. It does not, and it also characterises Riemann integrability: a bounded function is Riemann integrable exactly when its discontinuities form a null set.

Let $X = [a,b]$, $\mu = \lambda$ on $\mathcal{M} = \{A \cap [a,b] \mid A \in \mathcal{M}(\lambda)\}$; this space is complete. Write $\int_a^b f\,dx := \int_X f\,d\lambda$, $f \in \mathcal{R}([a,b])$ if f is Riemann integrable, $\mathcal{R}\int_a^b f\,dx$ for the Riemann integral and $\underline{\mathcal{R}}\int$, $\overline{\mathcal{R}}\int$ for the lower and upper Riemann integrals. For a partition $P = \{x_0, \dots, x_n\}$, $\mathrm{mesh}(P) := \max_i |x_i - x_{i-1}|$.

Theorem 1.60

(i) $f \in \mathcal{R}([a,b]) \Rightarrow f \in \mathcal{L}([a,b])$ and $\int_a^b f\,dx = \mathcal{R}\int_a^b f\,dx$.

(ii) For bounded f: $f \in \mathcal{R}([a,b]) \iff f$ is continuous a.e. on $[a,b]$.

Proof.

Step 1 (step functions). Let f be bounded (Riemann integrable functions are bounded by definition). Choose partitions $P_k \subset P_{k+1}$ with $\mathrm{mesh}(P_k) \le 1/k$ such that the lower and upper sums satisfy

$$L(P_k, f) \to \underline{\mathcal{R}}\int_a^b f\,dx, \qquad U(P_k, f) \to \overline{\mathcal{R}}\int_a^b f\,dx. \tag{1.20}$$

For $P_k = \{x_0, \dots, x_{n(k)}\}$ define step functions L_k, U_k by $L_k(a) = U_k(a) = f(a)$ and, on $(x_{i-1}, x_i]$, $L_k := m_i = \inf_{[x_{i-1},x_i]} f$, $U_k := M_i = \sup_{[x_{i-1},x_i]} f$. Then

$$L(P_k,f) = \int_a^b L_k\,dx, \qquad U(P_k,f) = \int_a^b U_k\,dx, \qquad L_1 \le L_2 \le \cdots \le f \le \cdots \le U_2 \le U_1. \tag{1.21}$$

Step 2 (limits). $L := \lim_k L_k$ and $U := \lim_k U_k$ exist and are measurable. $\max(|L_1|, |U_1|)$ is bounded, hence an integrable majorant, so the DCT, (1.21) and (1.20) give

$$\int_a^b L\,dx = \underline{\mathcal{R}}\int_a^b f\,dx, \qquad \int_a^b U\,dx = \overline{\mathcal{R}}\int_a^b f\,dx. \tag{1.22}$$

Step 3 (i). Since $L \le f \le U$,

$$f \in \mathcal{R}([a,b]) \iff \int_a^b L\,dx = \int_a^b U\,dx \iff L = U \text{ a.e.} \iff L = f = U \text{ a.e.}, \tag{1.23}$$

where the middle step is Lemma 1.62(i) applied to $U - L \ge 0$. By completeness (Remark 1.55) f is then measurable, and $\int_a^b f\,dx = \int_a^b L\,dx = \mathcal{R}\int_a^b f\,dx$ by Remark 1.51 and (1.22).

Step 4 (ii). Off the countable (hence null) set $\bigcup_k P_k$,

$$U(x) = L(x) \iff f \text{ is continuous at } x. \tag{1.24}$$

[Supplement] Indeed, such an x is interior to an interval (x_{i-1}, x_i) of each P_k, of length $\le 1/k$. If f is continuous at x and $\varepsilon > 0$, then for large k this interval lies in a δ-neighbourhood on which $|f - f(x)| < \varepsilon$, so $U_k(x) - L_k(x) \le 2\varepsilon$. Conversely, if $U(x) = L(x)$, choose k with $U_k(x) - L_k(x) < \varepsilon$; then $|f(y) - f(x)| \le M_i - m_i < \varepsilon$ on the open interval around x. Combining (1.23) and (1.24): $f \in \mathcal{R}([a,b]) \iff L = U$ a.e. $\iff f$ is continuous a.e. ∎

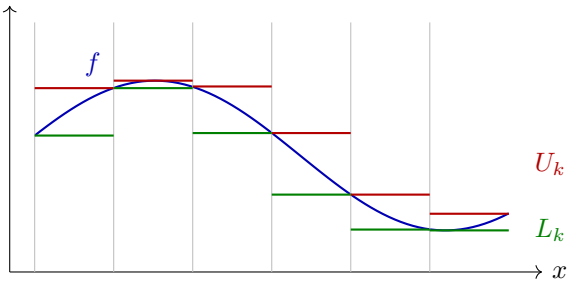


Figure. *Lower and upper step functions of a partition. Their integrals are the lower and upper sums, and refining the partition squeezes them towards f at every continuity point.*

1.5 Some fundamental inequalities and the L^p-spaces

Motivation. Integrals of powers $|f|^p$ measure the size of a function. Markov, Jensen, Hölder and Minkowski control these sizes, and Minkowski's inequality makes $\|f\|_p$ a (semi-)norm. The payoff is completeness: L^p is a Banach space, the natural home for limits of random variables.

Again $(X, \mathcal{M}, \mu)$ is an arbitrary measure space.

Proposition 1.61 (Markov ($p = 1$), Chebyshev ($p = 2$))

For measurable f, $p \in (0, \infty)$ and $\varepsilon > 0$:

$$\mu(\{|f| \ge \varepsilon\}) \le \frac{1}{\varepsilon^p} \int_X |f|^p \, d\mu.$$

Proof. $\mathbb{1}_{\{|f| \ge \varepsilon\}} = \mathbb{1}_{\{|f|^p \ge \varepsilon^p\}} \le \varepsilon^{-p} |f|^p$. Integrate, using (1.16) and Remark 1.46(iv) (whose proof for $h \ge 0$ does not need integrability). ∎

Lemma 1.62

Let f be measurable. (i) If $f \ge 0$ μ-a.e. and $\int_X f \, d\mu = 0$, then $f = 0$ μ-a.e. (ii) If $\int_A f \, d\mu = 0$ for every $A \in \mathcal{M}$, then $f = 0$ μ-a.e.

Proof [Supplement]. (i) $|f| = f$ a.e., so $\int |f| \, d\mu = 0$ by Remark 1.51. Markov gives $\mu(\{|f| \ge \frac{1}{n}\}) \le n \int |f| = 0$, and $\{f \ne 0\} = \bigcup_n \{|f| \ge \frac{1}{n}\}$.

(ii) With $A := \{f > 0\}$, $f \mathbb{1}_A \ge 0$ and $\int_X f \mathbb{1}_A \, d\mu = \int_A f \, d\mu = 0$, so $f \mathbb{1}_A = 0$ a.e. by (i); since $f \mathbb{1}_A > 0$ on A, $\mu(A) = 0$. Likewise $\mu(\{f < 0\}) = 0$. ∎

Theorem 1.63 (Jensen's inequality)

Let $(X, \mathcal{M}, \mu)$ be a probability space, $f \in \mathcal{L}(X, \mu)$ real-valued with $f(x) \in (a, b)$ for all x, and

$\varphi:(a,b)\to\mathbb{R}$ convex. Then $(\varphi\circ f)^-\in\mathcal{L}(X,\mu)$ and

$$\varphi\Big(\int_X f\,d\mu\Big)\le\int_X \varphi\circ f\,d\mu\in(-\infty,\infty].$$

Proof.

Step 1. A convex φ is continuous, so $\varphi\circ f$ is measurable. Put $t:=\int_X f\,d\mu\in(a,b)$ (since $a<f<b$ and $\mu(X)=1$). Convexity gives the supporting line at t, $\varphi(y)\ge\varphi(t)+\beta_+(y-t)$ for $y\in(a,b)$, where β_+ is the right derivative of φ at t. Hence

$$\varphi\circ f\ge g:=\varphi(t)+\beta_+(f-t)\in\mathcal{L}(X,\mu). \tag{1.25}$$

Step 2. From (1.25), $(\varphi\circ f)^-\le g^-$, so $(\varphi\circ f)^-\in\mathcal{L}(X,\mu)$ and $\int\varphi\circ f$ is defined in $(-\infty,\infty]$.

Step 3. Since g is finite a.e., (1.25) reads $(\varphi\circ f)^+ + g^-\ge(\varphi\circ f)^- + g^+\ge 0$ a.e. Integrating (monotonicity and Theorem 1.54 for nonnegative functions) and rearranging the finite terms $\int(\varphi\circ f)^-$, $\int g^-$:

$$\varphi\Big(\int_X f\,d\mu\Big)=\int_X g\,d\mu\le\int_X\varphi\circ f\,d\mu.$$

■

Example 1.64 (AM–GM)

Let $X=\{p_1,\dots,p_n\}$, $\mu(\{p_i\})=\alpha_i>0$ with $\sum\alpha_i=1$, $f(p_i)=x_i$, $\varphi=\exp$. Jensen gives $e^{\sum\alpha_i x_i}\le\sum\alpha_i e^{x_i}$, i.e. with $y_i=e^{x_i}$,

$$\prod_{i=1}^n y_i^{\alpha_i}\le\sum_{i=1}^n\alpha_i y_i\qquad(\alpha_i,y_i>0,\ \textstyle\sum\alpha_i=1).$$

Theorem 1.65 (Hölder and Minkowski inequalities)

Let $f,g:X\to[0,\infty]$ be measurable and $p,q\in(1,\infty)$ with $\frac1p+\frac1q=1$. Then

$$\int_X fg\,d\mu\le\Big(\int_X f^p\,d\mu\Big)^{1/p}\Big(\int_X g^q\,d\mu\Big)^{1/q}, \tag{1.26}$$

$$\Big(\int_X(f+g)^p\,d\mu\Big)^{1/p}\le\Big(\int_X f^p\,d\mu\Big)^{1/p}+\Big(\int_X g^p\,d\mu\Big)^{1/p}. \tag{1.27}$$

Proof.

Step 1 (Hölder, trivial cases). Let A,B be the two factors on the right of (1.26). If $A=0$, then $f=0$ a.e. (Lemma 1.62), so $fg=0$ a.e. and (1.26) holds; similarly if $B=0$. If $A,B>0$ and one of them is ∞, (1.26) is trivial. So let $A,B\in(0,\infty)$, $F:=f/A$, $G:=g/B$; then $\int F^p=\int G^q=1$.

Step 2 (Young). Where $F(x),G(x)\in(0,\infty)$ write $F=e^{s/p}$, $G=e^{t/q}$; convexity of exp gives $FG=e^{s/p+t/q}\le\frac1p e^s+\frac1q e^t=\frac1pF^p+\frac1qG^q$. This also holds when F or $G\in\{0,\infty\}$. Integrating: $\int FG\le\frac1p+\frac1q=1$, which is (1.26).

Step 3 (Minkowski). $(f+g)^p=f(f+g)^{p-1}+g(f+g)^{p-1}$; apply (1.26) to each term and use $(p-1)q=p$:

$$\int(f+g)^p\le\Big[\Big(\int f^p\Big)^{1/p}+\Big(\int g^p\Big)^{1/p}\Big]\Big(\int(f+g)^p\Big)^{1/q}. \tag{1.28}$$

Step 4. We may assume $\int(f+g)^p>0$ and $\int f^p+\int g^p<\infty$. Convexity of $t\mapsto t^p$ gives $(\frac{f+g}{2})^p\le\frac12(f^p+g^p)$, so $C:=(\int(f+g)^p)^{1/q}\in(0,\infty)$. Dividing (1.28) by C gives (1.27) since $1-\frac1q=\frac1p$. ■

Definition 1.66 ($\mathcal{L}^p$)

For $p \in [1,\infty)$ let $\mathcal{L}^p(X,\mu) := \{f : X \to \mathbb{R} \text{ measurable} \mid \|f\|_p := \left(\int_X |f|^p \, d\mu\right)^{1/p} < \infty\}$.

Theorem 1.67

(i) For $p \in [1,\infty)$, $\mathcal{L}^p(X,\mu)$ is a vector space and $\|\cdot\|_p$ a semi-norm on it. If (f_n) is Cauchy in $\mathcal{L}^p$, there are $f \in \mathcal{L}^p(X,\mu)$ and a subsequence (f_{n_k}) with $\|f - f_n\|_p \to 0$ and $f_{n_k} \to f$ μ-a.e.

(ii) If $p,q \in (1,\infty)$, $\frac{1}{p} + \frac{1}{q} = 1$, $f \in \mathcal{L}^p$, $g \in \mathcal{L}^q$, then $fg \in \mathcal{L}^1$ and

$$\|fg\|_1 \le \|f\|_p \|g\|_q. \tag{1.29}$$

Proof.

Step 1 (semi-norm). $\|\alpha f\|_p = |\alpha| \|f\|_p$ is clear, and $|f+g| \le |f| + |g|$ with Minkowski gives $\|f+g\|_p \le \|f\|_p + \|g\|_p$ (for $p = 1$ by linearity). So $\mathcal{L}^p$ is a vector space. $\|f\|_p = 0$ only gives $|f|^p = 0$ a.e. (Lemma 1.62(i)), i.e. $f = 0$ a.e.; hence a semi-norm only.

Step 2 (a.e. convergent subsequence). Choose $n_1 < n_2 < \dots$ with $\|f_{n_{k+1}} - f_{n_k}\|_p < 2^{-k}$ and let $g_k := f_{n_{k+1}} - f_{n_k}$. By the MCT and Minkowski,

$$\Big\| \sum_k |g_k| \Big\|_p = \lim_n \Big\| \sum_{k \le n} |g_k| \Big\|_p \le \limsup_n \sum_{k \le n} \|g_k\|_p < 1.$$

By Theorem 1.52(iii) there is a null set N with $\sum_k |g_k(x)| < \infty$ off N. There $f_{n_{k+1}}(x) = f_{n_1}(x) + \sum_{l \le k} g_l(x)$ converges. Put $f := \lim_k f_{n_k} \mathbb{1}_{X \setminus N}$, which is measurable with $f_{n_k} \to f$ a.e.

Step 3 (L^p convergence). Given $\varepsilon > 0$ take N_0 with $\|f_m - f_n\|_p < \varepsilon$ for $m,n \ge N_0$. For $n_k \ge N_0$, Fatou gives $\|f - f_{n_k}\|_p \le \liminf_l \|f_{n_l} - f_{n_k}\|_p \le \varepsilon$. Hence $f = (f - f_{n_k}) + f_{n_k} \in \mathcal{L}^p$ and $f_{n_k} \to f$ in $\mathcal{L}^p$; a Cauchy sequence with a convergent subsequence converges, so $f_n \to f$ in $\mathcal{L}^p$.

(ii) is (1.26) applied to $|f|, |g|$. ∎

Remark 1.68

(i) Identifying functions that agree μ-a.e. turns $\mathcal{L}^p$ into a normed space: $L^p(X,\mu) := \mathcal{L}^p(X,\mu)/\|\cdot\|_p^{-1}(0)$, where $\|\cdot\|_p^{-1}(0) = \{f \text{ measurable} \mid f = 0 \ \mu\text{-a.e.}\}$. For the class $\tilde{f}$ of f put

$$\|\tilde{f}\|_p := \|f\|_p; \tag{1.30}$$

now $\|\tilde{f}\|_p = 0 \iff f = 0$ a.e. $\iff \tilde{f} = 0$. In particular $L^2(X,\mu)$ is an inner product space with

$$\langle \tilde{f}, \tilde{g} \rangle_2 := \int_X fg \, d\mu, \qquad \|\tilde{f}\|_2 = \sqrt{\langle \tilde{f}, \tilde{f} \rangle_2}, \tag{1.31}$$

for arbitrary representatives f, g.

(ii) In practice we work with representatives and keep in mind the identification $f \sim g \iff f = g$ μ-a.e.

(iii) $\mathcal{L}^p$ uses real-valued functions so that sums are always defined. A measurable $g : X \to [0,\infty]$ with $\int g^p < \infty$ is finite a.e. (Theorem 1.52(iii)) and is tacitly replaced by a finite version.

Example 1.69

For $X = \mathbb{N}$, $\mathcal{M} = \mathcal{P}(\mathbb{N})$, $\mu = \sum_k \delta_k$ (counting measure), $\emptyset$ is the only null set and

$$L^p(X,\mu) = \mathcal{L}^p(X,\mu) = \ell^p := \Big\{ x = (x_k) \in \mathbb{R}^{\mathbb{N}} \;\Big|\; \sum_k |x_k|^p < \infty \Big\}.$$

Theorem 1.70 (Riesz–Fischer)

For $p \in [1,\infty)$, $(L^p(X,\mu), \|\cdot\|_p)$ is a Banach space; in particular $L^2(X,\mu)$ is a Hilbert space with the inner product (1.31).

Proof. By Remark 1.68(i) it is a normed space (an inner product space for $p = 2$). If $(\tilde{f}_n)$ is Cauchy, the representatives (f_n) are Cauchy in $\mathcal{L}^p$ by (1.30). Theorem 1.67(i) gives $f \in \mathcal{L}^p$ with $\|f - f_n\|_p \to 0$, i.e. $\|\tilde{f} - \tilde{f}_n\|_p \to 0$. ∎

Chapter 2. Basic Probability Theory

2.1 Random variables and integration

Motivation. Probability theory is measure theory on a space of total mass one, with its own vocabulary: measurable functions become random variables, integrals become expectations, "a.e." becomes "a.s.". This section sets up the dictionary so that every result of Chapter 1 can be used directly.

Let $(\Omega, \mathcal{A})$ be a measurable space. $\mathbb{R}$ carries $\mathcal{B}(\mathbb{R})$ and $\overline{\mathbb{R}} = \mathbb{R} \cup \{\pm\infty\}$ carries $\mathcal{B}(\overline{\mathbb{R}}) = \{B \subset \overline{\mathbb{R}} \mid B \cap \mathbb{R} \in \mathcal{B}(\mathbb{R})\}$, unless said otherwise.

Definition 2.1 (Random variable)

A **random variable** (r.v.) on $(\Omega, \mathcal{A})$ is an $\mathcal{A}/\mathcal{B}(\overline{\mathbb{R}})$-measurable map $X : \Omega \to \overline{\mathbb{R}}$. For real-valued X it suffices to check $\mathcal{A}/\mathcal{B}(\mathbb{R})$-measurability. We write $\{X \le c\} := \{\omega \mid X(\omega) \le c\}$, $\{X < c\}$, etc.

Facts 2.2

Mostly restatements of Section 1.3:

(i) $X : \Omega \to \overline{\mathbb{R}}$ is a r.v. iff $\{X \le c\} \in \mathcal{A}$ for all $c \in \mathbb{R}$ (since $\sigma(\{[-\infty, c]\}) = \mathcal{B}(\overline{\mathbb{R}})$); for real-valued X likewise with $(-\infty, c]$. It suffices to take c in a countable dense $S \subset \mathbb{R}$, e.g. $\mathbb{Q}$, because $\{X > c\} = \bigcup_{q \in S,\, q > c} \{X > q\}$.

(ii) If $\mathcal{A} = \mathcal{P}(\Omega)$, every function $\Omega \to \overline{\mathbb{R}}$ is a r.v.

(iii) If X is a r.v. and $h : \mathbb{R} \to \mathbb{R}$ (or $\overline{\mathbb{R}} \to \overline{\mathbb{R}}$) is Borel measurable, then $h(X)$ is a r.v. (preimages compose); e.g. $|X|$, X^2, $|X|^p$, e^X.

(iv) For r.v.'s $X_1, X_2, \ldots$ and $\alpha_i \in \mathbb{R}$, the following are r.v.'s: $\sum_{i \le n} \alpha_i X_i$ whenever no $\infty - \infty$ occurs; $\sum_{i=1}^{\infty} \alpha_i X_i$ if all $\alpha_i, X_i \ge 0$; $\sup_i X_i$, $\inf_i X_i$, $X_1 \wedge X_2 := \min$, $X_1 \vee X_2 := \max$; $\limsup_i X_i$, $\liminf_i X_i$, and $\lim_i X_i$ if it exists.

(v) $\mathbb{1}_A$ is a r.v. for $A \in \mathcal{A}$. An **elementary** (simple) r.v. is $X = \sum_{i=1}^{n} c_i \mathbb{1}_{A_i}$ with $c_i \in \mathbb{R}$ and disjoint $A_i \in \mathcal{A}$. Every r.v. is a limit of elementary ones (Theorem 1.40): split $X = X^+ - X^-$ with $X^+ := X \vee 0$, $X^- := -(X \wedge 0)$, and for $X \ge 0$ use

$$X_n := \sum_{i=0}^{n2^n - 1} \frac{i}{2^n} \mathbb{1}_{\{\frac{i}{2^n} \le X < \frac{i+1}{2^n}\}} + n \mathbb{1}_{\{X \ge n\}} \nearrow X.$$

On a probability space $(\Omega, \mathcal{A}, P)$ the integral of Section 1.4 reads: $\int \mathbb{1}_A \, dP = P(A)$; $\int \sum c_i \mathbb{1}_{A_i} \, dP = \sum c_i P(A_i)$ for disjoint A_i; for $X \ge 0$, $\int X \, dP = \lim_n \int X_n \, dP \in [0, \infty]$ with elementary $X_n \nearrow X$ (MCT); and in general, if $\min(\int X^+ dP, \int X^- dP) < \infty$,

$$\mathbb{E}[X] := \int_\Omega X \, dP := \int X^+ dP - \int X^- dP \in \overline{\mathbb{R}}$$

is the **expectation** of X. One also writes $\int X \, dP$ or $\int X(\omega) \, P(d\omega)$. Up to Theorem 2.8, everything is a direct translation of Section 1.4.

Definition 2.3 ($\mathcal{L}^p$ and almost sure)

(i) $\mathcal{L}^1 := \mathcal{L}^1(\Omega, \mathcal{A}, P) := \{X : \Omega \to \mathbb{R} \text{ r.v.} \mid \mathbb{E}[|X|] < \infty\}$, and for $p \in [1, \infty)$, $\mathcal{L}^p := \{X \text{ real r.v.} \mid \mathbb{E}[|X|^p] < \infty\}$.

(ii) A property holds **P-almost surely** (P-a.s.) if it holds outside some $N \in \mathcal{A}$ with $P(N) = 0$.

With $\mathcal{N}_0 := \{X \mid X = 0 \ P\text{-a.s.}\}$, $L^p := \mathcal{L}^p/\mathcal{N}_0$ is a Banach space for $\|X\|_p := \mathbb{E}[|X|^p]^{1/p}$ (Theorem 1.70). The expectation is monotone ($0 \le X \le Y$ a.s. $\Rightarrow 0 \le \mathbb{E}[X] \le \mathbb{E}[Y]$) and linear on $\mathcal{L}^1$ (Theorem 1.54).

2.2 Inequalities

Motivation. Most estimates in probability reduce to three inequalities: Jensen (convex functions and expectations), Markov–Chebyshev (tails from moments), and Hölder–Minkowski (products and sums). They are the measure-theoretic results of Section 1.5 specialised to $P(\Omega) = 1$.

Throughout, $(\Omega, \mathcal{A}, P)$ is fixed.

Proposition 2.4 (Jensen's inequality)

Let φ be convex on an open interval $I \subset \mathbb{R}$ and $X \in \mathcal{L}^1$ with $X(\Omega) \subset I$. Then $\mathbb{E}[X] \in I$, $(\varphi \circ X)^- \in \mathcal{L}^1$ and $\varphi(\mathbb{E}[X]) \le \mathbb{E}[\varphi(X)]$.

Proof. This is Theorem 1.63 with $(X, \mathcal{M}, \mu) = (\Omega, \mathcal{A}, P)$; its proof works verbatim when an endpoint of I is $\pm\infty$. ∎

Example 2.5

For $X \in \mathcal{L}^1$: $\mathbb{E}[X]^2 \le \mathbb{E}[X^2]$ ($\varphi(x) = x^2$). Moreover, for $1 \le p \le q$ and any r.v. X (integrability not needed),

$$\|X\|_p = \mathbb{E}[|X|^p]^{1/p} \le \mathbb{E}[|X|^q]^{1/q} = \|X\|_q.$$

Proof. $\varphi(x) = |x|^{q/p}$ is convex. $(|X| \wedge n)^p$ is bounded, hence in $\mathcal{L}^1$, so Jensen gives $\mathbb{E}[(|X| \wedge n)^p]^{q/p} \le \mathbb{E}[(|X| \wedge n)^q]$. Take the power $1/q$ and let $n \to \infty$ (MCT on both sides).

Corollary 2.6

Let X be a r.v.

(i) (Markov) $P(|X| \ge c) \le \frac{1}{c}\mathbb{E}[|X|]$ for $c > 0$. In particular $\mathbb{E}[|X|] = 0 \Rightarrow X = 0$ a.s., and $\mathbb{E}[|X|] < \infty \Rightarrow |X| < \infty$ a.s.

(ii) (Chebyshev) For $X \in \mathcal{L}^1$, $P(|X - \mathbb{E}[X]| \ge c) \le \frac{\mathrm{Var}(X)}{c^2}$, where $\mathrm{Var}(X) := \mathbb{E}[(X - \mathbb{E}[X])^2]$ is the **variance**. More generally $P(|X| \ge c) \le c^{-p}\mathbb{E}[|X|^p]$ for $p > 0$.

Proof. The inequalities are Proposition 1.61 (applied to X, resp. $X - \mathbb{E}[X]$). The two consequences are Lemma 1.62(i) and Theorem 1.52(iii). ∎

Theorem 2.7

(i) (Hölder) For $p, q \in (1,\infty)$, $\frac{1}{p}+\frac{1}{q}=1$, $X \in \mathcal{L}^p$, $Y \in \mathcal{L}^q$: $\|XY\|_1 \le \|X\|_p \|Y\|_q$. For $p=q=2$ this is the Cauchy–Schwarz inequality in the Hilbert space L^2. (ii) (Minkowski) For $p \ge 1$ and $X, Y \in \mathcal{L}^p$: $\|X+Y\|_p \le \|X\|_p + \|Y\|_p$.

Proof. Theorem 1.67(ii) and Step 1 of the proof of Theorem 1.67(i). ∎

2.3 Convergence of random variables

Motivation. A sequence of random variables can converge in several inequivalent ways. Almost sure convergence is about individual sample paths, convergence in probability about the size of the exceptional event, L^p convergence about moments. The Borel–Cantelli lemma is the bridge from probabilities of events to almost sure statements.

Theorem 2.8 (Limit theorems)

(i) (MCT) $0 \le X_1 \le X_2 \le \dots \Rightarrow \lim_n \mathbb{E}[X_n] = \mathbb{E}[\lim_n X_n]$.

(ii) (Fatou) $X_n \ge 0 \Rightarrow \mathbb{E}[\liminf_n X_n] \le \liminf_n \mathbb{E}[X_n]$.

(iii) (DCT) If $|X_n| \le Y \in \mathcal{L}^1$ a.s. for all n and $\lim_n X_n$ exists a.s., then $\mathbb{E}[\lim_n X_n] = \lim_n \mathbb{E}[X_n]$. Here $\lim_n X_n$ means any r.v. X with $X = \lim_n X_n$ a.s., e.g. $X = \lim_n X_n \mathbb{1}_{\Omega\setminus N}$ with $P(N)=0$.

Proof. Theorems 1.53, 1.57, 1.58 together with Remark 1.59; the a.s. limit $\lim_n X_n \mathbb{1}_{\Omega\setminus N}$ is measurable by Corollary 1.36(ii). ∎

Events. For $A_n \in \mathcal{A}$: $\bigcup_i A_i$ = "at least one A_i occurs", $\bigcap_i A_i$ = "all occur",

$$\limsup_n A_n := \bigcap_n \bigcup_{m\ge n} A_m = \text{"infinitely many } A_n \text{ occur" } (A_n \text{ i.o.}),$$

$$\liminf_n A_n := \bigcup_n \bigcap_{m\ge n} A_m = \text{"all but finitely many } A_n \text{ occur"},$$

and $\liminf_n A_n \subset \limsup_n A_n$.

Lemma 2.9 (Borel–Cantelli, part 1)

$\sum_n P(A_n) < \infty \;\Rightarrow\; P(\limsup_n A_n) = 0$.

Proof. $\bigcup_{m\ge n} A_m \searrow \limsup_n A_n$, so by continuity from above and subadditivity $P(\limsup_n A_n) = \lim_n P(\bigcup_{m\ge n} A_m) \le \lim_n \sum_{m\ge n} P(A_m) = 0$. ∎

Lemma 2.10 (Fast convergence in probability)

If $\sum_n P(|X_n| \ge \varepsilon) < \infty$ for every $\varepsilon > 0$, then $P(\lim_n X_n = 0) = 1$.

Proof.

Step 1. For $\varepsilon > 0$ let $N_\varepsilon := \limsup_n \{|X_n| \ge \varepsilon\}$; Borel–Cantelli gives $P(N_\varepsilon) = 0$. For $\omega \notin N_{1/k}$,

$$|X_m(\omega)| < \tfrac{1}{k} \quad \text{for all but finitely many } m. \tag{2.1}$$

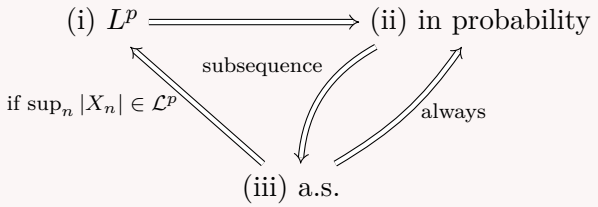
Step 2. $N := \bigcup_k N_{1/k}$ is null, and for $\omega \notin N$, (2.1) holds for every k, i.e. $X_n(\omega) \to 0$. ∎

Definition 2.11 (Modes of convergence)

For real r.v.'s $X, X_1, X_2, \ldots$: (i) **L^p-convergence** ($p \geq 1$): $\|X_n - X\|_p \to 0$; (ii) **convergence in probability**: $P(|X_n - X| \geq \varepsilon) \to 0$ for every $\varepsilon > 0$; (iii) **P-a.s. convergence**: $P(\lim_n X_n = X) = 1$.

Theorem 2.12 (Comparison of the modes)

(i)⇒(ii); (iii)⇒(ii); (ii)⇒(iii) along a subsequence; (iii)⇒(i) if $\sup_n |X_n| \in \mathcal{L}^p$.



Proof.

Step (i)⇒(ii). Chebyshev: $P(|X_n - X| \geq \varepsilon) \leq \varepsilon^{-p}\mathbb{E}[|X_n - X|^p]$.

Step (iii)⇒(ii). $\{\lim_n X_n = X\} = \bigcap_k A_k$ with $A_k := \bigcup_m \bigcap_{n \geq m} \{|X_n - X| \leq \frac{1}{k}\}$. So $P(A_k) = 1$, and continuity from below gives

$$1 = P(A_k) = \lim_m P\Big(\bigcap_{n \geq m} \{|X_n - X| \leq \tfrac{1}{k}\} \Big) \leq \liminf_m P(|X_m - X| \leq \tfrac{1}{k}) \leq 1.$$

Hence $P(|X_m - X| > \frac{1}{k}) \to 0$ for every k.

Step (iii)⇒(i). $Y := \sup_n |X_n| \in \mathcal{L}^p$ and $|X| \leq Y$ a.s., so $|X_n - X|^p \leq 2^p Y^p \in \mathcal{L}^1$; since $|X_n - X|^p \to 0$ a.s., the DCT gives $\mathbb{E}[|X_n - X|^p] \to 0$.

Step (ii)⇒(iii) along a subsequence. Choose $n_1 < n_2 < \ldots$ with $P(|X_{n_k} - X| \geq \frac{1}{k}) < 2^{-k}$. For $\varepsilon > 0$, the finitely many k with $\frac{1}{k} \geq \varepsilon$ contribute a finite amount, and for $\frac{1}{k} < \varepsilon$, $P(|X_{n_k} - X| \geq \varepsilon) \leq 2^{-k}$. Hence $\sum_k P(|X_{n_k} - X| \geq \varepsilon) < \infty$, and Lemma 2.10 gives $X_{n_k} \to X$ a.s. ∎

2.4 Image measure and distribution of random variables

Motivation. What matters about a random variable is not the function X on an abstract Ω but how it distributes probability over $\mathbb{R}$. The image measure P_X (equivalently, the distribution function F_X) is that information; expectations of $\Phi(X)$ can be computed on $\mathbb{R}$ alone.

Let $(\Omega, \mathcal{A})$ and $(\tilde{\Omega}, \tilde{\mathcal{A}})$ be measurable spaces.

Proposition 2.13 (Image measure)

If $T : \Omega \to \tilde{\Omega}$ is $\mathcal{A}/\tilde{\mathcal{A}}$-measurable and P a probability measure on $(\Omega, \mathcal{A})$, then $\tilde{P}(\tilde{A}) := P(T \in \tilde{A}) = P(T^{-1}(\tilde{A}))$, $\tilde{A} \in \tilde{\mathcal{A}}$, is a probability measure on $(\tilde{\Omega}, \tilde{\mathcal{A}})$, the **image** (push-forward, induced) measure, written $P \circ T^{-1}$, $T(P)$ or P_T.

Proof. $\tilde{P} \geq 0$, $\tilde{P}(\emptyset) = 0$, $\tilde{P}(\tilde{\Omega}) = 1$. For disjoint $\tilde{A}_i$ the preimages $T^{-1}(\tilde{A}_i)$ are disjoint and $T^{-1}(\bigcup_i \tilde{A}_i) = \bigcup_i T^{-1}(\tilde{A}_i)$, so σ-additivity of P gives $\tilde{P}(\bigcup_i \tilde{A}_i) = \sum_i \tilde{P}(\tilde{A}_i)$. ∎

Definition 2.14 (Distribution, distribution function)

The **distribution** of a real r.v. X on $(\Omega, \mathcal{A}, P)$ is $P_X(B) := P(X \in B)$, $B \in \mathcal{B}(\mathbb{R})$. Its **distribution function** is

$$F_X(t) := P(X \le t) = P_X((-\infty, t]), \qquad t \in \mathbb{R}. \tag{2.2}$$

For a probability measure μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, $F(x) := \mu((-\infty, x])$ is its distribution function. (For extended real-valued X one may use $B \in \mathcal{B}(\overline{\mathbb{R}})$ for the distribution, but distribution functions need real-valued X.)

Remark 2.15

(i) F is increasing, right continuous and normalised ($F(-\infty) = 0$, $F(+\infty) = 1$). Monotonicity is clear; the rest is continuity from above and below of μ, since $b_n \searrow a \Rightarrow (-\infty, b_n] \searrow (-\infty, a]$, $(-\infty, a] \searrow \emptyset$ as $a \searrow -\infty$ and $(-\infty, b] \nearrow \mathbb{R}$ as $b \nearrow \infty$.

Conversely, every such F is the distribution function of a unique probability measure μ on $\mathcal{B}(\mathbb{R})$, so every probability measure on $\mathbb{R}$ is the distribution of a r.v. (namely $X(x) = x$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu)$).

[Supplement] *Proof.* Uniqueness: the intervals $(-\infty, t]$ form a $\cap$-stable generator of $\mathcal{B}(\mathbb{R})$ (Theorem 1.9). Existence: on $((0,1), \mathcal{B}, \lambda)$ let $G(u) := \inf\{t \mid F(t) \ge u\} \in \mathbb{R}$ (finite by normalisation). By right continuity the infimum is attained, so $G(u) \le t \iff u \le F(t)$. Hence $\lambda(G \le t) = \lambda((0, F(t)]) = F(t)$, and $\mu := \lambda \circ G^{-1}$ works.

(ii) In probability a r.v. is known through its distribution. For Borel $\Phi : \mathbb{R} \to \mathbb{R}$ with $\Phi(X) \in \mathcal{L}^1$ or $\Phi \ge 0$,

$$\mathbb{E}[\Phi(X)] = \int_\Omega \Phi(X)\, dP = \int_{\mathbb{R}} \Phi\, dP_X = \int_{\mathbb{R}} \Phi(s)\, dF_X(s) = \int_{\mathbb{R}} \Phi(s) F_X'(s)\, ds.$$

The third equality is notation. The last holds if F_X is continuously differentiable (more generally, absolutely continuous).

[Supplement] *Second equality (transformation theorem).* For $\Phi = \mathbb{1}_B$ both sides equal $P(X \in B)$; by linearity it holds for simple $\Phi \ge 0$, by the MCT for $\Phi \ge 0$, and by splitting $\Phi = \Phi^+ - \Phi^-$ in general.

Last equality. If $F_X \in C^1$, then $F_X(x) - F_X(-n) = \int_{-n}^x F_X'(t)\, dt$ by the fundamental theorem of calculus and Theorem 1.60. Letting $n \to \infty$ (MCT, $F_X' \ge 0$) gives $F_X(x) = \int_{-\infty}^x F_X'\, dt$, so P_X has density F_X' (Remark 2.18(ii)). Then $\int \Phi\, dP_X = \int \Phi F_X'\, ds$ follows by the same indicator–simple–MCT argument.

Remark 2.16

(i) The **jump** of F at x is $F(x) - F(x-) = \lim_n \mu((x - \frac{1}{n}, x]) = \mu(\{x\})$. Hence F is continuous iff $\mu(\{x\}) = 0$ for all x ("μ is continuous").

(ii) An increasing bounded F has at most countably many discontinuities. [Supplement] *Reason:* there are at most k points with jump $\ge 1/k$, since the jumps add up to at most 1.

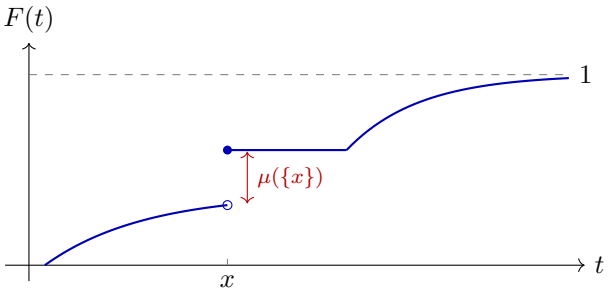


Figure. *A distribution function is increasing and right continuous. Its jump at x is the mass $\mu(\{x\})$ of the atom at x; between atoms it may grow continuously or stay constant.*

Definition 2.17 (Discrete and absolutely continuous distributions)

(i) F (or μ) is **discrete** if $\mu(S) = 1$ for a countable $S \subset \mathbb{R}$. Then μ is determined by the weights $\mu(\{x\})$, $x \in S$, and $F(x) = \sum_{y \in S,\, y \le x} \mu(\{y\})$ is a step function.

(ii) F (or μ) is **absolutely continuous** if there is a measurable **density** $f \ge 0$ with

$$F(x) = \int_{-\infty}^{x} f(t)\, dt, \tag{2.3}$$

respectively

$$\mu(A) = \int_A f(t)\, dt = \int_{-\infty}^{\infty} \mathbb{1}_A f\, dt \quad \text{for all } A \in \mathcal{B}(\mathbb{R}). \tag{2.4}$$

In particular $\int_{-\infty}^{\infty} f\, dt = 1$.

Remark 2.18

(i) Every measurable $f \ge 0$ with $\int f\, dt = 1$ defines a probability measure $A \mapsto \int_A f\, dt$ (Theorem 1.47). (ii) (2.3)⇒(2.4), because F determines a probability measure uniquely on the $\cap$-stable generator $\{(-\infty, x]\}$ of $\mathcal{B}(\mathbb{R})$ (Theorem 1.9); the converse is trivial.

Example 2.19

(i) (*Uniform on $[a,b]$*) $f = \frac{1}{b-a}\mathbb{1}_{[a,b]}$, $F(x) = 0$ for $x \le a$, $\frac{x-a}{b-a}$ on $[a,b]$, 1 for $x \ge b$.

(ii) (*Exponential with parameter $\alpha > 0$*) $f(x) = \alpha e^{-\alpha x}\mathbb{1}_{[0,\infty)}(x)$, $F(x) = (1-e^{-\alpha x})\mathbb{1}_{[0,\infty)}(x)$. It is the continuous analogue of the geometric distribution: $\int_k^{k+1} f = e^{-\alpha k}(1-e^{-\alpha}) = (1-p)^k p$ with $p := 1 - e^{-\alpha}$.

(iii) (*Normal $N(m,\sigma^2)$*) Density $f_{m,\sigma^2}(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-m)^2/(2\sigma^2)}$. The distribution function has no closed form, but the substitution $z = (y-m)/\sigma$ gives $F_{m,\sigma^2}(x) = F_{0,1}\big(\frac{x-m}{\sigma}\big)$. If $X \sim N(m,\sigma^2)$, then $\mathbb{E}[X] = m$ and $\operatorname{Var}(X) = \sigma^2$. [Supplement] *Reason:* after substituting z, $\mathbb{E}[X] - m = \sigma \int z f_{0,1}(z)\, dz = 0$ by symmetry, and $\operatorname{Var}(X) = \sigma^2 \int z^2 f_{0,1}\, dz = \sigma^2$ by integrating by parts ($\int z \cdot z e^{-z^2/2} dz = \int e^{-z^2/2} dz$). The central moments are

$$\mathbb{E}[|X-m|^p] = 2\int_0^\infty x^p \frac{e^{-x^2/(2\sigma^2)}}{\sqrt{2\pi\sigma^2}}\, dx = \frac{2^{p/2}\sigma^p}{\sqrt{\pi}}\,\Gamma\Big(\frac{p+1}{2}\Big) \qquad (y = x^2/(2\sigma^2)),$$

proportional to σ^p.

(iv) (*Binomial*) Repeat an experiment with finite state space S n times: $\Omega = S^n$ with the uniform distribution, and call $x_i \in S_0 \subset S$ a success, $p := |S_0|/|S|$. The event $A_k =$ "exactly k successes" has $P(A_k) = \binom{n}{k}|S_0|^k|S \setminus S_0|^{n-k}/|S|^n = \binom{n}{k}p^k(1-p)^{n-k}$. The **binomial distribution** is $B(n,p) := \sum_{k=0}^n \binom{n}{k}p^k(1-p)^{n-k}\delta_k$.

(v) (*Poisson limit*) With $p_n = \lambda/n$,

$$\binom{n}{k}\left(\frac{\lambda}{n}\right)^k\left(1-\frac{\lambda}{n}\right)^{n-k} = \frac{\lambda^k}{k!}\underbrace{\frac{n(n-1)\cdots(n-k+1)}{n^k}}_{\to 1}\underbrace{\left(1-\frac{\lambda}{n}\right)^{n-k}}_{\to e^{-\lambda}} \to \frac{\lambda^k}{k!}e^{-\lambda}.$$

So for large n and small p, $B(n,p)$ is close to the **Poisson distribution** $\pi_\lambda := e^{-\lambda}\sum_{j\geq 0}\frac{\lambda^j}{j!}\delta_j$ with $\lambda = np$.

2.5 Variance and covariance

Motivation. The expectation is the best constant prediction of X; the variance measures the risk of that prediction. Covariance measures how two variables fluctuate together, and uncorrelatedness is exactly what makes variances add.

Example 2.20

For the uniform distribution on $\{\omega_1,\dots,\omega_n\}$, $\mathbb{E}[X] = \frac{1}{n}\sum_i X(\omega_i)$ is the arithmetic mean.

Definition 2.21 (Variance)

For $X \in \mathcal{L}^1$, $\operatorname{Var}(X) := \sigma^2(X) := \mathbb{E}[(X-\mathbb{E}[X])^2] \in [0,\infty]$ (the mean square prediction error).

Remark 2.22

For $X \in \mathcal{L}^1$: (i) $\operatorname{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$; (ii) $\operatorname{Var}(X) = 0 \iff P(X = \mathbb{E}[X]) = 1$; (iii) $\operatorname{Var}(X) < \infty \iff X \in \mathcal{L}^2$.
[Supplement] *Proof.*
(i) Expand $(X-m)^2 = X^2 - 2mX + m^2$ with $m = \mathbb{E}[X]$; if $\mathbb{E}[X^2] = \infty$ both sides are ∞.
(ii) Corollary 2.6(i) applied to $(X-m)^2$.
(iii) Use (i) and $X^2 \leq 2(X-m)^2 + 2m^2$.

Definition 2.23 (Covariance)

For $X, Y \in \mathcal{L}^2$ (so $XY \in \mathcal{L}^1$ by Cauchy–Schwarz)

$$\operatorname{Cov}(X,Y) := \mathbb{E}[(X-\mathbb{E}[X])(Y-\mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]. \tag{2.5}$$

Remark 2.24

(i) $\operatorname{Var}(aX+b) = a^2\operatorname{Var}(X)$ for $X \in \mathcal{L}^1$. (ii) $\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X,Y)$ for $X, Y \in \mathcal{L}^2$. Both follow by expanding the square, using linearity.

Definition 2.25 (Uncorrelated)

$X, Y \in \mathcal{L}^2$ are **uncorrelated** if

$$\operatorname{Cov}(X,Y) = 0 \quad (\iff \operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y)). \tag{2.6}$$

Remark 2.26 (Bienaymé)

If $X_1, \dots, X_n \in \mathcal{L}^2$ are pairwise uncorrelated, then $\operatorname{Var}\left(\sum_i X_i\right) = \sum_i \operatorname{Var}(X_i)$, by expanding $\operatorname{Var}(\sum_i X_i) = \sum_{i,j} \operatorname{Cov}(X_i, X_j)$.

2.6 The monotone class theorem

Motivation. To prove an identity for all bounded measurable functions it is often enough to prove it for a small, multiplicatively closed class (e.g. continuous functions or products of indicators) and check that the identity survives monotone limits. The monotone class theorem makes this precise; it plays the role that Stone–Weierstrass plays in analysis.

Definition 2.27 (Monotone vector space)

A vector space $\mathcal{H}$ of bounded real functions on Ω is a **monotone vector space** (MVS) if (i) $1 = \mathbb{1}_\Omega \in \mathcal{H}$, and (ii) whenever $f_n \in \mathcal{H}$ with $0 \le f_1 \le f_2 \le \cdots \nearrow f$ and f bounded, then $f \in \mathcal{H}$.

Lemma 2.28

An MVS is closed under uniform convergence.

Proof. Let $f_n \in \mathcal{H}$ with $\|f - f_n\|_\infty := \sup_\omega |f(\omega) - f_n(\omega)| \to 0$.

Step 1. We may assume $f \ge 0$: otherwise replace f, f_n by $f + \|f\|_\infty$, $f_n + \|f\|_\infty$, using that constants lie in $\mathcal{H}$.

Step 2. Choose a subsequence with $\varepsilon_k := \|f_{n_k} - f_{n_{k+1}}\|_\infty$ summable and put $a_k := \sum_{n \ge k} \varepsilon_n \searrow 0$ and $g_k := f_{n_k} - a_k + 2a_1 \in \mathcal{H}$. Then $g_{k+1} - g_k = f_{n_{k+1}} - f_{n_k} + \varepsilon_k \ge 0$, $g_k \nearrow f + 2a_1$ (bounded), and $g_1 = f_{n_1} + a_1 \ge f \ge 0$ because $f - f_{n_1} = \sum_k (f_{n_{k+1}} - f_{n_k}) \le a_1$.

Step 3. By Definition 2.27(ii), $f + 2a_1 \in \mathcal{H}$, hence $f \in \mathcal{H}$. ∎

Notation. For a class $\mathcal{M}$ of functions on Ω, $\sigma(\mathcal{M})$ is the smallest σ-algebra making all $f \in \mathcal{M}$ measurable, and $\sigma(\mathcal{M})_b$ the bounded $\sigma(\mathcal{M})$-measurable functions.

Theorem 2.29 (Monotone class theorem, algebra form)

Let $\mathcal{M}$ be a multiplicative class of bounded functions ($f, g \in \mathcal{M} \Rightarrow fg \in \mathcal{M}$) and $\mathcal{H}$ an MVS with $\mathcal{M} \subset \mathcal{H}$. Then $\sigma(\mathcal{M})_b \subset \mathcal{H}$.

Proof.

Step 1 (algebra). $\mathcal{M}_0 := \operatorname{span}(\mathcal{M}, 1) \subset \mathcal{H}$ is an algebra with $\sigma(\mathcal{M}_0) = \sigma(\mathcal{M})$. Its uniform closure $\overline{\mathcal{M}_0}$ is again an algebra and a linear subspace, satisfies $\overline{\mathcal{M}_0} \subset \mathcal{H}$ (Lemma 2.28), and $\sigma(\overline{\mathcal{M}_0}) = \sigma(\mathcal{M})$ since uniform limits of measurable functions are measurable.

Step 2 (truncation). $f \in \overline{\mathcal{M}_0} \Rightarrow f \wedge \alpha \in \overline{\mathcal{M}_0}$ for $\alpha \in \mathbb{R}$. With $L := \|f\|_\infty$, Weierstrass gives polynomials P_ε with $|x \wedge \alpha - P_\varepsilon(x)| < \varepsilon$ on $[-L, L]$, so $\|f \wedge \alpha - P_\varepsilon(f)\|_\infty < \varepsilon$ and $P_\varepsilon(f) \in \overline{\mathcal{M}_0}$.

Step 3 (reduction to indicators). Every $f \in \sigma(\mathcal{M})_b$ is a uniform limit of $\sigma(\mathcal{M})$-elementary functions (Note 1.41), and $\mathcal{H}$ is closed under uniform limits. So it suffices that $\mathcal{S} := \{A \in \sigma(\overline{\mathcal{M}_0}) \mid \mathbb{1}_A \in \mathcal{H}\}$ equals $\sigma(\overline{\mathcal{M}_0})$.

Step 4 ($\mathcal{S}$ is a Dynkin system). $\Omega \in \mathcal{S}$ since $1 \in \mathcal{H}$; $\mathbb{1}_{A^c} = 1 - \mathbb{1}_A$; for disjoint A_i, $\sum_{i \le n} \mathbb{1}_{A_i} \nearrow \mathbb{1}_{\bigcup A_i}$, so MVS (ii) applies.

Step 5 (a $\cap$-stable generator). Let $\mathcal{E} := \{A \in \sigma(\overline{\mathcal{M}_0}) \mid \exists f_n \in \overline{\mathcal{M}_0},\ f_n \geq 0,\ f_n \nearrow \mathbb{1}_A\}$. $\mathcal{E}$ is $\cap$-stable (if $f_n \nearrow \mathbb{1}_A$, $g_n \nearrow \mathbb{1}_B$, then $f_n g_n \nearrow \mathbb{1}_{A\cap B}$), and $\mathcal{E} \subset \mathcal{S}$ by MVS (ii). Theorem 1.8 gives

$$\sigma(\mathcal{E}) = \mathcal{D}(\mathcal{E}) \subset \mathcal{S} \subset \sigma(\overline{\mathcal{M}_0}). \tag{2.7}$$

Step 6. For $f \in \overline{\mathcal{M}_0}$ and $\alpha \in \mathbb{R}$, $(n(f-\alpha)^+) \wedge 1 \in \overline{\mathcal{M}_0}$ by Step 2 (note $(f-\alpha)^+ = (f-\alpha) - ((f-\alpha) \wedge 0)$), and it increases to $\mathbb{1}_{\{f>\alpha\}}$. So $\{f > \alpha\} \in \mathcal{E}$; these sets generate $\sigma(\overline{\mathcal{M}_0})$, hence equality holds throughout (2.7). ■

Example 2.30

Let μ_1, μ_2 be probability measures on a metric space (S, d) with $\int f\, d\mu_1 = \int f\, d\mu_2$ for all $f \in C_b(S)$ (bounded continuous). Then $\mu_1 = \mu_2$.
Proof. $\mathcal{H} := \{f \in \mathcal{B}(S)_b \mid \int f\, d\mu_1 = \int f\, d\mu_2\}$ is an MVS (MCT) containing the multiplicative class $C_b(S)$, so $\mathcal{H} \supset \sigma(C_b(S))_b$. Clearly $\sigma(C_b(S)) \subset \mathcal{B}(S)$. Conversely, for closed F the function $f := d(\cdot, F) \wedge 1$ is Lipschitz, bounded, and $F = \{f = 0\} \in \sigma(C_b(S))$; hence $\mathcal{B}(S) \subset \sigma(C_b(S))$, all $\mathbb{1}_A$, $A \in \mathcal{B}(S)$, lie in $\mathcal{H}$, and $\mu_1 = \mu_2$.

2.7 Conditional expectation

Motivation. If only the information in a sub-σ-algebra $\mathcal{A}$ is available, what is the best guess for X? For $X \in L^2$ the answer is the orthogonal projection onto $L^2(\mathcal{A})$, characterised by $\mathbb{E}[X\mathbb{1}_A] = \mathbb{E}[Y\mathbb{1}_A]$ for $A \in \mathcal{A}$. That characterisation makes sense for integrable (and for nonnegative) X and defines the conditional expectation.

Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\mathcal{A} \subset \mathcal{F}$ a sub-σ-algebra.

Heuristics. For $X \in L^2(\Omega, \mathcal{F}, P)$ seek $Y \in L^2(\Omega, \mathcal{A}, P)$ with

$$\mathbb{E}[(X-Y)^2] \leq \mathbb{E}[(X-Y_0)^2] \qquad \forall Y_0 \in L^2(\Omega, \mathcal{A}, P). \tag{2.8}$$

If Y satisfies (2.8), then for $Z \in L^2(\mathcal{A})$ and $\varepsilon \neq 0$, $\mathbb{E}[(X-Y)^2] \leq \mathbb{E}[(X-Y-\varepsilon Z)^2]$ gives $2\varepsilon\mathbb{E}[(X-Y)Z] \leq \varepsilon^2\mathbb{E}[Z^2]$; letting $\varepsilon \to 0$ from both sides, $\mathbb{E}[XZ] = \mathbb{E}[YZ]$ for all $\mathcal{A}$-measurable $Z \in L^2$. Such Y exists and is a.s. unique: it is the orthogonal projection of X onto the closed subspace $L^2(\Omega, \mathcal{A}, P)$.

Definition 2.31 (Conditional expectation)

For $X \in \mathcal{L}^1(\Omega, \mathcal{F}, P)$, a **conditional expectation of X given $\mathcal{A}$** is any r.v. Y with (i) Y is $\mathcal{A}$-measurable ("$Y \in \mathcal{A}$"), and (ii) $\mathbb{E}[X\mathbb{1}_A] = \mathbb{E}[Y\mathbb{1}_A]$ for all $A \in \mathcal{A}$. Such Y exists, is integrable and is a.s. unique; we write $Y = \mathbb{E}[X \mid \mathcal{A}]$.

Integrability. With $A = \{Y > 0\} \in \mathcal{A}$: $\mathbb{E}[Y^+] = \mathbb{E}[Y\mathbb{1}_A] = \mathbb{E}[X\mathbb{1}_A] \leq \mathbb{E}[|X|]$ and $\mathbb{E}[Y^-] = \mathbb{E}[-X\mathbb{1}_{A^c}] \leq \mathbb{E}[|X|]$.

Uniqueness. If Y' also satisfies (i), (ii), let $A_n := \{Y - Y' \geq \frac{1}{n}\} \in \mathcal{A}$. Then $0 = \mathbb{E}[(Y-Y')\mathbb{1}_{A_n}] \geq \frac{1}{n}P(A_n)$, so $P(Y > Y') \leq \sum_n P(A_n) = 0$; symmetrically $P(Y' > Y) = 0$.

Existence could be obtained from the L^2 projection and truncation; we derive it from the Radon–Nikodym theorem, which exhibits $\mathbb{E}[X \mid \mathcal{A}]$ as a density.

Definition 2.32 (Absolute continuity)

For finite measures ν, μ on $(\Omega, \mathcal{A})$, ν is **absolutely continuous** w.r.t. μ, $\nu \ll \mu$, if $\mu(N) = 0 \Rightarrow \nu(N) = 0$ for $N \in \mathcal{A}$.

Theorem 2.33 (Radon–Nikodym)

Let ν, μ be finite measures on $(\Omega, \mathcal{A})$ with $\nu \ll \mu$. Then there is $\varphi \in \mathcal{L}^1(\Omega, \mathcal{A}, \mu)$, $\varphi \geq 0$, with

$$\nu(A) = \int_A \varphi \, d\mu \qquad \text{for all } A \in \mathcal{A},$$

and φ is μ-a.e. unique. It is the **Radon–Nikodym derivative** $\frac{d\nu}{d\mu}$. (The theorem extends to σ-finite ν, μ, without integrability of the density.)

Proof [Supplement] (von Neumann).

Step 1 (a representing function). Put $\rho := \mu + \nu$ (finite). For $h \in L^2(\rho)$, Cauchy–Schwarz gives $\left|\int h \, d\nu\right| \leq \int |h| \, d\rho \leq \rho(\Omega)^{1/2} \|h\|_{L^2(\rho)}$, so $h \mapsto \int h \, d\nu$ is a bounded linear functional on the Hilbert space $L^2(\rho)$ (Theorem 1.70). By the Riesz representation theorem there is $g \in L^2(\rho)$ with

$$\int h \, d\nu = \int hg \, d\rho \qquad \forall h \in L^2(\rho). \tag{2.9}$$

Step 2 ($0 \leq g \leq 1$). With $h = \mathbb{1}_{\{g<0\}}$: $0 \leq \nu(g < 0) = \int_{\{g<0\}} g \, d\rho \leq 0$, so $\rho(g < 0) = 0$. With $h = \mathbb{1}_{\{g>1\}}$: $\nu(g > 1) = \int_{\{g>1\}} g \, d\rho \geq \rho(g > 1) \geq \nu(g > 1)$, forcing $\rho(g > 1) = 0$. Modify g on a ρ-null set so that $0 \leq g \leq 1$ everywhere.

Step 3. (2.9) reads $\int h(1-g) \, d\nu = \int hg \, d\mu$ for bounded measurable h, hence for all measurable $h \geq 0$ by the MCT. Let $N := \{g = 1\}$. With $h = \mathbb{1}_N$: $\mu(N) = \int_N (1-g) \, d\nu = 0$, so $\nu(N) = 0$ by $\nu \ll \mu$.

Step 4. For $A \in \mathcal{A}$ take $h = \mathbb{1}_A(1 + g + \cdots + g^n)$: $\int_A (1 - g^{n+1}) \, d\nu = \int_A g(1 + g + \cdots + g^n) \, d\mu$. As $n \to \infty$, $1 - g^{n+1} \nearrow \mathbb{1}_{N^c}$ and $g(1 + \cdots + g^n) \nearrow \varphi := \frac{g}{1-g} \mathbb{1}_{N^c}$. By the MCT, $\nu(A \cap N^c) = \int_A \varphi \, d\mu$, and $\nu(A \cap N^c) = \nu(A)$. Taking $A = \Omega$ shows $\int \varphi \, d\mu = \nu(\Omega) < \infty$.

Step 5 (uniqueness). If φ, φ' both work, $\int_A (\varphi - \varphi') \, d\mu = 0$ for all A, so $\varphi = \varphi'$ μ-a.e. by Lemma 1.62(ii). ∎

Existence of $\mathbb{E}[X \mid \mathcal{A}]$. For $X \in \mathcal{L}^1(\Omega, \mathcal{F}, P)$ let $\nu^{\pm}(A) := \int_A X^{\pm} \, dP$, $A \in \mathcal{A}$. These are finite measures on $(\Omega, \mathcal{A})$ with $\nu^{\pm} \ll P|_{\mathcal{A}}$, so Theorem 2.33 yields $\mathcal{A}$-measurable densities $\frac{d\nu^{\pm}}{dP|_{\mathcal{A}}} \geq 0$ in $\mathcal{L}^1(\Omega, \mathcal{A}, P)$. With $Y := \frac{d\nu^+}{dP|_{\mathcal{A}}} - \frac{d\nu^-}{dP|_{\mathcal{A}}}$, for every $A \in \mathcal{A}$

$$\int_A X \, dP = \nu^+(A) - \nu^-(A) = \int_A \frac{d\nu^+}{dP|_{\mathcal{A}}} \, dP - \int_A \frac{d\nu^-}{dP|_{\mathcal{A}}} \, dP = \int_A Y \, dP,$$

so $Y = \mathbb{E}[X \mid \mathcal{A}]$. With $A = \Omega$: $\mathbb{E}[\mathbb{E}[X \mid \mathcal{A}]] = \mathbb{E}[X]$.

Interpretation. $\mathcal{A}$ is the available information (for each $A \in \mathcal{A}$ we can tell whether A occurred), and $\mathbb{E}[X \mid \mathcal{A}]$ is the best guess for X given that information.

Example 2.34

Let $X \in \mathcal{L}^1(\Omega, \mathcal{F}, P)$.

(i) X $\mathcal{A}$-measurable $\Rightarrow \mathbb{E}[X \mid \mathcal{A}] = X$.

(ii) $\mathcal{A} = \{\emptyset, \Omega\} \Rightarrow \mathbb{E}[X \mid \mathcal{A}] = \mathbb{E}[X]$.

(iii) $\mathcal{A} = \{\emptyset, A, A^c, \Omega\}$ with $0 < P(A) < 1 \Rightarrow \mathbb{E}[X \mid \mathcal{A}] = \frac{\mathbb{E}[X\mathbb{1}_A]}{P(A)}\mathbb{1}_A + \frac{\mathbb{E}[X\mathbb{1}_{A^c}]}{P(A^c)}\mathbb{1}_{A^c}$.

(iv) If X is independent of $\mathcal{A}$ (i.e. $\mathbb{1}_A$ and X are independent for all $A \in \mathcal{A}$, Section 3.4), then $\mathbb{E}[X\mathbb{1}_A] = \mathbb{E}[X]\mathbb{E}[\mathbb{1}_A] = \mathbb{E}[\mathbb{E}[X]\mathbb{1}_A]$, so $\mathbb{E}[X \mid \mathcal{A}] = \mathbb{E}[X]$: no additional information.

In each case, check (i),
(ii) of Definition 2.31 directly.

Theorem 2.35 (Properties of conditional expectation)

Let $X, Y \in \mathcal{L}^1(\Omega, \mathcal{F}, P)$; (in)equalities hold P-a.s.

(i) (Linearity) $\mathbb{E}[\alpha X + \beta Y \mid \mathcal{A}] = \alpha\mathbb{E}[X \mid \mathcal{A}] + \beta\mathbb{E}[Y \mid \mathcal{A}]$.

(ii) (Monotonicity) $X \le Y \Rightarrow \mathbb{E}[X \mid \mathcal{A}] \le \mathbb{E}[Y \mid \mathcal{A}]$.

(iii) (Pulling out what is known) Y $\mathcal{A}$-measurable, $XY \in \mathcal{L}^1 \Rightarrow \mathbb{E}[XY \mid \mathcal{A}] = Y\mathbb{E}[X \mid \mathcal{A}]$.

(iv) (Tower property) For $\mathcal{A}_1 \subset \mathcal{A}_2 \subset \mathcal{F}$: $\mathbb{E}[\mathbb{E}[X \mid \mathcal{A}_1] \mid \mathcal{A}_2] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{A}_2] \mid \mathcal{A}_1] = \mathbb{E}[X \mid \mathcal{A}_1]$.

(v) (Contraction) $|\mathbb{E}[X \mid \mathcal{A}]| \le \mathbb{E}[|X| \mid \mathcal{A}]$.

(vi) (DCT) If $|X_n| \le Y \in \mathcal{L}^1$ and $X_n \to X$ a.s., then $\mathbb{E}[X_n \mid \mathcal{A}] \to \mathbb{E}[X \mid \mathcal{A}]$ in $L^1(\Omega, \mathcal{A}, P)$ and a.s.

(vii) (Jensen) $\varphi : \mathbb{R} \to \mathbb{R}$ convex, $\varphi(X) \in \mathcal{L}^1 \Rightarrow \varphi(\mathbb{E}[X \mid \mathcal{A}]) \le \mathbb{E}[\varphi(X) \mid \mathcal{A}]$.

Proof. (i) For $A \in \mathcal{A}$, $\mathbb{E}[(\alpha X + \beta Y)\mathbb{1}_A] = \alpha\mathbb{E}[\mathbb{E}[X \mid \mathcal{A}]\mathbb{1}_A] + \beta\mathbb{E}[\mathbb{E}[Y \mid \mathcal{A}]\mathbb{1}_A] = \mathbb{E}[(\alpha\mathbb{E}[X \mid \mathcal{A}] + \beta\mathbb{E}[Y \mid \mathcal{A}])\mathbb{1}_A]$, and the bracket is $\mathcal{A}$-measurable.

(ii) $A := \{\mathbb{E}[X \mid \mathcal{A}] > \mathbb{E}[Y \mid \mathcal{A}]\} \in \mathcal{A}$ and $0 \ge \mathbb{E}[(X - Y)\mathbb{1}_A] = \mathbb{E}[(\mathbb{E}[X \mid \mathcal{A}] - \mathbb{E}[Y \mid \mathcal{A}])\mathbb{1}_A]$ by (i); the integrand is > 0 on A, so $P(A) = 0$.

(iii) First X^+Y^+: take $\mathcal{A}$-elementary $Y_n^+ = \sum_i c_{i,n}\mathbb{1}_{A_{i,n}} \nearrow Y^+$. For $A \in \mathcal{A}$, the MCT twice gives

$$\mathbb{E}[X^+Y^+\mathbb{1}_A] = \lim_n \sum_i c_{i,n}\mathbb{E}[X^+\mathbb{1}_{A_{i,n}\cap A}] = \lim_n \sum_i c_{i,n}\mathbb{E}[\mathbb{E}[X^+ \mid \mathcal{A}]\mathbb{1}_{A_{i,n}\cap A}] = \mathbb{E}[\mathbb{E}[X^+ \mid \mathcal{A}]Y^+\mathbb{1}_A].$$

So $\mathbb{E}[X^+Y^+ \mid \mathcal{A}] = Y^+\mathbb{E}[X^+ \mid \mathcal{A}]$; the same holds for $X^\pm Y^\mp$, X^-Y^-, and linearity gives the claim.

(iv) $\mathbb{E}[X \mid \mathcal{A}_1]$ is $\mathcal{A}_2$-measurable, so the first expression equals $\mathbb{E}[X \mid \mathcal{A}_1]$ by Example 2.34(i). For $A \in \mathcal{A}_1 \subset \mathcal{A}_2$: $\mathbb{E}[\mathbb{1}_A\mathbb{E}[\mathbb{E}[X \mid \mathcal{A}_2] \mid \mathcal{A}_1]] = \mathbb{E}[\mathbb{1}_A\mathbb{E}[X \mid \mathcal{A}_2]] = \mathbb{E}[\mathbb{1}_A X] = \mathbb{E}[\mathbb{1}_A\mathbb{E}[X \mid \mathcal{A}_1]]$.

(v) $\pm X \le |X|$ with (i), (ii).

(vi) $X \in \mathcal{L}^1$ by the DCT. By (v) and (i), $\mathbb{E}|\mathbb{E}[X_n \mid \mathcal{A}] - \mathbb{E}[X \mid \mathcal{A}]| \le \mathbb{E}[\mathbb{E}[|X_n - X| \mid \mathcal{A}]] = \mathbb{E}|X_n - X| \to 0$, which is L^1 convergence. [Supplement] *A.s. convergence:* $Z_n := \sup_{k \ge n} |X_k - X| \le 2Y$ decreases to 0 a.s., and $|\mathbb{E}[X_n \mid \mathcal{A}] - \mathbb{E}[X \mid \mathcal{A}]| \le \mathbb{E}[Z_n \mid \mathcal{A}]$. By (ii), $\mathbb{E}[Z_n \mid \mathcal{A}]$ decreases a.s. to some $Z \ge 0$. Then $\mathbb{E}[Z] \le \mathbb{E}[\mathbb{E}[Z_n \mid \mathcal{A}]] = \mathbb{E}[Z_n] \to 0$ by the DCT, so $Z = 0$ a.s.

(vii) [Supplement] A convex φ on $\mathbb{R}$ is the supremum of countably many affine functions $\ell_k(x) = a_k x + b_k$ (supporting lines at rational points). For each k, $\ell_k(X) \le \varphi(X)$ gives by (i),

(ii) $\ell_k(\mathbb{E}[X \mid \mathcal{A}]) = \mathbb{E}[\ell_k(X) \mid \mathcal{A}] \le \mathbb{E}[\varphi(X) \mid \mathcal{A}]$ outside a null set N_k. Off $\bigcup_k N_k$ take the supremum over k. ∎

Conditional expectations also make sense for nonnegative, not necessarily integrable r.v.'s.

Definition 2.36 (Conditional expectation of nonnegative r.v.'s)

For $\mathcal{F}$-measurable $X : \Omega \to [0,\infty]$, $\mathbb{E}[X \mid \mathcal{A}]$ is any Y with (i) Y $\mathcal{A}$-measurable and (ii) $\mathbb{E}[X\mathbb{1}_A] = \mathbb{E}[Y\mathbb{1}_A]$ for all $A \in \mathcal{A}$. Such Y exists, is a.s. unique and a.s. $[0,\infty]$-valued.

Existence. $X_n := X \wedge n \in \mathcal{L}^1$ increases, so $\mathbb{E}[X_n \mid \mathcal{A}] \le \mathbb{E}[X_m \mid \mathcal{A}]$ a.s. for $n \le m$ by Theorem 2.35(ii). Set $Y := \lim_n \max_{k\le n} \mathbb{E}[X_k \mid \mathcal{A}]$, an everywhere increasing limit and hence $\mathcal{A}$-measurable. Since $\max_{k\le n} \mathbb{E}[X_k \mid \mathcal{A}] = \mathbb{E}[X_n \mid \mathcal{A}]$ a.s., the MCT gives for $A \in \mathcal{A}$

$$\mathbb{E}[X\mathbb{1}_A] = \lim_n \mathbb{E}[X_n\mathbb{1}_A] = \lim_n \mathbb{E}[\mathbb{E}[X_n \mid \mathcal{A}]\mathbb{1}_A] = \lim_n \mathbb{E}\big[\max_{k\le n} \mathbb{E}[X_k \mid \mathcal{A}]\mathbb{1}_A\big] = \mathbb{E}[Y\mathbb{1}_A].$$

Uniqueness and positivity [Supplement]. If Y, Y' both qualify and $a < b$ are rational, let $A := \{Y' \le a < b \le Y\} \in \mathcal{A}$. Then $bP(A) \le \mathbb{E}[Y\mathbb{1}_A] = \mathbb{E}[Y'\mathbb{1}_A] \le aP(A) < \infty$, so $P(A) = 0$. The union over $a < b$ gives $Y \le Y'$ a.s., and by symmetry $Y = Y'$ a.s. Similarly $A_n := \{Y < -\frac{1}{n}\}$ has $0 \le \mathbb{E}[X\mathbb{1}_{A_n}] = \mathbb{E}[Y\mathbb{1}_{A_n}] \le -\frac{1}{n}P(A_n)$, so $Y \ge 0$ a.s.

As before one checks for $\mathcal{F}$-measurable $X, Y : \Omega \to [0,\infty]$: $\mathbb{E}[\mathbb{E}[X \mid \mathcal{A}]] = \mathbb{E}[X]$; $X \in \mathcal{A} \Rightarrow \mathbb{E}[X \mid \mathcal{A}] = X$; $X = Y$ a.s. $\Rightarrow$ equal conditional expectations; linearity with coefficients $\alpha, \beta \ge 0$; and $X \le Y$ a.s. $\Rightarrow \mathbb{E}[X \mid \mathcal{A}] \le \mathbb{E}[Y \mid \mathcal{A}]$.

Theorem 2.37

Let $X_n : \Omega \to [0,\infty]$ be $\mathcal{F}$-measurable; (in)equalities hold a.s.

(1) (Beppo Levi) $X_n \nearrow \Rightarrow \mathbb{E}[\sup_n X_n \mid \mathcal{A}] = \sup_n \mathbb{E}[X_n \mid \mathcal{A}]$.

(2) (Fatou) $\mathbb{E}[\liminf_n X_n \mid \mathcal{A}] \le \liminf_n \mathbb{E}[X_n \mid \mathcal{A}]$.

Proof [Supplement]. (1) $Y_n := \mathbb{E}[X_n \mid \mathcal{A}]$ increases a.s. by monotonicity, so $Y := \sup_n Y_n$ is $\mathcal{A}$-measurable and $Y_n \nearrow Y$ a.s. For $A \in \mathcal{A}$ the MCT twice gives $\mathbb{E}[\sup_n X_n\mathbb{1}_A] = \lim_n \mathbb{E}[X_n\mathbb{1}_A] = \lim_n \mathbb{E}[Y_n\mathbb{1}_A] = \mathbb{E}[Y\mathbb{1}_A]$, so $Y = \mathbb{E}[\sup_n X_n \mid \mathcal{A}]$.

(2) $g_n := \inf_{k\ge n} X_k \nearrow \liminf_n X_n$ and $\mathbb{E}[g_n \mid \mathcal{A}] \le \mathbb{E}[X_k \mid \mathcal{A}]$ for $k \ge n$. By (1), $\mathbb{E}[\liminf_n X_n \mid \mathcal{A}] = \lim_n \mathbb{E}[g_n \mid \mathcal{A}] \le \lim_n \inf_{k\ge n} \mathbb{E}[X_k \mid \mathcal{A}]$. ∎

Chapter 3. Sequences of Random Variables

3.1 Infinitely many fair coin tosses

Motivation. Finitely many coin tosses need only counting. To talk about limits (relative frequencies, waiting times, "infinitely many heads"), all tosses must live on one probability space. We build it from Lebesgue measure on $[0,1]$ via binary expansions, which also shows why uncountable sample spaces force us into measure theory.

A **random experiment** is one whose outcome is known only after it is performed; Ω is the set of possible outcomes (finite, countable or uncountable), and probability attaches a number to every event.

One toss. $\Omega = \{0,1\}$, $\mathcal{A} = \mathcal{P}(\Omega)$, $P(\{0\}) = P(\{1\}) = \frac{1}{2}$.

n **tosses.** $\Omega = \{0,1\}^n$, $\mathcal{A} = \mathcal{P}(\Omega)$ and $P(A) = |A|/2^n$: the **uniform distribution**, each binary word having probability 2^{-n}.

Infinitely many tosses. $\Omega = \{0,1\}^{\mathbb{N}} = \{(x_i)_{i\in\mathbb{N}} \mid x_i \in \{0,1\}\}$ is uncountable; $(x_i) \mapsto \sum_i x_i 2^{-i}$ maps it onto $[0,1]$ (not injectively). Let $\mathcal{A} := \sigma(\mathcal{A}_0)$ with the *cylinder sets*

$$\mathcal{A}_0 := \{B \subset \Omega \mid B = B_0 \times \{0,1\} \times \{0,1\} \times \cdots \text{ for some } n \text{ and } B_0 \subset \{0,1\}^n\},$$

the events that depend on finitely many tosses. We want $P(x_1 = \bar{x}_1, \dots, x_n = \bar{x}_n) = 2^{-n}$ and must show that such a P extends to $\mathcal{A}$.

Existence via Lebesgue measure. Let $(\tilde{\Omega}, \tilde{\mathcal{A}}, \tilde{P}) = ([0,1], \mathcal{B}([0,1]), \lambda)$. On $\Omega = \{0,1\}^{\mathbb{N}}$ let $X_i((x_n)_n) := x_i$ (the i-th toss) and $\mathcal{A} := \sigma(\{X_i = 1\} \mid i \in \mathbb{N})$, which equals $\sigma(\mathcal{A}_0)$. The non-terminating binary expansion of $\tilde{\omega} \in [0,1]$ is

$$\tilde{\omega} = \sum_{i=1}^{\infty} T_i(\tilde{\omega}) 2^{-i}, \qquad T_i := \sum_{k=1}^{2^{i-1}} \mathbb{1}_{\left(\frac{2k-1}{2^i}, \frac{2k}{2^i}\right]},$$

and defines $T : \tilde{\Omega} \to \Omega$, $T(\tilde{\omega}) := (T_1(\tilde{\omega}), T_2(\tilde{\omega}), \dots)$. T is not surjective: sequences ending in infinitely many zeros (other than $(0,0,\dots)$) are missed; they correspond to the dyadic rationals in $(0,1]$, a countable set.

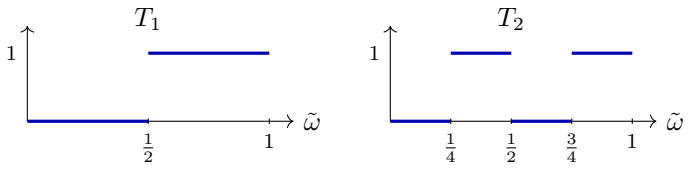


Figure. *The binary digit maps T_1, T_2 (and similarly $T_3, T_4, \dots$). Each $\{T_i = 1\}$ is a finite union of intervals of total length $\frac{1}{2}$, and fixing n digits pins $\tilde{\omega}$ down to an interval of length 2^{-n}.*

Since $T_i = X_i \circ T$, $T^{-1}(\{X_i = 1\}) = \{T_i = 1\}$ is a finite union of intervals, so T is $\tilde{\mathcal{A}}/\mathcal{A}$-measurable (Remark 1.32). $P := \tilde{P} \circ T^{-1}$ is a probability measure on $(\Omega, \mathcal{A})$ (Proposition 2.13), and for $(x_1, \dots, x_n) \in \{0,1\}^n$

$$P(X_1 = x_1, \dots, X_n = x_n) = \tilde{P}\Big(\bigcap_{i \le n} \{T_i = x_i\}\Big) = \tilde{P}(\text{an interval of length } 2^{-n}) = 2^{-n}.$$

So P extends the model for n tosses; uniqueness follows from Theorem 1.9, since the cylinder sets form a $\cap$-stable generator.

Some expectations.

- The i-th toss: $\mathbb{E}[X_i] = 1 \cdot P(X_i = 1) + 0 \cdot P(X_i = 0) = \frac{1}{2}$.
- Number of successes $S_n := X_1 + \cdots + X_n$: $P(S_n = k) = \binom{n}{k} 2^{-n}$, so $\mathbb{E}[S_n] = \sum_k k\binom{n}{k}2^{-n} = \frac{n}{2}$; more simply, $\mathbb{E}[S_n] = \sum_i \mathbb{E}[X_i] = \frac{n}{2}$ by linearity.
- Waiting time $S := \min\{n \mid X_n = 1\}$ ($\min \emptyset := \infty$), measurable since $\{S = k\} = \{X_1 = \cdots = X_{k-1} = 0, X_k = 1\}$: $P(S = k) = 2^{-k}$, $P(S = \infty) = 0$, and

$$\mathbb{E}[S] = \sum_{k\geq 1} k2^{-k} = \frac{1}{2}\sum_{k\geq 1} k\Big(\frac{1}{2}\Big)^{k-1} = 2, \qquad \text{using } \sum_{k\geq 1} kq^{k-1} = \frac{d}{dq}\frac{q}{1-q} = \frac{1}{(1-q)^2}.$$

- (*Doubling strategy*) A player bets on a fair coin, getting twice the stake on success, and doubles the stake until the first success, starting from 1 unit. The stake in round n is $I_n = 2^{n-1}\mathbb{1}_{\{T>n-1\}}$ with T the waiting time for the first 1. Then $\mathbb{E}[I_n] = 2^{n-1}P(T > n-1) = 2^{n-1}2^{-(n-1)} = 1$ for all n, while $I_n \to 0$ a.s. (for every $\omega \neq (0, 0, \dots)$). Expectations need not follow pointwise limits.

3.2 Infinitely many biased coin tosses

Motivation. Changing the success probability from $\frac{1}{2}$ to p gives the general Bernoulli scheme. Its moments are computed here from the joint law alone; existence and uniqueness of the measure follow from Section 3.6.

With $\Omega = \{0,1\}^{\mathbb{N}}$, X_i and $\mathcal{A}$ as before, there is a unique probability measure $P = P_p$ on $(\Omega, \mathcal{A})$ with

$$P_p(X_{i_1} = x_1, \dots, X_{i_n} = x_n) = p^{\sum_i x_i}(1-p)^{n-\sum_i x_i} \qquad (1 \leq i_1 < \cdots < i_n).$$

(Existence for $p = \frac{1}{2}$ was shown above; the general case and uniqueness come in Section 3.6 and Example 3.13.) Then $P_p(X_i = 1) = p$ and $P_p(X_i = X_j = 1) = p^2$ for $i \neq j$, so

$$\mathbb{E}_p[X_i] = p, \qquad \mathrm{Var}(X_i) = p - p^2 = p(1-p), \qquad \mathrm{Cov}(X_i, X_j) = \mathbb{E}_p[X_iX_j] - p^2 = 0 \ (i \neq j).$$

The X_i are pairwise uncorrelated (in fact independent, Example 3.13). For $S_n = X_1 + \cdots + X_n$: $\mathbb{E}_p[S_n] = np$, $\mathrm{Var}(S_n) = np(1-p)$ (Bienaymé).

- $X := \sum_n 2^{-n}X_n$ has $\mathbb{E}_p[X] = \sum_n 2^{-n}p = p$ by Theorem 1.56. The partial sums are Cauchy in L^2, and uncorrelatedness gives $\mathrm{Var}(X) = \sum_n 4^{-n}p(1-p) = \frac{1}{3}p(1-p)$.
- The waiting time S for the first success is **geometric**: $P_p(S = n) = (1-p)^{n-1}p$. Hence $\mathbb{E}_p[S] = \sum_n n(1-p)^{n-1}p = \frac{1}{p}$ and $\mathrm{Var}(S) = \frac{1-p}{p^2}$. [Supplement] For the variance, $\mathbb{E}_p[S(S-1)] = p(1-p)\sum_n n(n-1)(1-p)^{n-2} = \frac{2(1-p)}{p^2}$ (second derivative of the geometric series), so $\mathrm{Var}(S) = \frac{2(1-p)}{p^2} + \frac{1}{p} - \frac{1}{p^2} = \frac{1-p}{p^2}$.

3.3 The (strong and the weak) law of large numbers

Motivation. Repeating an experiment and averaging should remove the randomness: the empirical mean S_n/n should approach the true mean. Under uncorrelatedness and bounded variances this holds in L^2, in probability (weak law) and almost surely (strong law); the passage to a.s. convergence is the subsequence trick plus Borel–Cantelli.

Throughout, $(\Omega, \mathcal{A}, P)$ is a probability space and $X_1, X_2, \cdots \in \mathcal{L}^2$ are uncorrelated ($\mathrm{Cov}(X_i, X_j) = 0$ for $i \neq j$) with uniformly bounded variances $\sup_i \sigma_i^2 < \infty$, $\sigma_i^2 := \mathrm{Var}(X_i)$. Let $S_n := X_1 + \cdots + X_n$; $S_n(\omega)/n$ is the *empirical mean* of the first n observations. We want $\frac{S_n}{n} \approx \frac{\mathbb{E}[S_n]}{n}$ (resp. $\approx m$ if $\mathbb{E}[X_i] \equiv m$) for large n.

Remark 3.1

We may assume $\mathbb{E}[X_i] = 0$: the centred $\tilde{X}_i := X_i - \mathbb{E}[X_i]$ are in $\mathcal{L}^2$, uncorrelated, have the same variances, and $\frac{\tilde{S}_n}{n} - \frac{\mathbb{E}[\tilde{S}_n]}{n} = \frac{S_n}{n} - \frac{\mathbb{E}[S_n]}{n}$.

Theorem 3.2 (L^2 law of large numbers)

$\lim_n \mathbb{E}\Big[\Big(\frac{S_n}{n} - \frac{\mathbb{E}[S_n]}{n}\Big)^2\Big] = 0$ (resp. $\mathbb{E}[(\frac{S_n}{n} - m)^2] \to 0$ if $\mathbb{E}[X_i] \equiv m$).

Proof. By Bienaymé (Remark 2.26),

$$\mathbb{E}\Big[\Big(\frac{S_n}{n} - \frac{\mathbb{E}[S_n]}{n}\Big)^2\Big] = \operatorname{Var}\Big(\frac{S_n}{n}\Big) = \frac{1}{n^2}\sum_{i=1}^{n}\sigma_i^2 \le \frac{1}{n}\sup_i \sigma_i^2 \to 0.$$

■

Remark 3.3

This is a Hilbert space fact: in L^2 with $(X, Y) = \mathbb{E}[XY]$, the arithmetic mean of uniformly bounded orthogonal vectors tends to 0, since $\|S_n/n\|^2 = n^{-2}\sum_{i\le n}\|X_i\|^2 \to 0$.

Theorem 3.4 (Weak law of large numbers)

If $X_i \in \mathcal{L}^2$ are uncorrelated with uniformly bounded variances and $\mathbb{E}[X_i] = m$, then for every $\varepsilon > 0$, $\lim_n P(|\frac{S_n}{n} - m| \ge \varepsilon) = 0$, i.e. $S_n/n \to m$ in probability.

Proof. Chebyshev: $P(|\frac{S_n}{n} - m| \ge \varepsilon) \le \varepsilon^{-2}\operatorname{Var}(\frac{S_n}{n}) \to 0$ by Theorem 3.2. ■

Example 3.5 (Bernoulli)

For Bernoulli experiments with parameter p, $\mathbb{E}_p[X_i] = p$ and $\operatorname{Var}(X_i) = p(1-p) \le \frac{1}{4}$, so

$$P_p\Big(\Big|\frac{S_n}{n} - p\Big| \ge \varepsilon\Big) \to 0. \tag{3.1}$$

The relative frequency of "1" approaches p (J. Bernoulli, *Ars Conjectandi*), which justifies reading p as a long-run relative frequency.

Example 3.6 (Bernstein polynomials)

Every $f \in C([0,1])$ is a uniform limit of polynomials. For $p \in [0,1]$ the transformation theorem (Remark 2.15) gives

$$B_n(p) := \sum_{k=0}^{n} f\Big(\frac{k}{n}\Big)\binom{n}{k}p^k(1-p)^{n-k} = \sum_k f\Big(\frac{k}{n}\Big)P_p(S_n = k) = \mathbb{E}_p\Big[f\Big(\frac{S_n}{n}\Big)\Big].$$

Given $\varepsilon > 0$, uniform continuity gives $\delta > 0$ with $|f(x) - f(y)| \le \varepsilon$ for $|x - y| \le \delta$. Splitting on $\{|\frac{S_n}{n} - p| \le \delta\}$ and using Chebyshev with $p(1-p) \le \frac{1}{4}$:

$$|B_n(p) - f(p)| \le \mathbb{E}_p\Big|f\Big(\frac{S_n}{n}\Big) - f(p)\Big| \le \varepsilon + 2\|f\|_\infty P_p\Big(\Big|\frac{S_n}{n} - p\Big| > \delta\Big) \le \varepsilon + \frac{2\|f\|_\infty}{4\delta^2 n}.$$

Hence $\limsup_n \|B_n - f\|_\infty \le \varepsilon$ for every ε, i.e. $\|B_n - f\|_\infty \to 0$.

From convergence in probability we now pass to almost sure convergence.

Theorem 3.7 (Strong law of large numbers)

Let $X_1, X_2, \dots \in \mathcal{L}^2$ be uncorrelated with $\sup_i \sigma^2(X_i) = c < \infty$. Then $\frac{S_n}{n} - \frac{\mathbb{E}[S_n]}{n} \to 0$ P-a.s. (resp. $\frac{S_n}{n} \to m$ a.s. if $\mathbb{E}[X_i] \equiv m$).

Proof. By Remark 3.1 let $\mathbb{E}[X_i] = 0$.

Step 1 (the subsequence $n_k = k^2$). For $\varepsilon > 0$, Chebyshev gives $P(|S_{k^2}|/k^2 > \varepsilon) \le \frac{\mathrm{Var}(S_{k^2})}{\varepsilon^2 k^4} \le \frac{c}{\varepsilon^2 k^2}$, which is summable. By Lemma 2.10, $S_{k^2}/k^2 \to 0$ outside a null set N_1.

Step 2 (fluctuations between squares). Let $D_k := \max_{k^2 \le l < (k+1)^2} |S_l - S_{k^2}|$. Since $\mathrm{Var}(S_l - S_{k^2}) = \sum_{k^2 < i \le l} \sigma_i^2 \le 2kc$ and there are $2k$ indices l,

$$P\Big(\frac{D_k}{k^2} \ge \varepsilon\Big) \le \sum_{l=k^2+1}^{k^2+2k} P(|S_l - S_{k^2}| \ge \varepsilon k^2) \le 2k \cdot \frac{2kc}{\varepsilon^2 k^4} = \frac{4c}{\varepsilon^2 k^2}.$$

Lemma 2.10 gives $D_k/k^2 \to 0$ outside a null set N_2.

Step 3. For $k^2 \le n < (k+1)^2$: $\frac{|S_n|}{n} \le \frac{|S_{k^2}| + D_k}{k^2} \to 0$ on $(N_1 \cup N_2)^c$. ∎

Example 3.8

For Bernoulli experiments, $\frac{1}{n}\sum_{i \le n} X_i \to p$ P_p-a.s. (É. Borel 1909, strengthening Bernoulli's (3.1)). For a fair coin, $Y_i := 2X_i - 1 \in \{\pm 1\}$ and $S_n = Y_1 + \cdots + Y_n$ is the position of a **simple random walk** on $\mathbb{Z}$; refined, rescaled and linearly interpolated it converges to Brownian motion (Section 4.1). The strong law gives $S_n/n \to 0$ a.s.: fluctuations grow slower than linearly. Their exact order is given by the *law of the iterated logarithm*:

$$\limsup_n \frac{S_n}{\sqrt{2n \log\log n}} = +1, \qquad \liminf_n \frac{S_n}{\sqrt{2n \log\log n}} = -1 \qquad P\text{-a.s.}$$

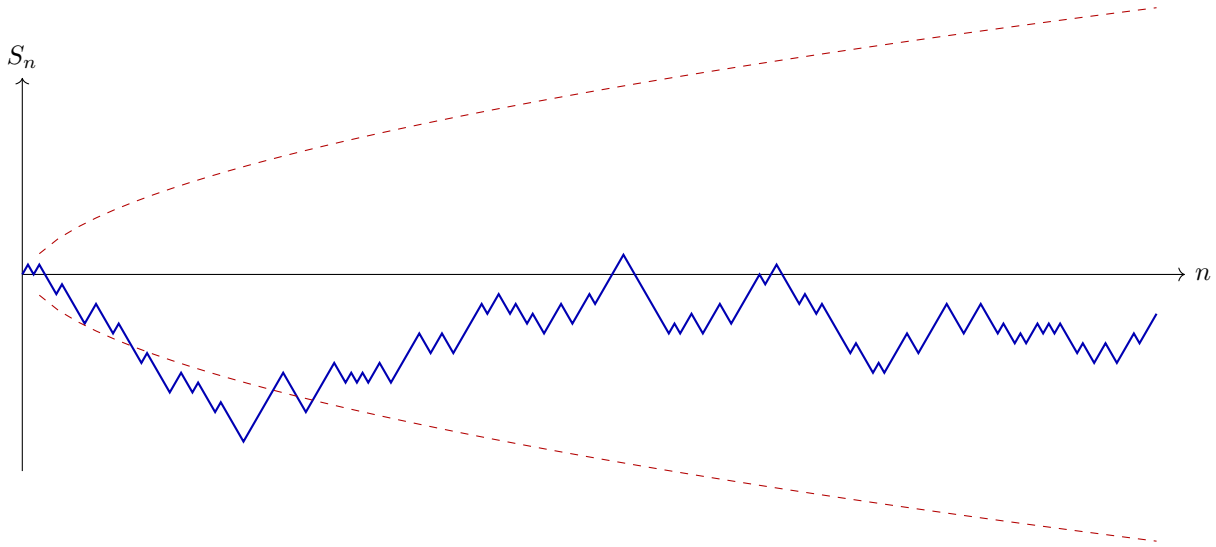


Figure. *A simple random walk path (200 steps). It is much smaller than n (strong law) but keeps returning to the order of $\sqrt{n \log\log n}$ (dashed: $\pm\sqrt{2n \log\log n}$, the scale in the law of the iterated logarithm).*

3.4 Independence

Motivation. Independence is the structural assumption behind most limit theorems. It is defined through products of probabilities on finitely many events at a time. With the π–λ theorem it passes from generators to σ-algebras, which gives Kolmogorov's 0–1 law and the second Borel–Cantelli lemma.

3.4.1 Independent events

Let $(\Omega, \mathcal{A}, P)$ be a probability space.

Definition 3.9 (Independence)

Events $A_i \in \mathcal{A}$, $i \in I$, are **independent** if $P(\bigcap_{j\in J} A_j) = \prod_{j\in J} P(A_j)$ for every finite $J \subset I$. Families $\mathcal{B}_i \subset \mathcal{A}$, $i \in I$, are **independent** if this holds for all finite $J \subset I$ and all choices $A_j \in \mathcal{B}_j$.

Theorem 3.10

Let $\mathcal{B}_i$, $i \in I$, be independent $\cap$-stable families. Then (i) $\sigma(\mathcal{B}_i)$, $i \in I$, are independent; (ii) for a partition $I = \bigcup_{k\in K} J_k$, the σ-algebras $\sigma(\bigcup_{i\in J_k} \mathcal{B}_i)$, $k \in K$, are independent.

Proof. (i) Let $J = \{j_1, \dots, j_n\}$ and $A_{j_\nu} \in \sigma(\mathcal{B}_{j_\nu})$; we show

$$P(A_{j_1} \cap \cdots \cap A_{j_n}) = P(A_{j_1}) \cdots P(A_{j_n}). \tag{3.2}$$

Step 1. Fix $A_{j_2} \in \mathcal{B}_{j_2}, \dots, A_{j_n} \in \mathcal{B}_{j_n}$. Then $\mathcal{D}_{j_1} := \{A \in \sigma(\mathcal{B}_{j_1}) \mid P(A \cap A_{j_2} \cap \cdots \cap A_{j_n}) = P(A)P(A_{j_2}) \cdots P(A_{j_n})\}$ is a Dynkin system containing $\mathcal{B}_{j_1}$, so $\mathcal{D}_{j_1} = \sigma(\mathcal{B}_{j_1})$ by Theorem 1.8.

Step 2. Now $\sigma(\mathcal{B}_{j_1}), \mathcal{B}_{j_2}, \dots, \mathcal{B}_{j_n}$ are independent $\cap$-stable families; repeat Step 1 for j_2, then j_3, and so on, to get (3.2).

(ii) For $k \in K$ let $\mathcal{C}_k := \{\bigcap_{j\in J} A_j \mid J \subset J_k \text{ finite}, \ A_j \in \mathcal{B}_j\}$. Each $\mathcal{C}_k$ is $\cap$-stable, and the $\mathcal{C}_k$ are independent: for $k_1, \dots, k_n$ and finite $J_\nu \subset J_{k_\nu}$, independence of the $\mathcal{B}_i$ gives $P(\bigcap_\nu \bigcap_{i\in J_\nu} A_i) = \prod_\nu P(\bigcap_{i\in J_\nu} A_i)$. By (i), the $\sigma(\mathcal{C}_k)$ are independent. Finally $\bigcup_{i\in J_k} \mathcal{B}_i \subset \mathcal{C}_k \subset \sigma(\bigcup_{i\in J_k} \mathcal{B}_i)$, so $\sigma(\mathcal{C}_k) = \sigma(\bigcup_{i\in J_k} \mathcal{B}_i)$. ∎

Example 3.11

If A_i, $i \in I$, are independent, then the σ-algebras $\sigma(\{A_i\}) = \{\emptyset, A_i, A_i^c, \Omega\}$ are independent (Theorem 3.10(i) with $\mathcal{B}_i = \{A_i\}$). The collection of events $\{A_i, A_i^c \mid i \in I\}$, however, is in general not independent: $P(A_i \cap A_i^c) = 0 \neq P(A_i)P(A_i^c)$ unless $P(A_i) \in \{0, 1\}$.

Remark 3.12

Pairwise independence does not imply independence. Toss a fair coin twice, $\Omega = \{0,1\}^2$ uniform, and let $A =$ "first toss 1", $B =$ "second toss 1", $C =$ "both tosses equal". All have probability $\frac{1}{2}$ and every pairwise intersection has probability $\frac{1}{4}$, so they are pairwise independent; but $P(A \cap B \cap C) = \frac{1}{4} \neq \frac{1}{8} = P(A)P(B)P(C)$.

Example 3.13 (Independent 0-1 experiments)

On $\Omega = \{0,1\}^{\mathbb{N}}$ with $\mathcal{A} = \sigma(\{X_i = 1\}, i \in \mathbb{N})$ let P_p satisfy (i) $P_p(X_i = 1) = p$ and (ii) the events $\{X_i = 1\}$, $i \in \mathbb{N}$, are independent. By (ii) and Example 3.11, for $i_1 < \cdots < i_n$ and $x_j \in \{0,1\}$

$$P_p(X_{i_1} = x_1, \dots, X_{i_n} = x_n) = \prod_j P_p(X_{i_j} = x_j) = p^k(1-p)^{n-k}, \qquad k = \textstyle\sum_j x_j.$$

These cylinder probabilities form a $\cap$-stable generator, so P_p is uniquely determined by (i) and (ii) (Theorem 1.9).

Theorem 3.14 (Kolmogorov's 0–1 law)

Let $\mathcal{B}_n$, $n \in \mathbb{N}$, be independent σ-algebras and $\mathcal{B}_\infty := \bigcap_{n=1}^\infty \sigma(\bigcup_{m \geq n} \mathcal{B}_m)$ the **tail σ-algebra**. Then $P(A) \in \{0,1\}$ for all $A \in \mathcal{B}_\infty$.

Illustration. For 0-1 experiments with $\mathcal{B}_i = \sigma(\{X_i = 1\})$, $\mathcal{B}_\infty$ contains the events of the remote future, e.g. $\limsup_i \{X_i = 1\}$ ("infinitely many 1's") and $\{\lim_n \frac{1}{n} \sum_{i \leq n} X_i \text{ exists}\}$.

Proof.

Step 1. By Theorem 3.10(ii), for every $n \geq 2$ the σ-algebras $\mathcal{B}_1, \dots, \mathcal{B}_{n-1}, \sigma(\bigcup_{m \geq n} \mathcal{B}_m)$ are independent. Since $\mathcal{B}_\infty \subset \sigma(\bigcup_{m \geq n} \mathcal{B}_m)$, also $\mathcal{B}_1, \dots, \mathcal{B}_{n-1}, \mathcal{B}_\infty$ are independent for all n, i.e. $\mathcal{B}_\infty, \mathcal{B}_1, \mathcal{B}_2, \dots$ are independent.

Step 2. Theorem 3.10(ii) again: $\sigma(\bigcup_n \mathcal{B}_n)$ and $\mathcal{B}_\infty$ are independent. As $\mathcal{B}_\infty \subset \sigma(\bigcup_n \mathcal{B}_n)$, $\mathcal{B}_\infty$ is independent of itself; conclude with Lemma 3.15. ∎

Lemma 3.15

If a σ-algebra $\mathcal{B} \subset \mathcal{A}$ is independent of itself, then $P(A) \in \{0,1\}$ for all $A \in \mathcal{B}$.

Proof. $P(A) = P(A \cap A) = P(A)^2$. ∎

In particular, for independent events A_n, $A_\infty := \limsup_n A_n$ has $P(A_\infty) \in \{0,1\}$: the σ-algebras $\mathcal{B}_n := \sigma(\{A_n\})$ are independent (Example 3.11) and $A_\infty \in \mathcal{B}_\infty$. Which value occurs is decided by:

Lemma 3.16 (Borel–Cantelli, part 2)

If $A_i \in \mathcal{A}$ are independent and $\sum_i P(A_i) = \infty$, then $P(\limsup_i A_i) = 1$.

Proof. It suffices that $P(\bigcap_{m \geq n} A_m^c) = 0$ for every n. By continuity from above, independence of the complements (Example 3.11) and $1 - \alpha \leq e^{-\alpha}$,

$$P\Big(\bigcap_{m \geq n} A_m^c\Big) = \lim_k P\Big(\bigcap_{m=n}^{n+k} A_m^c\Big) = \lim_k \prod_{m=n}^{n+k} (1 - P(A_m)) \leq \lim_k \exp\Big(-\sum_{m=n}^{n+k} P(A_m)\Big) = 0.$$

∎

Example 3.17 (A text appears infinitely often)

Let $p \in (0,1)$ and fix a binary word $(x_1, \dots, x_N)$ with K ones. Cut $\omega = (y_n)$ into consecutive blocks of length N and let A_i = "the word occurs as the i-th block". By Theorem 3.10(ii) the A_i are independent, each with probability $\alpha := p^K(1-p)^{N-K} > 0$. So $\sum_i P_p(A_i) = \infty$

and Borel–Cantelli 2 gives $P_p(\limsup_i A_i) = 1$: the word occurs in infinitely many blocks. Moreover the $\mathbb{1}_{A_i}$ are independent, hence uncorrelated, with bounded variances, so the strong law gives $\frac{1}{n}\sum_{i\le n}\mathbb{1}_{A_i} \to \alpha > 0$ P_p-a.s.: the word has a strictly positive relative frequency.

3.4.2 Independent random variables

Definition 3.18 (Independent random variables)

(i) For a r.v. $X : (\Omega, \mathcal{A}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, $\sigma(X) := X^{-1}(\mathcal{B}(\mathbb{R})) = \{\{X \in A\} \mid A \in \mathcal{B}(\mathbb{R})\} \subset \mathcal{A}$ is the σ-algebra **generated by** X.

(ii) R.v.'s X_i, $i \in I$, are **independent** if the $\sigma(X_i)$ are independent, i.e. $P(\bigcap_{j\in J}\{X_j \in A_j\}) = \prod_{j\in J} P(X_j \in A_j)$ for finite J and Borel A_j.

(iii) X_i, $i \in I$, are **independent of** a sub-σ-algebra $\tilde{\mathcal{A}}$ if $\sigma(X_i \mid i \in I)$ (the smallest σ-algebra making all X_i measurable) is independent of $\tilde{\mathcal{A}}$.

Remark 3.19

If X_i, $i \in I$, are independent and $h_i : \mathbb{R} \to \mathbb{R}$ are Borel, then $Y_i := h_i(X_i)$ are independent, because $\sigma(Y_i) \subset \sigma(X_i)$.

Theorem 3.20 (Product rule)

If $X_1, \dots, X_n \ge 0$ are independent, then $\mathbb{E}[X_1 \cdots X_n] = \mathbb{E}[X_1] \cdots \mathbb{E}[X_n]$.

Proof.

Step 1 (reduction to $n = 2$). $X_1 \cdots X_{n-1}$ is measurable w.r.t. $\sigma(\sigma(X_1) \cup \cdots \cup \sigma(X_{n-1}))$, which is independent of $\sigma(X_n)$ by Theorem 3.10(ii); induct. So let $X, Y \ge 0$ be independent; we show

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]. \tag{3.3}$$

Step 2 (simple functions). For $X_n = \sum_i \alpha_i \mathbb{1}_{A_i}$, $Y_n = \sum_j \beta_j \mathbb{1}_{B_j}$ with $\alpha_i, \beta_j \ge 0$, $A_i \in \sigma(X)$, $B_j \in \sigma(Y)$:

$$\mathbb{E}[X_n Y_n] = \sum_{i,j} \alpha_i \beta_j P(A_i \cap B_j) = \sum_{i,j} \alpha_i \beta_j P(A_i) P(B_j) = \mathbb{E}[X_n]\mathbb{E}[Y_n].$$

Step 3. Choose $\sigma(X)$- resp. $\sigma(Y)$-measurable simple $X_n \nearrow X$, $Y_n \nearrow Y$ (Facts 2.2(v)); then $X_n Y_n \nearrow XY$, and the MCT turns Step 2 into (3.3). ∎

Corollary 3.21

X, Y independent and in $\mathcal{L}^1 \Rightarrow XY \in \mathcal{L}^1$ and $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.

Proof. For $\varepsilon_1, \varepsilon_2 \in \{+, -\}$, X^{ε_1} and Y^{ε_2} are independent (Remark 3.19) and nonnegative, so $\mathbb{E}[X^{\varepsilon_1} Y^{\varepsilon_2}] = \mathbb{E}[X^{\varepsilon_1}]\mathbb{E}[Y^{\varepsilon_2}] < \infty$. Hence $XY = X^+Y^+ + X^-Y^- - X^+Y^- - X^-Y^+ \in \mathcal{L}^1$ and expanding gives $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$. ∎

Remark 3.22

The converse fails. For $X \sim N(0,1)$ and $Y = X^2$, X, Y are dependent but $\mathbb{E}[XY] = \mathbb{E}[X^3] = 0 = \mathbb{E}[X]\mathbb{E}[Y]$. Uncorrelatedness is also not preserved under functions: $\mathbb{E}[X^+Y^+] \neq \mathbb{E}[X^+]\mathbb{E}[Y^+]$. [Supplement] Indeed $\mathbb{E}[X^+Y^+] = \int_0^\infty x^3\varphi(x)\,dx = \frac{2}{\sqrt{2\pi}}$ (since $\int_0^\infty x^3 e^{-x^2/2}dx = 2$), while $\mathbb{E}[X^+]\mathbb{E}[Y^+] = \frac{1}{\sqrt{2\pi}} \cdot 1$.
Obviously $X, Y \in \mathcal{L}^2$ independent $\Rightarrow$ uncorrelated, since $\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0$.

Corollary 3.23 (Strong law for independent variables)

If $X_1, X_2, \cdots \in \mathcal{L}^2$ are independent with $\sup_i \mathrm{Var}(X_i) < \infty$, then $\frac{1}{n}\sum_{i\le n}(X_i - \mathbb{E}[X_i]) \to 0$ a.s.; if $\mathbb{E}[X_i] \equiv m$, then $\frac{1}{n}\sum_{i\le n} X_i \to m$ a.s.

Proof. Independent $\mathcal{L}^2$ variables are uncorrelated (Remark 3.22); apply Theorem 3.7. ∎

3.5 Kolmogorov's law of large numbers

Motivation. The strong law of Section 3.3 needs second moments. For identically distributed variables a first moment suffices, and pairwise independence is enough. The proof truncates (Borel–Cantelli removes the truncation), uses Chebyshev along geometric subsequences, and interpolates by monotonicity.

(X_i) is **identically distributed** (i.d.) if $P \circ X_i^{-1} = P \circ X_1^{-1}$ for all i; i.i.d. means independent and identically distributed.

Theorem 3.24 (Kolmogorov 1930)

If $X_1, X_2, \cdots \in \mathcal{L}^1$ are i.i.d. with $m = \mathbb{E}[X_1]$, then $\frac{1}{n}\sum_{i=1}^n X_i \to m$ P-a.s.

Proof. Independent variables are in particular pairwise independent; apply Theorem 3.25. ∎

It follows from the more general:

Theorem 3.25 (Etemadi 1981)

If $X_1, X_2, \cdots \in \mathcal{L}^1$ are pairwise independent and identically distributed with $m = \mathbb{E}[X_1]$, then $\frac{1}{n}\sum_{i=1}^n X_i \to m$ P-a.s.

Proof. We may assume $X_i \ge 0$; otherwise treat (X_i^+) and (X_i^-), which are again in $\mathcal{L}^1$, pairwise independent and identically distributed.

Step 1 (truncation). Let $\tilde{X}_i := X_i \mathbb{1}_{\{X_i < i\}} = h_i(X_i)$ with $h_i(x) = x\mathbb{1}_{\{x<i\}}$; the $\tilde{X}_i$ are pairwise independent (Remark 3.19). It suffices to show $\tilde{S}_n/n \to m$ a.s. for $\tilde{S}_n := \sum_{i\le n}\tilde{X}_i$, because

$$\sum_n P(X_n \neq \tilde{X}_n) = \sum_n P(X_1 \ge n) = \sum_n \sum_{k\ge n} P(X_1 \in [k, k+1))$$
$$= \sum_k k\, P(X_1 \in [k, k+1)) \le \mathbb{E}[X_1] < \infty,$$

so by Borel–Cantelli $X_n = \tilde{X}_n$ for all but finitely many n, a.s.

Step 2 (geometric subsequence). Fix $\alpha > 1$ and $k_n := \lfloor \alpha^n \rfloor$. Suppose (Step 3)

$$\frac{\tilde{S}_{k_n} - \mathbb{E}[\tilde{S}_{k_n}]}{k_n} \to 0 \quad \text{a.s.} \tag{3.4}$$

Since $\mathbb{E}[\tilde{X}_i] = \mathbb{E}[X_1 \mathbb{1}_{\{X_1<i\}}] \nearrow m$, the Cesàro means $\frac{1}{k_n}\mathbb{E}[\tilde{S}_{k_n}] \to m$, so $\tilde{S}_{k_n}/k_n \to m$ outside a null set N_α. For $k_n \le l < k_{n+1}$, monotonicity of $l \mapsto \tilde{S}_l$ gives

$$\frac{k_n}{k_{n+1}} \cdot \frac{\tilde{S}_{k_n}}{k_n} \le \frac{\tilde{S}_l}{l} \le \frac{\tilde{S}_{k_{n+1}}}{k_{n+1}} \cdot \frac{k_{n+1}}{k_n},$$

and $k_{n+1}/k_n \to \alpha$. Hence $\frac{m}{\alpha} \le \liminf_l \frac{\tilde{S}_l}{l} \le \limsup_l \frac{\tilde{S}_l}{l} \le \alpha m$ off N_α. Taking $\alpha_j \searrow 1$ and the null set $\bigcup_j N_{\alpha_j}$ gives $\tilde{S}_l/l \to m$ a.s.

Step 3 (proof of (3.4)). By Lemma 2.10 it suffices to show $\sum_n P(|\tilde{S}_{k_n} - \mathbb{E}[\tilde{S}_{k_n}]| \ge \varepsilon k_n) < \infty$. Pairwise independent $\Rightarrow$ pairwise uncorrelated, so Chebyshev gives the bound $\frac{1}{\varepsilon^2 k_n^2}\sum_{i\le k_n}\mathbb{E}[\tilde{X}_i^2]$, and it is enough that

$$s := \sum_n \frac{1}{k_n^2}\sum_{i\le k_n}\mathbb{E}[\tilde{X}_i^2] = \sum_i \Big(\sum_{n:\, k_n \ge i}\frac{1}{k_n^2}\Big)\mathbb{E}[\tilde{X}_i^2] < \infty.$$

Step 4 (a geometric bound). We claim there is c with

$$\sum_{n:\, k_n \ge i}\frac{1}{k_n^2} \le \frac{c}{i^2}. \tag{3.5}$$

Indeed $k_n > \alpha^n - 1 \ge \alpha^n - \alpha^{n-1} = c_\alpha \alpha^n$ with $c_\alpha = \frac{\alpha-1}{\alpha}$. If n_i is the least n with $k_n \ge i$, then $\alpha^{n_i} \ge i$ and

$$\sum_{n\ge n_i}\frac{1}{k_n^2} \le c_\alpha^{-2}\sum_{n\ge n_i}\alpha^{-2n} = \frac{c_\alpha^{-2}\alpha^{-2n_i}}{1-\alpha^{-2}} \le \frac{c_\alpha^{-2}}{1-\alpha^{-2}}\cdot\frac{1}{i^2}.$$

Step 5. With (3.5), $\mathbb{E}[\tilde{X}_i^2] = \mathbb{E}[X_1^2\mathbb{1}_{\{X_1<i\}}] \le \sum_{l\le i} l^2 P(X_1 \in [l-1,l))$ and $\sum_{i\ge l}\frac{1}{i^2} \le \frac{1}{l^2} + \sum_{i>l}\left(\frac{1}{i-1} - \frac{1}{i}\right) \le \frac{2}{l}$:

$$\begin{aligned} s &\le c\sum_i \frac{1}{i^2}\sum_{l\le i} l^2 P(X_1 \in [l-1,l)) = c\sum_l l^2\Big(\sum_{i\ge l}\frac{1}{i^2}\Big)P(X_1 \in [l-1,l)) \\ &\le 2c\sum_l \mathbb{E}\big[l\mathbb{1}_{\{X_1\in[l-1,l)\}}\big] \le 2c(\mathbb{E}[X_1]+1). \end{aligned}$$

■

Corollary 3.26

If $X_1, X_2, \cdots \ge 0$ are pairwise independent and identically distributed, then $\frac{1}{n}\sum_{i\le n} X_i \to \mathbb{E}[X_1] \in [0,\infty]$ a.s.

Proof. If $\mathbb{E}[X_1] < \infty$ this is Theorem 3.25. If $\mathbb{E}[X_1] = \infty$, apply Theorem 3.25 to $X_i \wedge N$:

$$\frac{1}{n}\sum_{i\le n} X_i \ge \frac{1}{n}\sum_{i\le n}(X_i \wedge N) \to \mathbb{E}[X_1 \wedge N] \quad \text{a.s.},$$

and $\mathbb{E}[X_1 \wedge N] \nearrow \mathbb{E}[X_1] = \infty$ as $N \to \infty$ (MCT). Hence $\liminf_n \frac{1}{n}\sum_{i\le n} X_i = \infty$ a.s. ■

3.6 Finite products of probability measures

Motivation. Independent experiments are modelled on a product space whose measure multiplies the probabilities of rectangles. We construct it by integrating sections, which gives Fubini's theorem at the same time: integrals over a product can be computed as iterated integrals in any order.

Let $(S_1, \mathcal{S}_1, \mu_1)$, $(S_2, \mathcal{S}_2, \mu_2)$ be probability spaces, $\Omega := S_1 \times S_2$, $X_i(x_1, x_2) = x_i$, and

$$\mathcal{A} := \mathcal{S}_1 \otimes \mathcal{S}_2 := \sigma(X_1, X_2) = \sigma(\{A_1 \times A_2 \mid A_i \in \mathcal{S}_i\}).$$

We seek a probability measure $P =: \mu_1 \otimes \mu_2$ (the **product measure**) with

$$P(A_1 \times A_2) = \mu_1(A_1)\mu_2(A_2) \qquad (A_i \in \mathcal{S}_i). \tag{3.6}$$

Such P is unique, since the rectangles $A_1 \times A_2$ form a $\cap$-stable generator of $\mathcal{A}$ (Theorem 1.9).

Theorem 3.27 (Fubini for probability measures)

(i) For every $\mathcal{A}$-measurable $f : \Omega \to \overline{\mathbb{R}}$ and $x_1 \in S_1$, $x_2 \in S_2$, the sections $f_{x_1} := f(x_1, \cdot)$ and $f_{x_2} := f(\cdot, x_2)$ are measurable. If moreover $f \geq 0$, then $x_1 \mapsto \int_{S_2} f_{x_1}\, d\mu_2$ and $x_2 \mapsto \int_{S_1} f_{x_2}\, d\mu_1$ are $\mathcal{S}_1$- resp. $\mathcal{S}_2$-measurable.

(ii) There is a unique probability measure $P = \mu_1 \otimes \mu_2$ on $(\Omega, \mathcal{A})$ such that for all $\mathcal{A}$-measurable $f \geq 0$

$$\int_\Omega f\, dP = \int_{S_1} \Big(\int_{S_2} f(x_1, x_2)\, \mu_2(dx_2) \Big) \mu_1(dx_1), \tag{3.7}$$

and the order may be interchanged:

$$\int_\Omega f\, dP = \int_{S_2} \Big(\int_{S_1} f(x_1, x_2)\, \mu_1(dx_1) \Big) \mu_2(dx_2). \tag{3.8}$$

In particular, with the sections $A_{x_1} := \{x_2 \mid (x_1, x_2) \in A\}$, $A_{x_2} := \{x_1 \mid (x_1, x_2) \in A\}$,

$$P(A) = \int_{S_1} \mu_2(A_{x_1})\, \mu_1(dx_1) = \int_{S_2} \mu_1(A_{x_2})\, \mu_2(dx_2) \qquad (A \in \mathcal{A}). \tag{3.9}$$

Proof.

Step 1 (sections are measurable). $\phi_{x_1}(x_2) := (x_1, x_2)$ is measurable, because $\phi_{x_1}^{-1}(A_1 \times A_2)$ is A_2 or $\emptyset$ and rectangles generate $\mathcal{A}$ (Remark 1.32). So $f_{x_1} = f \circ \phi_{x_1}$ is measurable; likewise f_{x_2}.

Step 2 (indicators). For $f \geq 0$, the map

$$x_1 \mapsto \int_{S_2} f(x_1, x_2)\, \mu_2(dx_2) \in [0, \infty] \tag{3.10}$$

is well defined. For $f = \mathbb{1}_A$ it equals $\mu_2(A_{x_1})$. Let $\mathcal{D} := \{A \in \mathcal{A} \mid x_1 \mapsto \mu_2(A_{x_1}) \text{ is } \mathcal{S}_1\text{-measurable}\}$. $\mathcal{D}$ is a Dynkin system: $\Omega_{x_1} = S_2$; $\mu_2((A^c)_{x_1}) = 1 - \mu_2(A_{x_1})$; for disjoint A_n the sections are disjoint and $\mu_2((\bigcup A_n)_{x_1}) = \sum_n \mu_2((A_n)_{x_1})$ is a series of measurable functions. $\mathcal{D}$ contains the rectangles, since $\mu_2((A_1 \times A_2)_{x_1}) = \mathbb{1}_{A_1}(x_1)\mu_2(A_2)$. By Theorem 1.8, $\mathcal{D} = \mathcal{A}$.

Step 3 (measure-theoretic induction). By linearity (3.10) is measurable for simple $f \geq 0$; for general $f \geq 0$ take simple $f_n \nearrow f$ (Facts 2.2(v)). The MCT makes (3.10) a pointwise limit of measurable functions. The other order is identical.

Step 4 (existence). Define $P(A) := \int_{S_1} \mu_2(A_{x_1})\, \mu_1(dx_1)$. Then $P(\Omega) = 1$, and for disjoint A_n the MCT (Theorem 1.56) gives $P(\bigcup_n A_n) = \int \sum_n \mu_2((A_n)_{x_1})\, d\mu_1 = \sum_n P(A_n)$. Also $P(A_1 \times A_2) = \int \mathbb{1}_{A_1} \mu_2(A_2)\, d\mu_1 = \mu_1(A_1)\mu_2(A_2)$, so (3.6) holds and P is unique. (3.7) holds for indicators by definition and extends to $f \geq 0$ by linearity and the MCT.

Step 5 (interchanging the order). Repeating Steps 2–4 with the coordinates swapped, $Q(A) := \int_{S_2} \mu_1(A_{x_2})\, \mu_2(dx_2)$ is a probability measure with $Q(A_1 \times A_2) = \mu_1(A_1)\mu_2(A_2)$, so $Q = P$ by Theorem 1.9. Hence (3.8) holds for indicators and, by linearity and the MCT, for all $f \geq 0$; (3.9) is the case $f = \mathbb{1}_A$. ∎

Remark 3.28

If $f \in \mathcal{L}^1(\mu_1 \otimes \mu_2)$, Theorem 3.27 applies to f^+ and f^-, hence by linearity to f.

Definition 3.29 (Finite products of measurable spaces)

For measurable spaces $(S_i, \mathcal{S}_i)$, $1 \le i \le n$, $\Omega := S_1 \times \cdots \times S_n$ carries the **product σ-algebra**

$$\mathcal{A} := \mathcal{S}_1 \otimes \cdots \otimes \mathcal{S}_n = \bigotimes_{i=1}^{n} \mathcal{S}_i := \sigma(\{A_1 \times \cdots \times A_n \mid A_i \in \mathcal{S}_i\}).$$

Equivalently $\mathcal{A} = \sigma(X_1, \dots, X_n)$ with the coordinate projections X_i. [Supplement] *Reason:* $A_1 \times \cdots \times A_n = \bigcap_i X_i^{-1}(A_i)$, and conversely $X_i^{-1}(A_i)$ is the rectangle with A_i in slot i and S_j elsewhere.

Theorem 3.30 (Finite products of probability measures)

For probability spaces $(S_i, \mathcal{S}_i, \mu_i)$, $1 \le i \le n$, there is a unique probability measure $P = \mu_1 \otimes \cdots \otimes \mu_n = \bigotimes_{i=1}^n \mu_i$ on $(S_1 \times \cdots \times S_n, \mathcal{S}_1 \otimes \cdots \otimes \mathcal{S}_n)$ with $P(A_1 \times \cdots \times A_n) = \prod_i \mu_i(A_i)$.

Proof. Uniqueness: rectangles form a $\cap$-stable generator (Theorem 1.9). Existence by induction, $n = 2$ being Theorem 3.27. Given $P_{n-1} := \mu_1 \otimes \cdots \otimes \mu_{n-1}$ on $(\Omega_{n-1}, \mathcal{A}_{n-1})$, apply Theorem 3.27 to $(\Omega_{n-1}, \mathcal{A}_{n-1}, P_{n-1})$ and $(S_n, \mathcal{S}_n, \mu_n)$. Since $\Omega_{n-1} \times S_n = S_1 \times \cdots \times S_n$ and $\mathcal{A}_{n-1} \otimes \mathcal{S}_n = \mathcal{S}_1 \otimes \cdots \otimes \mathcal{S}_n$ (both are generated by the n-fold rectangles), $P := P_{n-1} \otimes \mu_n$ lives on the right space, and $P(A_1 \times \cdots \times A_n) = P_{n-1}(A_1 \times \cdots \times A_{n-1})\mu_n(A_n) = \prod_i \mu_i(A_i)$. ∎

Theorem 3.31 (Fubini for finite products)

With $P = \mu_1 \otimes \cdots \otimes \mu_n$ and $\mathcal{S}_1 \otimes \cdots \otimes \mathcal{S}_n$-measurable $f \ge 0$,

$$\int f \, dP = \int_{S_1} \cdots \int_{S_n} f(x_1, \dots, x_n) \, \mu_n(dx_n) \cdots \mu_1(dx_1), \tag{3.11}$$

and for every permutation π of $\{1, \dots, n\}$

$$\int f \, dP = \int_{S_{\pi(1)}} \cdots \int_{S_{\pi(n)}} f(x_1, \dots, x_n) \, \mu_{\pi(n)}(dx_{\pi(n)}) \cdots \mu_{\pi(1)}(dx_{\pi(1)}). \tag{3.12}$$

Proof.

Step 1 ((3.11)). Induction on n; $n = 2$ is Theorem 3.27. With $P = P_{n-1} \otimes \mu_n$ (proof of Theorem 3.30), the two-factor Fubini theorem gives $\int f \, dP = \int_{\Omega_{n-1}} \left(\int_{S_n} f \, d\mu_n \right) dP_{n-1}$. The inner integral is a nonnegative $\mathcal{A}_{n-1}$-measurable function, to which the induction hypothesis applies.

Step 2 ((3.12)) [Supplement]. Regrouping Ω as $\left(\prod_{i \ne \pi(n)} S_i\right) \times S_{\pi(n)}$ identifies the product σ-algebras (both are generated by rectangles). Hence, by Theorem 3.27(i) and induction, all iterated integrals in (3.12) are well defined and measurable at each stage. For $f = \mathbb{1}_A$ the right-hand side of (3.12), as a function of A, is a probability measure (MCT for disjoint unions) which gives rectangles the value $\prod_i \mu_i(A_i)$. By Theorem 1.9 it equals P. Linearity and the MCT extend this to all $f \ge 0$. ∎

Remark 3.32

If $f \in \mathcal{L}^1(\mu_1 \otimes \cdots \otimes \mu_n)$, Theorem 3.31 applies to $f^\pm$ and hence to f.

3.7 Weak convergence of probability measures

Motivation. Laws of averages such as the central limit theorem compare distributions, not random variables on one space. Requiring $\mu_n(A) \to \mu(A)$ for every set is far too strong ($\delta_{1/n} \not\to \delta_0$ on $\{0\}$). Weak convergence tests against bounded continuous functions, and the Portmanteau theorem lists equivalent, more concrete forms.

Definition 3.33 (Weak convergence)

Let Ω be a metric space with Borel σ-algebra $\mathcal{B}(\Omega)$, and μ, μ_n probability measures on it. $\mu_n \to \mu$ **weakly** if $\int f\,d\mu_n \to \int f\,d\mu$ for all $f \in C_b(\Omega)$ (bounded continuous functions).

Example 3.34

(i) $x_n \to x$ implies $\delta_{x_n} \to \delta_x$ weakly, since $f(x_n) \to f(x)$ for continuous f. (ii) $\mu_n := N(0, \frac{1}{n}) \to \delta_0$ weakly: substituting $x = y/\sqrt{n}$ and using the DCT,

$$\int f\,d\mu_n = \int f(x)\frac{e^{-nx^2/2}}{\sqrt{2\pi/n}}\,dx = \int f\Big(\frac{y}{\sqrt{n}}\Big)\frac{e^{-y^2/2}}{\sqrt{2\pi}}\,dy \to f(0) = \int f\,d\delta_0.$$

The following results also hold for finite measures with the obvious changes.

Theorem 3.35 (Portmanteau theorem)

For probability measures μ, μ_n on a metric space (Ω, d) the following are equivalent:

(i) $\mu_n \to \mu$ weakly;

(ii) $\int f\,d\mu_n \to \int f\,d\mu$ for all bounded uniformly continuous f;

(iii) $\limsup_n \mu_n(F) \le \mu(F)$ for all closed F;

(iv) $\liminf_n \mu_n(G) \ge \mu(G)$ for all open G;

(v) $\mu_n(A) \to \mu(A)$ for all μ-continuity sets A, i.e. $A \in \mathcal{B}(\Omega)$ with $\mu(\partial A) = 0$;

(vi) $\int f\,d\mu_n \to \int f\,d\mu$ for all bounded measurable f that are continuous μ-a.e.

Proof.

Step Easy implications. (i)⇒(ii) and (vi)⇒(i) are special cases; (iii)⇔(iv) by taking complements.

Step (ii)⇒(iii). For closed F the open sets $G_m := \{d(\cdot, F) < \frac{1}{m}\}$ decrease to F, so given ε there is m with $\mu(G_m) < \mu(F) + \varepsilon$. Let $\phi(x) = 1$ for $x \le 0$, $1 - x$ on $[0,1]$, 0 for $x \ge 1$, and $f := \phi(m\,d(\cdot, F))$; it is uniformly continuous (composition of Lipschitz maps) with $\mathbb{1}_F \le f \le \mathbb{1}_{G_m}$. Hence $\limsup_n \mu_n(F) \le \lim_n \int f\,d\mu_n = \int f\,d\mu \le \mu(G_m) < \mu(F) + \varepsilon$.

Step (iii)⇒(v). With interior $\mathring{A}$ and closure $\bar{A}$, $\mu(\partial A) = 0$ gives $\mu(\mathring{A}) = \mu(A) = \mu(\bar{A})$, and

$$\mu(A) = \mu(\mathring{A}) \overset{\text{(iv)}}{\le} \liminf_n \mu_n(\mathring{A}) \le \liminf_n \mu_n(A) \le \limsup_n \mu_n(A) \le \limsup_n \mu_n(\bar{A}) \overset{\text{(iii)}}{\le} \mu(\bar{A}) = \mu(A).$$

Step (v)⇒(vi). Let f be as in (vi) and $D := \{x \in \mathbb{R} \mid \mu(f = x) \ne 0\}$, the jumps of the distribution function of $\mu \circ f^{-1}$; it is countable (Remark 2.16), so $\mathbb{R} \setminus D$ is dense. Given $\varepsilon > 0$ pick $c_0 < \cdots < c_m$ in $\mathbb{R} \setminus D$ with $c_{k+1} - c_k \le \varepsilon$ and $c_0 \le f < c_m$, and set $A_k := f^{-1}([c_k, c_{k+1}))$. Each A_k is a μ-continuity set, because (see below)

$$\partial A_k \subset \{f = c_k\} \cup \{f = c_{k+1}\} \cup D_f, \qquad D_f := \{\omega \mid f \text{ is discontinuous at } \omega\}, \tag{3.13}$$

and all three sets are μ-null. With $g := \sum_k c_k \mathbb{1}_{A_k}$, $\|f-g\|_\infty \le \varepsilon$ and

$$\Big|\int f\,d\mu - \int f\,d\mu_n\Big| \le \int |f-g|\,d\mu + \Big|\int g\,d\mu - \int g\,d\mu_n\Big| + \int |g-f|\,d\mu_n$$
$$\le 2\varepsilon + \sum_k |c_k||\mu(A_k) - \mu_n(A_k)| \to 2\varepsilon.$$

Step Proof of (3.13). Let $\omega \in \partial A_k \setminus D_f$. For every $\varepsilon' > 0$ there is δ with $f(B(\omega,\delta)) \subset B(f(\omega),\varepsilon')$, and $B(\omega,\delta)$ meets both $f^{-1}([c_k,c_{k+1}))$ and its complement. Applying f, $B(f(\omega),\varepsilon')$ meets both $[c_k,c_{k+1})$ and its complement, for every ε'. So $f(\omega) \in \partial[c_k,c_{k+1}) = \{c_k, c_{k+1}\}$.

[Supplement] D_f is measurable: with the oscillation $\operatorname{osc}_f(\omega) := \lim_{\delta\to 0} \sup\{|f(x)-f(y)| \mid x,y \in B(\omega,\delta)\}$, $D_f = \bigcup_k \{\operatorname{osc}_f \ge \frac{1}{k}\}$, and each $\{\operatorname{osc}_f \ge \frac{1}{k}\}$ is closed (if $\operatorname{osc}_f(\omega) < \frac{1}{k}$, the same small ball works for nearby points). ∎

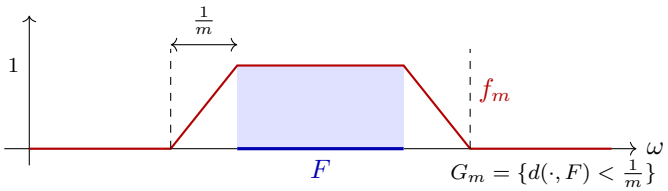


Figure. *The test function in (ii)⇒(iii): $f_m = \phi(m\,d(\cdot,F))$ is 1 on F, 0 off the $\frac{1}{m}$-neighbourhood G_m, and linear in between, so $\mathbb{1}_F \le f_m \le \mathbb{1}_{G_m} \searrow \mathbb{1}_F$.*

Definition 3.36 (Support, C_0)

For continuous f on a metric space Ω, $\operatorname{supp} f := \overline{\{f \neq 0\}}$, and $C_0(\Omega) := \{f \in C_b(\Omega) \mid \operatorname{supp} f \text{ compact}\}$.

Corollary 3.37

For probability measures μ, μ_n on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ with distribution functions F, F_n, the following are equivalent:

(i) $\mu_n \to \mu$ **vaguely**: $\int f\,d\mu_n \to \int f\,d\mu$ for all $f \in C_0(\mathbb{R})$;

(ii) $\mu_n \to \mu$ weakly;

(iii) $F_n(x) \to F(x)$ at every continuity point x of F;

(iv) $\mu_n((a,b]) \to \mu((a,b])$ whenever $\mu(\{a\}) = \mu(\{b\}) = 0$.

Proof.

Step (i)⇒(ii) [Supplement]. Let $f \in C_b(\mathbb{R})$, $\varepsilon > 0$. Choose R with $\mu([-R,R]) > 1-\varepsilon$ and $g \in C_0(\mathbb{R})$, $0 \le g \le 1$, $g = 1$ on $[-R,R]$. Then $\int g\,d\mu_n \to \int g\,d\mu > 1-\varepsilon$, so $\int (1-g)\,d\mu_n < \varepsilon$ for large n. Since $fg \in C_0(\mathbb{R})$,

$$\Big|\int f\,d\mu_n - \int f\,d\mu\Big| \le \Big|\int fg\,d\mu_n - \int fg\,d\mu\Big| + \|f\|_\infty\Big(\int (1-g)\,d\mu_n + \int (1-g)\,d\mu\Big),$$

whose lim sup is at most $2\|f\|_\infty\varepsilon$.

Step (ii)⇒(iii). If F is continuous at x, then $\mu(\{x\}) = 0 = \mu(\partial(-\infty,x])$, and Portmanteau (v) gives $F_n(x) = \mu_n((-\infty,x]) \to F(x)$.

Step (iii)⇒(iv). F is continuous at a and b, so $\mu((a,b]) = F(b) - F(a) = \lim_n (F_n(b) - F_n(a))$.

Step (iv)⇒(i). $D := \{x \mid \mu(\{x\}) \neq 0\}$ is countable, so $\mathbb{R}\setminus D$ is dense. Let $f \in C_0(\mathbb{R})$; it is uniformly continuous, so for $\varepsilon > 0$ there is δ with $|f(x) - f(y)| \leq \varepsilon$ if $|x - y| \leq \delta$. Choose $c_0 < \cdots < c_m$ in $\mathbb{R}\setminus D$ with $\operatorname{supp} f \subset (c_0, c_m]$ and $c_k - c_{k-1} \leq \delta$, and $g := \sum_{k=1}^{m} f(c_{k-1}) \mathbb{1}_{(c_{k-1}, c_k]}$. Then $\|f - g\|_\infty \leq \varepsilon$ and, as in the Portmanteau proof,

$$\Big| \int f \, d\mu - \int f \, d\mu_n \Big| \leq 2\varepsilon + \sum_k |f(c_{k-1})| |\mu((c_{k-1}, c_k]) - \mu_n((c_{k-1}, c_k])| \to 2\varepsilon.$$

■

3.8 Joint distribution and convolution

Motivation. For i.i.d. $X_i \in \mathcal{L}^1$, Kolmogorov's law and the DCT give $\mathbb{E}[f(S_n/n)] \to f(\mathbb{E}[X_1])$ for $f \in C_b$, i.e. the law of S_n/n converges weakly to $\delta_{\mathbb{E}[X_1]}$ (for $X_i \in \mathcal{L}^2$ simply because $\operatorname{Var}(S_n/n) \to 0$). The interesting scale is $S_n/\sqrt{n}$, whose variance stays fixed, and the central limit theorem will identify its limit law as normal. To compute laws of sums we need joint distributions, and for independent summands the law of the sum is a convolution.

Definition 3.38 (Joint distribution)

For real r.v.'s $X_1, \ldots, X_n$ on $(\Omega, \mathcal{A}, P)$, the law $P_{\bar{X}} := P \circ \bar{X}^{-1}$ of $\bar{X} := (X_1, \ldots, X_n) : \Omega \to \mathbb{R}^n$ is the **joint distribution** of $X_1, \ldots, X_n$.

Remark 3.39

(i) $P_{\bar{X}}(A) = P(\bar{X} \in A)$, $A \in \mathcal{B}(\mathbb{R}^n)$, is a probability measure (Proposition 2.13), because $\bar{X}$ is $\mathcal{A}/\mathcal{B}(\mathbb{R}^n)$-measurable. Indeed

$$\mathcal{B}(\mathbb{R}^n) = \bigotimes_{i=1}^{n} \mathcal{B}(\mathbb{R}) = \sigma(\{A_1 \times \cdots \times A_n \mid A_i = (-\infty, x_i]\}),$$

and $\bar{X}^{-1}(A_1 \times \cdots \times A_n) = \bigcap_i \{X_i \in A_i\} \in \mathcal{A}$ (Remark 1.32). [Supplement] *The equalities:* every open set of $\mathbb{R}^n$ is a countable union of open boxes (Remark 1.27(i)), and boxes are Borel; and $\sigma(\{(-\infty, x]\}) = \mathcal{B}(\mathbb{R})$ in each coordinate, with products of half-open intervals obtained from orthants by finite differences.

(ii) By Theorem 1.9, $P_{\bar{X}}$ is determined by $P_{\bar{X}}(A_1 \times \cdots \times A_n) = P(X_1 \in A_1, \ldots, X_n \in A_n)$, even with $A_i = (-\infty, x_i]$.

Remark 3.40 (Transformation theorem)

For Borel $\Phi : \mathbb{R}^n \to \mathbb{R}$ with $\Phi(\bar{X}) \in \mathcal{L}^1$ or $\Phi \geq 0$: $\mathbb{E}[\Phi(\bar{X})] = \int_\Omega \Phi(\bar{X}) \, dP = \int_{\mathbb{R}^n} \Phi \, dP_{\bar{X}}$ (same proof as in Remark 2.15).

Example 3.41

(i) Let X, Y be uniform on $[0,1]$. If they are independent, their joint law is uniform on $[0,1]^2$ (Theorem 3.42). If $X = Y$, it is the "uniform distribution on the diagonal", the image of $\lambda|_{[0,1]}$ under $x \mapsto (x, x)$. (ii) For independent $X, Y \sim N(m, \sigma^2)$, Theorem 3.42 gives the joint density $f(x,y) = \frac{1}{2\pi\sigma^2} \exp\big(-\frac{(x-m)^2+(y-m)^2}{2\sigma^2}\big)$, a two-dimensional normal distribution.

Theorem 3.42

Let $X_1, \dots, X_n$ have laws $\mu_1, \dots, \mu_n$ and joint law $P_{\bar{X}}$. Then $X_1, \dots, X_n$ are independent $\iff$ $P_{\bar{X}} = \bigotimes_i \mu_i$, i.e. $P_{\bar{X}}(A_1 \times \cdots \times A_n) = \prod_i \mu_i(A_i)$. In this case:

(i) $P_{\bar{X}}$ is determined by $\mu_1, \dots, \mu_n$;

(ii) for every permutation π and $\phi \geq 0$ (or $P_{\bar{X}}$-integrable), $\int \phi \, dP_{\bar{X}}$ equals the iterated integral in the order π;

(iii) if $\mu_i(dx_i) = f_i(x_i)\, dx_i$, then $P_{\bar{X}}(d\bar{x}) = \bar{f}(\bar{x})\, d\bar{x}$ with $\bar{f}(\bar{x}) = \prod_i f_i(x_i)$.

Proof. The equivalence is Definition 3.18 rewritten, since $\mu_i(A_i) = P(X_i \in A_i)$.

(i) Theorem 3.30.

(ii) Theorem 3.31 and Remark 3.32.

(iii) $\bar{f} \geq 0$ is Borel with $\int \bar{f}\, d\bar{x} = \prod_i \int f_i = 1$ (Fubini for Lebesgue measure, which is the product of one-dimensional Lebesgue measures on boxes and hence everywhere by Theorem 1.9), so $\bar{\mu}(A) := \int_A \bar{f}\, d\bar{x}$ is a probability measure. On rectangles, $\bar{\mu}(A_1 \times \cdots \times A_n) = \prod_i \int_{A_i} f_i = \prod_i \mu_i(A_i) = P_{\bar{X}}(A_1 \times \cdots \times A_n)$, so $\bar{\mu} = P_{\bar{X}}$ by Theorem 1.9. ∎

For independent $X_1, \dots, X_n$, how is the law of $S_n = X_1 + \cdots + X_n$ obtained from the laws of the X_i? Let $T_x(y) := x + y$ be translation by x.

Theorem 3.43 (Convolution)

Let X_1, X_2 be independent with laws μ_1, μ_2.

(i) $P_{X_1+X_2} = \mu_1 * \mu_2 := \int \mu_2 \circ T_{x_1}^{-1}(\cdot)\, \mu_1(dx_1)$, i.e. for $A \in \mathcal{B}(\mathbb{R})$

$$\mu_1 * \mu_2(A) = \iint \mathbb{1}_A(x_1 + x_2)\, \mu_1(dx_1)\, \mu_2(dx_2) = \int \mu_2(A - x_1)\, \mu_1(dx_1),$$

where $A - x_1 := \{x - x_1 \mid x \in A\}$.

(ii) If μ_2 has density f_2, then $\mu_1 * \mu_2$ has density $f(x) = \int f_2(x - x_1)\, \mu_1(dx_1)$, which equals $(f_1 * f_2)(x) := \int f_1(x_1) f_2(x - x_1)\, dx_1$ if $\mu_1(dx_1) = f_1(x_1)\, dx_1$.

Proof. (i) With $\bar{A} := \{(x_1, x_2) \mid x_1 + x_2 \in A\} \in \mathcal{B}(\mathbb{R}^2)$ and Theorem 3.42,

$$P(X_1 + X_2 \in A) = (\mu_1 \otimes \mu_2)(\bar{A}) = \int \Big(\int \mathbb{1}_{A - x_1}(x_2)\, \mu_2(dx_2) \Big) \mu_1(dx_1) = \int \mu_2(A - x_1)\, \mu_1(dx_1).$$

(ii) Substituting $x = x_1 + x_2$ and using Fubini (Theorem 3.42(ii)):

$$\begin{aligned}(\mu_1 * \mu_2)(A) &= \int \int_{A - x_1} f_2(x_2)\, dx_2\, \mu_1(dx_1) = \int \int_A f_2(x - x_1)\, dx\, \mu_1(dx_1) \\ &= \int_A \Big(\int f_2(x - x_1)\, \mu_1(dx_1) \Big) dx.\end{aligned}$$

∎

Example 3.44

(i) If $X_i \sim N(m_i, \sigma_i^2)$ are independent, then $X_1 + X_2 \sim N(m_1 + m_2, \sigma_1^2 + \sigma_2^2)$, i.e. $f_{m_1,\sigma_1^2} * f_{m_2,\sigma_2^2} = f_{m_1+m_2,\sigma_1^2+\sigma_2^2}$. [Supplement] After shifting, let $m_i = 0$, $a = \sigma_1^2$, $b = \sigma_2^2$. Completing

the square in y,

$$\int e^{-\frac{y^2}{2a}-\frac{(x-y)^2}{2b}}\,dy = e^{-\frac{x^2}{2(a+b)}}\int e^{-\frac{a+b}{2ab}(y-\frac{ax}{a+b})^2}\,dy = e^{-\frac{x^2}{2(a+b)}}\sqrt{\frac{2\pi ab}{a+b}},$$

and multiplying by $\frac{1}{2\pi\sqrt{ab}}$ gives $f_{0,a+b}(x)$.
(ii) The **Gamma distribution** $\Gamma_{\alpha,p}$ has density $\gamma_{\alpha,p}(x) = \frac{\alpha^p}{\Gamma(p)}x^{p-1}e^{-\alpha x}$ for $x > 0$ (0 otherwise). If $X_i \sim \Gamma_{\alpha,p_i}$ are independent, then $X_1 + X_2 \sim \Gamma_{\alpha,p_1+p_2}$. [Supplement] For $x > 0$, substituting $y = xt$,

$$\begin{aligned}(\gamma_{\alpha,p_1} * \gamma_{\alpha,p_2})(x) &= \frac{\alpha^{p_1+p_2}e^{-\alpha x}}{\Gamma(p_1)\Gamma(p_2)}\int_0^x y^{p_1-1}(x-y)^{p_2-1}dy \\ &= \frac{\alpha^{p_1+p_2}e^{-\alpha x}x^{p_1+p_2-1}}{\Gamma(p_1)\Gamma(p_2)}B(p_1,p_2) = \gamma_{\alpha,p_1+p_2}(x),\end{aligned}$$

using $B(p_1,p_2) = \Gamma(p_1)\Gamma(p_2)/\Gamma(p_1+p_2)$. For $p_i = 1$: a sum $S_n = T_1 + \cdots + T_n$ of independent $\mathrm{Exp}(\alpha)$ variables has the Gamma density $\gamma_{\alpha,n}(x) = \frac{\alpha^n}{(n-1)!}x^{n-1}e^{-\alpha x}$, $x > 0$.

Example 3.45 (The waiting time paradox)

Let $T_1, T_2, \ldots$ be independent $\mathrm{Exp}(\alpha)$ waiting times (between calls at a call centre, or buses at a stop), so $\mathbb{E}[T_i] = \int_0^\infty x\alpha e^{-\alpha x}dx = \frac{1}{\alpha}$. Let $S_n := T_1 + \cdots + T_n$, $S_0 := 0$, fix $t > 0$, and let X be the time from the last event before t to t, and Y the time from t to the next event: $X = t - S_n$, $Y = S_{n+1} - t$ on $\{S_n \le t < S_{n+1}\}$.
Question: what is $\mathbb{E}[Y]$? The naive guess $\frac{1}{2\alpha}$ is wrong:

(i) $Y \sim \mathrm{Exp}(\alpha)$, so $\mathbb{E}[Y] = \frac{1}{\alpha}$;

(ii) X is $\mathrm{Exp}(\alpha)$ "compressed to $[0,t]$": $P(X \ge s) = e^{-\alpha s}$ for $0 \le s \le t$ and $P(X = t) = e^{-\alpha t}$, so $\mathbb{E}[X] = \int_0^t s\alpha e^{-\alpha s}ds + te^{-\alpha t} = \frac{1}{\alpha}(1 - e^{-\alpha t})$;

(iii) X and Y are independent.

The interval $[t-X, t+Y]$ containing t has mean length $\frac{1}{\alpha}(2 - e^{-\alpha t}) \approx \frac{2}{\alpha}$: a fixed time is more likely to fall into a long gap.
Proof. For $0 \le x \le t$, $y \ge 0$, decompose according to the number n of events in $[0,t]$; S_n and T_{n+1} are independent with densities $\gamma_{\alpha,n}$ and $\alpha e^{-\alpha r}$:

$$\begin{aligned}P(X \ge x, Y \ge y) &= P(T_1 \ge t + y) + \sum_{n\ge 1} P(S_n \le t - x,\ T_{n+1} \ge t + y - S_n) \\ &= e^{-\alpha(t+y)} + \sum_{n\ge 1}\int_0^{t-x} \gamma_{\alpha,n}(s)e^{-\alpha(t+y-s)}\,ds \\ &= e^{-\alpha(t+y)}\Big(1 + \int_0^{t-x} e^{\alpha s}\sum_{n\ge 1}\gamma_{\alpha,n}(s)\,ds\Big).\end{aligned}$$

Since $\sum_{n\ge1}\gamma_{\alpha,n}(s) = \alpha e^{-\alpha s}\sum_{n\ge1}\frac{(\alpha s)^{n-1}}{(n-1)!} = \alpha$, the bracket is $1 + (e^{\alpha(t-x)} - 1) = e^{\alpha(t-x)}$, and $P(X \ge x, Y \ge y) = e^{-\alpha y}e^{-\alpha x}$. Setting $x = 0$ gives (i), $y = 0$ gives (ii), and the product form gives (iii).

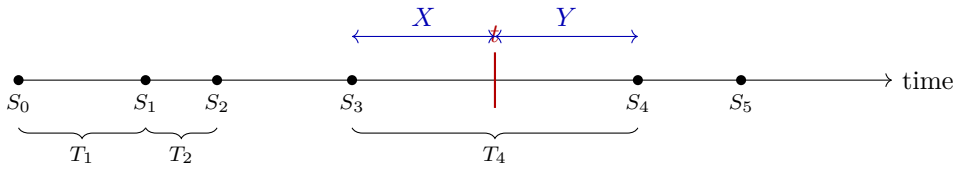


Figure. *The waiting time paradox. The gap containing the fixed time t (here T_4) is size-biased: on average it is about twice as long as a typical gap, and the residual waiting time Y has the same law as a full gap.*

3.9 Characteristic functions

Motivation. Convolution of laws is awkward; multiplication is easy. The characteristic function (Fourier transform of a law) turns sums of independent variables into products, determines the law uniquely, and turns weak convergence into pointwise convergence (Lévy). This is the machinery behind the central limit theorem.

Let $\langle \cdot, \cdot \rangle$ be the inner product on $\mathbb{R}^n$ and $\mathcal{M}_+^1(\mathbb{R}^n)$ the probability measures on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$. For $\mu \in \mathcal{M}_+^1(\mathbb{R}^n)$ the **characteristic function** is

$$\hat{\mu}(u) := \int e^{i\langle u,y\rangle} \mu(dy) := \int \cos\langle u,y\rangle \, \mu(dy) + i \int \sin\langle u,y\rangle \, \mu(dy), \qquad u \in \mathbb{R}^n.$$

Theorem 3.46

For $\mu \in \mathcal{M}_+^1(\mathbb{R}^n)$: (i) $|\hat{\mu}(u)| \le \hat{\mu}(0) = 1$; (ii) $\hat{\mu}(-u) = \overline{\hat{\mu}(u)}$; (iii) $\hat{\mu}$ is uniformly continuous; (iv) $\hat{\mu}$ is positive definite: $\sum_{j,k=1}^m c_j \bar{c}_k \hat{\mu}(u_j - u_k) \ge 0$ for $c_j \in \mathbb{C}$, $u_j \in \mathbb{R}^n$.

Proof [Supplement]. (i) $|\hat{\mu}(u)| \le \int |e^{i\langle u,y\rangle}| \, d\mu = 1 = \hat{\mu}(0)$.

(ii) Conjugate the integrand.

(iii) $|\hat{\mu}(u+h) - \hat{\mu}(u)| \le \int |e^{i\langle h,y\rangle} - 1| \, \mu(dy)$, which does not depend on u and tends to 0 as $h \to 0$ by the DCT.

(iv) $\sum_{j,k} c_j \bar{c}_k \hat{\mu}(u_j - u_k) = \int \left| \sum_j c_j e^{i\langle u_j, y\rangle} \right|^2 \mu(dy) \ge 0$. ∎

Theorem 3.47 (Uniqueness theorem)

If $\mu_1, \mu_2 \in \mathcal{M}_+^1(\mathbb{R}^n)$ and $\hat{\mu}_1 = \hat{\mu}_2$, then $\mu_1 = \mu_2$.

Proof [Supplement] (Gaussian smoothing).

Step 1 (a Fourier formula for the Gaussian). Let $p_\sigma(x) := (2\pi\sigma^2)^{-n/2} e^{-\|x\|^2/(2\sigma^2)}$. Substituting $y = \sigma u$ in Theorem 3.52 below (whose proof uses nothing from this section), with u replaced by x/σ, gives

$$p_\sigma(x) = (2\pi)^{-n} \int e^{i\langle x,u\rangle} e^{-\sigma^2\|u\|^2/2} \, du.$$

Step 2 (smoothed measure is determined by $\hat{\mu}$). For $\mu \in \mathcal{M}_+^1(\mathbb{R}^n)$ let $h_\sigma(x) := \int p_\sigma(x-y) \, \mu(dy)$. By Step 1 and Fubini (the integrand is bounded by $e^{-\sigma^2\|u\|^2/2}$, integrable w.r.t. $du \otimes \mu$),

$$h_\sigma(x) = (2\pi)^{-n} \int e^{i\langle x,u\rangle} e^{-\sigma^2\|u\|^2/2} \Big(\int e^{-i\langle y,u\rangle} \mu(dy) \Big) du = (2\pi)^{-n} \int e^{i\langle x,u\rangle} e^{-\sigma^2\|u\|^2/2} \hat{\mu}(-u) \, du,$$

which depends on μ only through $\hat{\mu}$.

Step 3 (letting $\sigma \to 0$). For $f \in C_b(\mathbb{R}^n)$, with γ the standard Gaussian on $\mathbb{R}^n$ and the substitution $x = y + \sigma z$,

$$\int f(x) h_\sigma(x)\,dx = \iint f(y+\sigma z)\,\gamma(dz)\,\mu(dy) \xrightarrow[\sigma\to 0]{} \int f\,d\mu$$

by the DCT. By Step 2 the left side is the same for μ_1 and μ_2, so $\int f\,d\mu_1 = \int f\,d\mu_2$ for all $f \in C_b(\mathbb{R}^n)$, and $\mu_1 = \mu_2$ by Example 2.30. ∎

Theorem 3.48 (Bochner)

If $\phi : \mathbb{R}^n \to \mathbb{C}$ is continuous, positive definite and $\phi(0) = 1$, there is exactly one $\mu \in \mathcal{M}_+^1(\mathbb{R}^n)$ with $\hat\mu = \phi$.

On the proof

Uniqueness is Theorem 3.47. Existence is not proved in the lectures; see Klenke, *Probability Theory*, Thm. 15.29. It is not used below.

Theorem 3.49 (Lévy's continuity theorem)

Let $\mu_m \in \mathcal{M}_+^1(\mathbb{R}^n)$.

(i) If $\mu_m \to \mu$ weakly, then $\hat\mu_m \to \hat\mu$ uniformly on compact sets.

(ii) If $\hat\mu_m \to \phi$ pointwise for some $\phi : \mathbb{R}^n \to \mathbb{C}$ continuous at 0, then there is a unique $\mu \in \mathcal{M}_+^1(\mathbb{R}^n)$ with $\hat\mu = \phi$, and $\mu_m \to \mu$ weakly.

Proof [Supplement] (part (ii) for $n = 1$, the case used in the CLT).

Step (i) Tightness. Given $\varepsilon > 0$ choose R with $\mu(\{\|y\| \ge R\}) < \varepsilon$. By Portmanteau (iii) for the closed set $\{\|y\| \ge R\}$, $\mu_m(\{\|y\| \ge R\}) < \varepsilon$ for $m \ge m_0$; enlarging R covers the finitely many $m < m_0$.

Step (i) Equicontinuity. Then $|\hat\mu_m(u) - \hat\mu_m(v)| \le \int_{\{\|y\|<R\}} |e^{i\langle u-v,y\rangle} - 1|\,d\mu_m + 2\varepsilon \le R\|u-v\| + 2\varepsilon$ for all m. Pointwise convergence ($\cos, \sin \in C_b$) plus this uniform equicontinuity gives uniform convergence on compacts, by covering a compact set with finitely many small balls.

Step (ii) Tightness from ϕ. For $\delta > 0$ and $\nu \in \mathcal{M}_+^1(\mathbb{R})$, Fubini gives

$$\frac{1}{\delta}\int_{-\delta}^{\delta}(1-\hat\nu(u))\,du = 2\int\Big(1 - \frac{\sin \delta x}{\delta x}\Big)\nu(dx) \ge \nu(\{|x| \ge 2/\delta\}),$$

since $1 - \frac{\sin s}{s} \ge \frac12$ for $|s| \ge 2$. As $\phi(0) = \lim \hat\mu_m(0) = 1$ and ϕ is continuous at 0, choose δ with $\frac{1}{\delta}\int_{-\delta}^{\delta}|1-\phi|\,du < \varepsilon$. By the DCT ($|\hat\mu_m| \le 1$), $\frac{1}{\delta}\int_{-\delta}^{\delta}(1-\hat\mu_m) \to \frac{1}{\delta}\int_{-\delta}^{\delta}(1-\phi)$. Hence $\mu_m(\{|x| \ge 2/\delta\}) < 2\varepsilon$ for large m: (μ_m) is tight.

Step (ii) Helly selection. Let F_m be the distribution functions. A diagonal argument yields a subsequence (m_k) along which $F_{m_k}(q) \to H(q)$ for every rational q. Put $G(x) := \inf_{q>x} H(q)$; G is increasing and right continuous. If G is continuous at x and $q < x < q'$ are rational, then $H(q) \le \liminf_k F_{m_k}(x) \le \limsup_k F_{m_k}(x) \le H(q')$. Letting $q \uparrow x$, $q' \downarrow x$ gives $F_{m_k}(x) \to G(x)$. Tightness forces $G(-\infty) = 0$ and $G(+\infty) = 1$, so G is the distribution function of some ν (Remark 2.15), and $\mu_{m_k} \to \nu$ weakly by Corollary 3.37.

Step (ii) Identification. By (i), $\hat\nu = \lim_k \hat\mu_{m_k} = \phi$. Every subsequence of (μ_m) is again tight, so it has a further subsequence converging weakly to some ν' with $\hat\nu' = \phi$; by Theorem 3.47, $\nu' = \nu =: \mu$. If $\mu_m \not\to \mu$ weakly, there would be $f \in C_b$, $\varepsilon > 0$ and a subsequence with $|\int f\,d\mu_m - \int f\,d\mu| \ge \varepsilon$, contradicting the above.

For $n \geq 2$ the same argument works with multivariate distribution functions in Helly's theorem; we do not need it. ■

For an $\mathcal{A}/\mathcal{B}(\mathbb{R}^n)$-measurable $\bar{X} = (X_1, \dots, X_n)$ on $(\Omega, \mathcal{A}, P)$,

$$\varphi_{\bar{X}}(u) := \hat{P}_{\bar{X}}(u) = \int e^{i\langle u, y\rangle} P_{\bar{X}}(dy) = \mathbb{E}[e^{i\langle u, \bar{X}\rangle}]$$

is the **characteristic function of** $\bar{X}$.

Remark 3.50

$X_1, \dots, X_n$ are independent iff $\varphi_{\bar{X}}(u_1, \dots, u_n) = \prod_j \varphi_{X_j}(u_j)$ for all $u \in \mathbb{R}^n$. "⇒" is Theorem 3.20 (applied to real and imaginary parts, via Corollary 3.21). For "⇐", the right side is the characteristic function of $\bigotimes_j P_{X_j}$ (by Fubini), so Theorem 3.47 gives $P_{\bar{X}} = \bigotimes_j P_{X_j}$, and Theorem 3.42 applies.

Theorem 3.51

If $X_1, \dots, X_n$ are independent, $\alpha \in \mathbb{R}$ and $S := \alpha \sum_k X_k$, then $\varphi_S(u) = \prod_{k=1}^n \varphi_{X_k}(\alpha u)$.

Proof. $\varphi_S(u) = \mathbb{E}[\prod_k e^{i\alpha u X_k}] = \prod_k \mathbb{E}[e^{i\alpha u X_k}]$ by independence (Corollary 3.21, applied to the bounded complex factors via real and imaginary parts and induction). ■

Theorem 3.52 (Gaussian characteristic function)

For all $u \in \mathbb{R}^n$: $(2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{i\langle u, y\rangle} e^{-\frac{1}{2}\|y\|^2} dy = e^{-\frac{1}{2}\|u\|^2}$.

Proof [Supplement]. By Fubini the integral factorises over coordinates, so let $n = 1$ and $\varphi(u) := (2\pi)^{-1/2} \int e^{iuy} e^{-y^2/2} dy$. Differentiating under the integral (dominated by $|y| e^{-y^2/2}$) and integrating by parts,

$$\varphi'(u) = (2\pi)^{-1/2} \int iy\, e^{iuy} e^{-y^2/2} dy = (2\pi)^{-1/2} \int i\, e^{iuy} \frac{d}{dy}\big(-e^{-y^2/2}\big) dy = -u\varphi(u).$$

With $\varphi(0) = 1$ this gives $\varphi(u) = e^{-u^2/2}$. ■

Example 3.53

(i) $\hat{\delta}_a(u) = e^{iua}$. (ii) For $\mu = \sum_i \alpha_i \delta_{a_i}$ ($\alpha_i \geq 0$, $\sum \alpha_i = 1$), $\hat{\mu}(u) = \sum_i \alpha_i e^{iua_i}$. In particular the binomial distribution $\beta_n^p = \sum_k \binom{n}{k} p^k q^{n-k} \delta_k$ ($q = 1 - p$) has $\hat{\beta}_n^p(u) = (q + pe^{iu})^n$, and the Poisson distribution $\pi_\alpha = \sum_n e^{-\alpha} \frac{\alpha^n}{n!} \delta_n$ has $\hat{\pi}_\alpha(u) = e^{-\alpha} \sum_n \frac{(\alpha e^{iu})^n}{n!} = e^{\alpha(e^{iu} - 1)}$.

3.10 Central limit theorem

Motivation. After centring and scaling by the standard deviation, a sum of many independent, individually negligible contributions is approximately standard normal, whatever their laws. Lindeberg's condition makes "individually negligible" precise. The proof compares $\prod_k \varphi_{X_k}(u/s_n)$ with $\exp(\sum_k (\varphi_{X_k}(u/s_n) - 1))$ using a second-order Taylor expansion.

Definition 3.54 (Central limit property)

Let $X_1, X_2, \cdots \in \mathcal{L}^2$ be independent with positive variances, $S_n := X_1 + \cdots + X_n$ and $S_n^* := \frac{S_n - \mathbb{E}[S_n]}{\sqrt{\mathrm{Var}(S_n)}}$ (so $\mathbb{E}[S_n^*] = 0$, $\mathrm{Var}(S_n^*) = 1$). (X_n) has the **central limit property** (CLP) if $P_{S_n^*} \to N(0,1)$ weakly, equivalently (Corollary 3.37)

$$\lim_n P(S_n^* \le b) = \Phi(b) := \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{b} e^{-x^2/2} dx \qquad \forall b \in \mathbb{R}.$$

Theorem 3.55 (Central limit theorem, Lindeberg)

Let $X_1, X_2, \cdots \in \mathcal{L}^2$ be independent, $\sigma_n^2 := \mathrm{Var}(X_n) > 0$, $s_n := (\sum_{k \le n} \sigma_k^2)^{1/2}$, and assume **Lindeberg's condition**

$$L_n(\varepsilon) := \sum_{k=1}^{n} \int_{\{|X_k - \mathbb{E}[X_k]| \ge \varepsilon s_n\}} \Big(\frac{X_k - \mathbb{E}[X_k]}{s_n} \Big)^2 dP \xrightarrow[n\to\infty]{} 0 \qquad \forall \varepsilon > 0. \tag{L}$$

Then (X_n) has the CLP.

Remark 3.56

(i) i.i.d. sequences in $\mathcal{L}^2$ satisfy (L): with $m = \mathbb{E}[X_1]$, $\sigma^2 = \mathrm{Var}(X_1)$, $s_n^2 = n\sigma^2$ and $L_n(\varepsilon) = \sigma^{-2} \int_{\{|X_1 - m| \ge \varepsilon \sqrt{n}\sigma\}} (X_1 - m)^2 dP \to 0$ by the DCT.

(ii) **Lyapunov's condition** (often with $\delta = 1$) implies (L):

$$\exists \delta > 0 : \quad \lim_n \sum_{k=1}^{n} \frac{\mathbb{E}|X_k - \mathbb{E}[X_k]|^{2+\delta}}{s_n^{2+\delta}} = 0. \tag{Lya}$$

Indeed $|X_k - \mathbb{E}X_k| \ge \varepsilon s_n$ implies $|X_k - \mathbb{E}X_k|^2 \le |X_k - \mathbb{E}X_k|^{2+\delta} / (\varepsilon s_n)^{\delta}$, so $L_n(\varepsilon) \le \varepsilon^{-\delta} \sum_k \mathbb{E}|X_k - \mathbb{E}X_k|^{2+\delta} / s_n^{2+\delta}$.

(iii) Uniformly bounded X_n with $s_n \to \infty$ satisfy (Lya) for every $\delta > 0$: if $|X_k| \le \alpha/2$, then $|X_k - \mathbb{E}X_k| \le \alpha$ and

$$\sum_k \frac{\mathbb{E}|X_k - \mathbb{E}X_k|^{2+\delta}}{s_n^{2+\delta}} \le \sum_k \frac{\alpha^{\delta} \mathbb{E}|X_k - \mathbb{E}X_k|^2}{s_n^2 s_n^{\delta}} = \Big(\frac{\alpha}{s_n} \Big)^{\delta} \to 0.$$

Lemma 3.57

Lindeberg's condition implies

$$\lim_n \max_{1 \le k \le n} \frac{\sigma_k}{s_n} = 0. \tag{3.14}$$

Proof. For $k \le n$ and $\varepsilon > 0$, splitting the integral on $\{|X_k - \mathbb{E}X_k| \ge \varepsilon s_n\}$:

$$\Big(\frac{\sigma_k}{s_n} \Big)^2 = \int \Big(\frac{X_k - \mathbb{E}X_k}{s_n} \Big)^2 dP \le L_n(\varepsilon) + \varepsilon^2.$$

■

Lemma 3.58

For $t \in \mathbb{R}$, $n \in \mathbb{N}$: $\left|e^{it} - \sum_{k=0}^{n-1} \frac{(it)^k}{k!}\right| \le \frac{|t|^n}{n!}$.

Proof. Taylor's formula with integral remainder for e^{it} (real and imaginary parts) gives the remainder $R_n(t) = \frac{1}{(n-1)!}\int_0^t (t-s)^{n-1} i^n e^{is} ds$, and $|R_n(t)| \le \frac{1}{(n-1)!}\int_0^{|t|} s^{n-1} ds = \frac{|t|^n}{n!}$. ∎

Theorem 3.59

For $X \in \mathcal{L}^2$, φ_X is twice continuously differentiable with $\varphi_X'(u) = i\int X e^{iuX} dP$, $\varphi_X''(u) = -\int X^2 e^{iuX} dP$; in particular $\varphi_X'(0) = i\mathbb{E}[X]$, $\varphi_X''(0) = -\mathbb{E}[X^2]$, $|\varphi_X''| \le \mathbb{E}[X^2]$. Moreover

$$\varphi_X(u) = 1 + iu\mathbb{E}[X] + \tfrac{1}{2}\theta(u)u^2\mathbb{E}[X^2] \qquad \text{with } \theta(u) \in \mathbb{C},\ |\theta(u)| \le 1.$$

Proof.

Step 1 (derivatives). $\frac{d}{du}e^{iuX} = iXe^{iuX}$, $\frac{d^2}{du^2}e^{iuX} = -X^2e^{iuX}$. By Lemma 3.58 with $n = 1$, the difference quotients are dominated by $|X|$, resp. X^2, which are integrable. The DCT allows differentiation under the integral and gives continuity of the derivatives.

Step 2 (expansion). Lemma 3.58 with $n = 2$, $t = uX$: $|e^{iuX} - 1 - iuX| \le \frac{1}{2}u^2X^2$, so $|\varphi_X(u) - 1 - iu\mathbb{E}[X]| \le \frac{1}{2}u^2\mathbb{E}[X^2]$. Put $\theta(u) := 0$ if $u^2\mathbb{E}[X^2] = 0$ and $\theta(u) := \frac{\varphi_X(u) - 1 - iu\mathbb{E}[X]}{\frac{1}{2}u^2\mathbb{E}[X^2]}$ otherwise. ∎

From now on $X_1, X_2, \cdots \in \mathcal{L}^2$ are independent with $\mathbb{E}[X_n] = 0$, $\sigma_n^2 = \mathrm{Var}(X_n) > 0$, $s_n = (\sum_{k\le n}\sigma_k^2)^{1/2}$.

Theorem 3.60

Assume

$$\text{(F)}\quad \lim_n \max_{k\le n} \frac{\sigma_k}{s_n} = 0 \ \text{(Feller)}, \qquad \text{(b)}\quad \lim_n \sum_{k=1}^n \left(\varphi_{X_k}\Big(\frac{u}{s_n}\Big) - 1\right) = -\frac{u^2}{2} \quad \forall u \in \mathbb{R}.$$

Then (X_n) has the CLP.

Proof.

Step 1 (reduction). By Theorem 3.51, $\varphi_{S_n^*}(u) = \prod_{k\le n}\varphi_{X_k}(u/s_n)$. If

$$\lim_n \prod_{k=1}^n \varphi_{X_k}\Big(\frac{u}{s_n}\Big) = e^{-u^2/2} \qquad \forall u, \tag{3.15}$$

then $\varphi_{S_n^*} \to e^{-u^2/2} = \widehat{N(0,1)}$ (Theorem 3.52), which is continuous at 0. Lévy's theorem 3.49 and Theorem 3.47 give $P_{S_n^*} \to N(0,1)$ weakly.

Step 2 (comparison with exponentials). By (b), $\prod_k \exp(\varphi_{X_k}(u/s_n) - 1) = \exp\left(\sum_k(\varphi_{X_k}(u/s_n) - 1)\right) \to e^{-u^2/2}$, so it suffices that the two products differ by $o(1)$. Since $|\varphi_{X_k}| \le 1$, $|\exp(\varphi_{X_k} - 1)| = \exp(\mathrm{Re}\,\varphi_{X_k} - 1) \le 1$. For a_k, b_k in the closed unit disc, telescoping gives $|\prod_k a_k - \prod_k b_k| \le \sum_k |a_k - b_k|$. Hence, with $z_k := \varphi_{X_k}(u/s_n) - 1$,

$$\Big|\prod_k \varphi_{X_k}\Big(\frac{u}{s_n}\Big) - \prod_k e^{z_k}\Big| \le D_n := \sum_{k=1}^n |z_k + 1 - e^{z_k}|.$$

Step 3. Fix $\varepsilon \in (0, \frac{1}{2}]$. For $|z| \le \varepsilon$, $|z + 1 - e^z| \le \sum_{j\ge 2}|z|^j/j! \le |z|^2 \le \varepsilon|z|$. By Theorem 3.59 with

$\mathbb{E}[X_k]=0$, $|z_k|\le \frac12\frac{u^2}{s_n^2}\sigma_k^2$, and by (F) this is $<\varepsilon$ for all $k\le n$ once $n\ge n_0$. Then

$$D_n \le \varepsilon\sum_k |z_k| \le \varepsilon\frac{u^2}{2}\sum_k \frac{\sigma_k^2}{s_n^2} = \varepsilon\frac{u^2}{2},$$

so $D_n\to 0$. ∎

Proof of Theorem 3.55.

Step 1. We may assume $\mathbb{E}[X_n]=0$. By Lemma 3.57, (L)⇒(F), so by Theorem 3.60 it remains to derive (b) from (L).

Step 2 (pointwise bounds). Fix u and let

$$Y_k := \Big|e^{i\frac{u}{s_n}X_k} - 1 - i\frac{u}{s_n}X_k + \frac12\frac{u^2}{s_n^2}X_k^2\Big|.$$

Lemma 3.58 ($n=3$) gives $Y_k\le \frac16|\frac{u}{s_n}X_k|^3$, and the triangle inequality with Lemma 3.58 ($n=2$) gives $Y_k\le \frac{u^2}{s_n^2}X_k^2$.

Step 3. Since $\mathbb{E}[X_k]=0$ and $\sum_k \sigma_k^2/s_n^2=1$,

$$\Big|\sum_{k=1}^n\Big(\varphi_{X_k}\Big(\frac{u}{s_n}\Big)-1\Big)+\frac{u^2}{2}\Big| = \Big|\sum_k \mathbb{E}\Big[e^{i\frac{u}{s_n}X_k}-1-i\frac{u}{s_n}X_k+\frac{u^2}{2s_n^2}X_k^2\Big]\Big| \le \sum_k \mathbb{E}[Y_k].$$

Step 4. For $\varepsilon>0$, use the second bound on $\{|X_k|\ge \varepsilon s_n\}$ and the first on the complement, where $\frac{1}{s_n^3}\int_{\{|X_k|<\varepsilon s_n\}}|X_k|^3dP \le \frac{\varepsilon}{s_n^2}\sigma_k^2$:

$$\sum_k \mathbb{E}[Y_k] \le u^2\sum_k\int_{\{|X_k/s_n|\ge\varepsilon\}}\Big(\frac{X_k}{s_n}\Big)^2dP + \frac{|u|^3}{6}\varepsilon\sum_k\frac{\sigma_k^2}{s_n^2} = u^2L_n(\varepsilon)+\frac{|u|^3}{6}\varepsilon.$$

Hence $\limsup_n \sum_k \mathbb{E}[Y_k]\le \frac{|u|^3}{6}\varepsilon$ for every $\varepsilon>0$, which proves (b). ∎

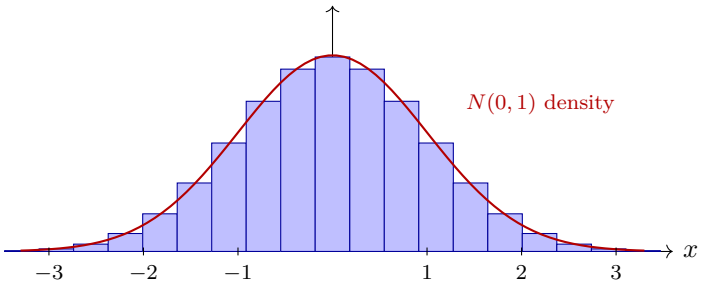


Figure. *The CLT for coin tossing: the law of the standardised number of heads S_n^* for $n=30$, $p=\frac12$, drawn as bars of area $P(S_n=k)$ and width $1/s_n$, against the standard normal density.*

Example 3.61 (Applications)

(i) Ruin probability. An insurer holds n contracts with i.i.d. claims $X_i\ge 0$ in $\mathcal{L}^2$, $m=\mathbb{E}[X_1]$, $\sigma^2=\mathrm{Var}(X_1)$, and charges the premium $\Pi := m+\lambda\sigma^2$ (average claim plus safety loading, $\lambda>0$). With initial capital K, no interest, and payments at the end of the period, ruin is $R:=\{S_n>K+n\Pi\}$. With $S_n^*=\frac{S_n-nm}{\sqrt{n}\sigma}$ the CLT gives

$$P(R)=P\Big(S_n^*>\frac{K+n\lambda\sigma^2}{\sqrt{n}\sigma}\Big)\approx 1-\Phi(c_n),\qquad c_n:=\frac{K+n\lambda\sigma^2}{\sqrt{n}\sigma}\to\infty.$$

As a function of n, c_n increases once $n > K/(\lambda\sigma^2)$, so the normal approximation suggests that the ruin probability eventually decreases as the portfolio grows.

Remark 3.62

Here the normal approximation is applied at levels $c_n \to \infty$, where the CLT alone gives no rate. If moreover $\mathbb{E}|X_1 - m|^3 < \infty$, the Berry–Esseen theorem (Klenke, Thm. 15.51) gives the uniform bound

$$\sup_x |P(S_n^* \le x) - \Phi(x)| \le \frac{C}{\sqrt{n}} \frac{\mathbb{E}|X_1 - m|^3}{\sigma^3},$$

so in particular $|P(S_n^* > c_n) - (1 - \Phi(c_n))| \le \frac{C}{\sqrt{n}} \mathbb{E}|X_1 - m|^3/\sigma^3$, which justifies the numbers below. With $n = 2000$, $\sigma = 60$, $\lambda = 0.5‰$: (a) $K = 0$ gives $P(R) \approx 1 - \Phi(1.342) \approx 9\%$; (b) $K = 1500$ gives $P(R) \approx 3\%$. For $P(R) \le 1‰$ with $K = 0$ one needs $\Phi(\sqrt{n}\lambda\sigma) \ge 0.999$, i.e. $n \ge 10\,611$.

(ii) Stirling's formula via the CLT (Weng 1977). Stirling (1730) proved

$$n! \approx \sqrt{2\pi}\, n^{n+\frac{1}{2}} e^{-n}, \tag{3.16}$$

and De Moivre used it for the CLT for Bernoulli trials (1733). Conversely, the CLT (whose proof above did not use Stirling) yields (3.16). Let X_i be i.i.d. Poisson(1); then $S_n \sim \pi_n$ with $\mathbb{E}[S_n] = \operatorname{Var}(S_n) = n$ and $S_n^* = t_n(S_n)$, $t_n(x) = \frac{x-n}{\sqrt{n}}$. For $f_\infty(x) := x^- = (-x) \vee 0$, a telescoping sum gives

$$\int f_\infty \, dP_{S_n^*} = e^{-n} \sum_{k=0}^{n} \frac{n^k}{k!} \frac{n-k}{\sqrt{n}} = \frac{e^{-n}}{\sqrt{n}} \Big(n + \sum_{k=1}^{n} \Big(\frac{n^{k+1}}{k!} - \frac{n^k}{(k-1)!} \Big) \Big) = \frac{e^{-n} n^{n+\frac{1}{2}}}{n!},$$

while $\int f_\infty \, dN(0,1) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{0} (-x) e^{-x^2/2} dx = \frac{1}{\sqrt{2\pi}}$. So (3.16) follows once

$$\int f_\infty \, dP_{S_n^*} \to \int f_\infty \, dN(0,1). \tag{3.17}$$

This is not immediate, since f_∞ is unbounded. With $f_m := f_\infty \wedge m \in C_b$, the CLT gives $\int f_m \, dP_{S_n^*} \to \int f_m \, dN(0,1)$. The remainder $g_m := f_\infty - f_m \ge 0$ is uniformly small:

$$\int g_m \, dP_{S_n^*} \le \int_{(-\infty, -m]} |x| \, dP_{S_n^*} \le \frac{1}{m} \int x^2 \, dP_{S_n^*} = \frac{1}{m} \operatorname{Var}(S_n^*) = \frac{1}{m},$$

and likewise $\int g_m \, dN(0,1) \le \frac{1}{m}$. A 3ε-argument gives (3.17).

Chapter 4. Brownian Motion

4.1 Definition and some properties

Motivation. Brownian motion is the continuous-time limit of the random walk of Example 3.8: Gaussian, independent increments and continuous paths. This section fixes the definition, records its two basic symmetries, and shows that a typical path has quadratic variation t and therefore infinite length on every interval, so it cannot be treated by classical calculus.

Let $(\Omega, \mathcal{F}, P)$ be a probability space and $I \subset \mathbb{R}_+ := [0, \infty)$ an index set, e.g. $\{0, \dots, n\}$, $\mathbb{N}$, $\mathbb{N}_0$, $[0,1]$, $[0,T]$, $[0,T)$, the positive dyadic rationals $\mathbb{Q}_2^+ := \{m2^{-n} \mid m, n \geq 0\}$, or $[0,\infty)$. A **stochastic process** is a family $(X_t)_{t\in I}$ of r.v.'s on $(\Omega, \mathcal{F}, P)$; for each ω, $t \mapsto X_t(\omega)$ is a **sample path**. More generally X_t may take values in a metric space S (the **state space**), measurably w.r.t. $\mathcal{B}(S)$. We read t as time and write $(X_t)_{t\geq 0}$ when $I = \mathbb{R}_+$.

Definition 4.1 (Brownian motion)

A real-valued process $(B_t)_{t\geq 0}$ is a (standard) **Brownian motion** (BM) if

(i) $P(B_0 = 0) = 1$;

(ii) for $0 \leq s < t$, $B_t - B_s \sim N(0, t-s)$, i.e. $P(B_t - B_s \leq b) = \frac{1}{\sqrt{2\pi(t-s)}} \int_{-\infty}^{b} e^{-x^2/(2(t-s))} dx$;

(iii) it has **independent increments**: for $0 \leq t_1 < \cdots < t_n$, the r.v.'s $B_{t_1} - B_0, B_{t_2} - B_{t_1}, \dots, B_{t_n} - B_{t_{n-1}}$ are independent;

(iv) almost all paths are continuous: $P(t \mapsto B_t \text{ is continuous on } [0,\infty)) = 1$.

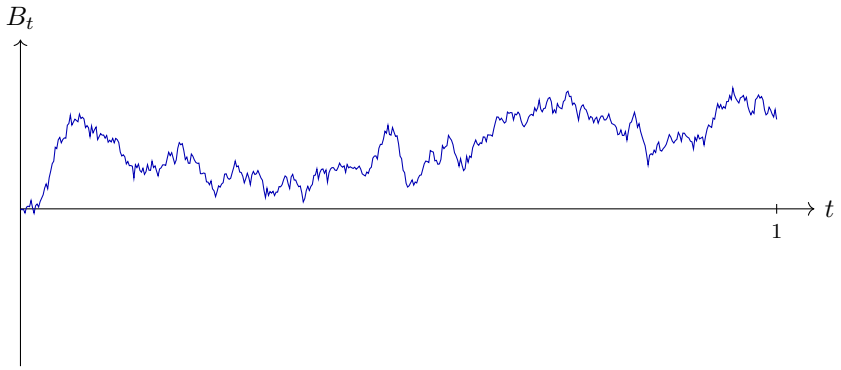


Figure. *A simulated Brownian path on $[0,1]$ (2000 Gaussian increments of variance $1/2000$). It is continuous but wiggles at every scale: its quadratic variation over $[0,t]$ is t (Theorem 4.3), so its total variation is infinite (Proposition 4.6).*

Proposition 4.2

Let $(B_t)_{t\geq 0}$ be a BM. (i) (Translation invariance) For fixed $t_0 \geq 0$, $\tilde{B}_t := B_{t+t_0} - B_{t_0}$ is a BM. (ii) (Scaling invariance) For $\lambda > 0$, $\tilde{B}_t := \frac{1}{\sqrt{\lambda}} B_{\lambda t}$ is a BM.

Proof. (i) Definition 4.1(i),(iv) are clear, and $\tilde{B}_t - \tilde{B}_s = B_{t+t_0} - B_{s+t_0} \sim N(0, t-s)$. For $0 \leq t_1 < \cdots < t_n$, the increments of $\tilde{B}$ are increments of B along $t_0 \leq t_1 + t_0 < \cdots < t_n + t_0$ (if $t_1 = 0$ the first one is the constant 0, which is independent of everything), hence independent by (iii) for B.

(ii) (i), (iii),

(iv) carry over directly. For (ii), substituting $y = x/\sqrt{\lambda}$,

$$P(\tilde{B}_t - \tilde{B}_s \le b) = P(B_{\lambda t} - B_{\lambda s} \le \sqrt{\lambda} b) = \int_{-\infty}^{\sqrt{\lambda} b} \frac{e^{-x^2/(2\lambda(t-s))}}{\sqrt{2\pi\lambda(t-s)}} dx = \int_{-\infty}^{b} \frac{e^{-y^2/(2(t-s))}}{\sqrt{2\pi(t-s)}} dy.$$

■

Let $\tau_n = \{0 = t_1^{(n)} < t_2^{(n)} < \cdots < t_{N(\tau_n)}^{(n)} < \infty\}$ be partitions of $[0, \infty)$ with

$$|\tau_n| := \max_i \left(t_{i+1}^{(n)} - t_i^{(n)}\right) \to 0 \quad \text{(mesh)}, \qquad t_{N(\tau_n)}^{(n)} \to \infty. \tag{$\star$}$$

Theorem 4.3 (Quadratic variation)

Let (B_t) be a BM and (τ_n) as in $(\star)$.

(i) For every $t \ge 0$, $S_t^n := \sum_{t_i^{(n)} \in \tau_n,\ t_{i+1}^{(n)} \le t} \left(B_{t_{i+1}^{(n)}} - B_{t_i^{(n)}}\right)^2 \to t$ in $L^2(\Omega, \mathcal{F}, P)$.

(ii) If $\tau_n \subset \tau_{n+1}$ for all n, or $\sum_n |\tau_n| < \infty$, then $P(\lim_n S_t^n = t \ \forall t \ge 0) = 1$.

Proof. (i) Write $\Delta_i := t_{i+1}^{(n)} - t_i^{(n)}$; the sums run over i with $t_{i+1}^{(n)} \le t$. Then $\mathbb{E}[S_t^n] = \sum \Delta_i \to t$ in $\mathbb{R}$, by $(\star)$. By independence of increments and $\mathbb{E}[N^4] = 3\sigma^4$ for $N \sim N(0, \sigma^2)$,

$$\operatorname{Var}(S_t^n) = \sum \operatorname{Var}\left((B_{t_{i+1}} - B_{t_i})^2\right) = \sum (3\Delta_i^2 - \Delta_i^2) = 2\sum \Delta_i^2 \le 2t|\tau_n| \to 0.$$

Hence $S_t^n - \mathbb{E}[S_t^n] \to 0$ in L^2, and with $\mathbb{E}[S_t^n] \to t$ we get $S_t^n \to t$ in L^2.

(ii) [Supplement] *Case* $\sum_n |\tau_n| < \infty$ (this includes the dyadic partitions $t_i^{(n)} = (i-1)2^{-n}$, the case treated in the lectures).

Step 1 (fixed t). By Chebyshev and the bound above, $\sum_n P(|S_t^n - \mathbb{E}S_t^n| \ge \varepsilon) \le \varepsilon^{-2} \sum_n 2t|\tau_n| < \infty$. By Lemma 2.10, $S_t^n - \mathbb{E}[S_t^n] \to 0$ a.s., hence $S_t^n \to t$ on a set Ω_t with $P(\Omega_t) = 1$.

Step 2 (all t at once). Let $\Omega' := \bigcap_{t \in \mathbb{Q}_2^+} \Omega_t$, so $P(\Omega') = 1$. For $\omega \in \Omega'$ and arbitrary t, take dyadic $t_k^- \nearrow t$, $t_k^+ \searrow t$. Since S_t^n is increasing in t,

$$t_k^- = \lim_n S_{t_k^-}^n(\omega) \le \liminf_n S_t^n(\omega) \le \limsup_n S_t^n(\omega) \le \lim_n S_{t_k^+}^n(\omega) = t_k^+,$$

and $k \to \infty$ gives the claim.

General nested partitions: see Revuz–Yor, *Continuous Martingales and Brownian Motion*, Ch. II, Prop. (2.12); not proved in the lectures. ■

Remark 4.4

Let Ω'' with $P(\Omega'') = 1$ be such that $t \mapsto B_t(\omega)$ is continuous for $\omega \in \Omega''$, and $\tilde{\Omega} := \Omega' \cap \Omega''$ with Ω' from the proof above. For $\tilde{\omega} \in \tilde{\Omega}$ the path is continuous and, along (τ_n) as in Theorem 4.3(ii),

$$\langle B \rangle_t(\tilde{\omega}) := \lim_n \sum_{t_{i+1}^{(n)} \le t} \left(B_{t_{i+1}^{(n)}}(\tilde{\omega}) - B_{t_i^{(n)}}(\tilde{\omega})\right)^2 = t \qquad \forall t \ge 0.$$

Such a path is a **typical path**, and $\langle B \rangle_t(\tilde{\omega})$ its **quadratic variation**.

Definition 4.5 (Finite variation)

$g:\mathbb{R}_+\to\mathbb{R}$ has **finite variation** on $[0,t]$ if

$$\operatorname{Var}_t g := \sup_{\tau} \sum_{t_i\in\tau,\ t_{i+1}\le t} |g(t_{i+1})-g(t_i)| < \infty, \tag{4.1}$$

the supremum over all partitions $0=t_1<\cdots<t_{N(\tau)}<\infty$; g has finite variation if this holds for every t.

Proposition 4.6

A typical path of BM is continuous and of unbounded variation on every interval $[0,T]$, $T>0$.

Proof. Suppose $\operatorname{Var}_T B_\cdot(\tilde\omega)<\infty$ for some $T>0$. The path is uniformly continuous on $[0,T]$, so for $\varepsilon>0$ the increments are $<\varepsilon$ once $|\tau_n|<\delta$, and

$$\sum_{t_{i+1}\le T} \left(B_{t_{i+1}}-B_{t_i}\right)^2(\tilde\omega) \le \max_i |B_{t_{i+1}}-B_{t_i}|(\tilde\omega)\cdot\sum_i |B_{t_{i+1}}-B_{t_i}|(\tilde\omega) \le \varepsilon \operatorname{Var}_T B_\cdot(\tilde\omega).$$

Hence $\langle B\rangle_T(\tilde\omega)=0$, contradicting $\langle B\rangle_T(\tilde\omega)=T>0$. ∎

4.2 Construction (hence existence) of Brownian motion

Motivation. Does Brownian motion exist? Kolmogorov's extension theorem builds a process with the prescribed Gaussian finite-dimensional laws on $\mathbb{R}^I$, but the set of continuous paths is not measurable there. We therefore build the process on the dyadic times, show with the Kolmogorov–Čentsov criterion that its paths are a.s. uniformly continuous, and extend them continuously.

For a process $(X_t)_{t\in I}$ and pairwise distinct $t_1,\dots,t_n\in I$ let

$$\mu_{t_1\dots t_n}(A_1\times\cdots\times A_n) := P\big((X_{t_1},\dots,X_{t_n})\in A_1\times\cdots\times A_n\big), \qquad A_i\in\mathcal{B}(\mathbb{R}). \tag{4.2}$$

By Theorem 1.9 this determines the law $P((X_{t_1},\dots,X_{t_n})\in A)$, $A\in\mathcal{B}(\mathbb{R}^n)$, because rectangles are a $\cap$-stable generator of $\mathcal{B}(\mathbb{R}^n)$. These **finite-dimensional distributions** are **consistent**:

$$\mu_{t_1\dots t_n}(A_1\times\cdots\times A_n) = \mu_{t_{\pi(1)}\dots t_{\pi(n)}}(A_{\pi(1)}\times\cdots\times A_{\pi(n)}) \quad \text{for permutations } \pi, \tag{4.3}$$

$$\mu_{t_1\dots t_n}(A_1\times\cdots\times A_{n-1}\times\mathbb{R}) = \mu_{t_1\dots t_{n-1}}(A_1\times\cdots\times A_{n-1}) \quad (n\ge 2). \tag{4.4}$$

Any family of probability measures $\mu_{t_1\dots t_n}$ on $\mathbb{R}^n$ satisfying (4.3), (4.4) is called consistent. Conversely, does every consistent family come from a process?

On $\mathbb{R}^I:=\{\omega: I\to\mathbb{R}\}$ let the **cylinder sets** be

$$\mathcal{C}_I := \big\{\{\omega\mid(\omega(t_1),\dots,\omega(t_n))\in A_1\times\cdots\times A_n\}\ \big|\ t_1<\cdots<t_n\in I,\ A_i\in\mathcal{B}(\mathbb{R}),\ n\in\mathbb{N}\big\}.$$

$\sigma(\mathcal{C}_I)$ is the smallest σ-algebra making all finite-dimensional projections $f_{t_1\dots t_n}(\omega) := (\omega(t_1),\dots,\omega(t_n))$ measurable, and, since $f^{-1}_{t_1\dots t_n}(A_1\times\cdots\times A_n)=\bigcap_i f_{t_i}^{-1}(A_i)$, even the smallest one making all f_t, $t\in I$, measurable.

Theorem 4.7 (Kolmogorov's extension theorem)

For every consistent family $\mu_{t_1\dots t_n}$ there is a unique probability measure P on $(\mathbb{R}^I,\sigma(\mathcal{C}_I))$ such that, with $X_t(\omega):=\omega(t)$, (4.2) holds, i.e. $\mu_{t_1\dots t_n}=P\circ f^{-1}_{t_1\dots t_n}$.

Proof [Supplement].

Step 1 (inner regularity). Every probability measure μ on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ satisfies $\mu(A) = \sup\{\mu(K) \mid K \subset A \text{ compact}\}$. The class of A such that for every ε there are F closed $\subset A \subset G$ open with $\mu(G \setminus F) < \varepsilon$ is a σ-algebra. It contains every closed F (take $G = \{d(\cdot, F) < \frac{1}{m}\}$ and continuity from above), hence all Borel sets. Intersecting F with a large closed ball gives a compact K.

Step 2 (a finitely additive P_0). The general cylinders $Z = \{\omega \mid f_J(\omega) \in A\}$, $J \subset I$ finite, $A \in \mathcal{B}(\mathbb{R}^J)$, form an algebra $\mathcal{Z}$ with $\sigma(\mathcal{Z}) = \sigma(\mathcal{C}_I)$. Put $P_0(Z) := \mu_J(A)$. This is well defined: two representations can be lifted to a common finite $J' \supset J$, and (4.3), (4.4), which extend from rectangles to all Borel sets by Theorem 1.9, show that $\mu_J(A) = \mu_{J'}(A \times \mathbb{R}^{J' \setminus J})$. Since finitely many cylinders can be written over one common J, P_0 is finitely additive with $P_0(\mathbb{R}^I) = 1$.

Step 3 (σ-additivity). By Theorem 1.15 it suffices that $Z_k \in \mathcal{Z}$, $Z_k \searrow \emptyset$ imply $P_0(Z_k) \to 0$. Suppose $P_0(Z_k) \geq \varepsilon > 0$ for all k, with $Z_k = \{f_{J_k} \in A_k\}$ and $J_1 \subset J_2 \subset \cdots$. By Step 1 choose compact $K_k \subset A_k$ with $\mu_{J_k}(A_k \setminus K_k) \leq \varepsilon 2^{-k-1}$, and put $E_k := \bigcap_{j \leq k} \{f_{J_j} \in K_j\} \subset Z_k$. Then $P_0(Z_k \setminus E_k) \leq \varepsilon/2$, so $P_0(E_k) \geq \varepsilon/2$ and $E_k \neq \emptyset$; pick $\omega_k \in E_k$. On the countable set $J := \bigcup_k J_k$, for each $t \in J_j$ the values $\omega_k(t)$, $k \geq j$, lie in the compact projection of K_j. A diagonal argument gives a subsequence with $\omega_{k_l}(t) \to \omega(t)$ for all $t \in J$; put $\omega(t) := 0$ off J. For each j, $f_{J_j}(\omega_{k_l}) \in K_j$ for $k_l \geq j$ and K_j is closed, so $f_{J_j}(\omega) \in K_j \subset A_j$. Thus $\omega \in \bigcap_j Z_j = \emptyset$, a contradiction.

Step 4. Carathéodory's Theorem 1.11 (on the algebra $\mathcal{Z}$, finite measure) extends P_0 uniquely to $\sigma(\mathcal{C}_I)$, and $P \circ f^{-1}_{t_1 \dots t_n} = \mu_{t_1 \dots t_n}$ by construction. Uniqueness: $\mathcal{Z}$ is a $\cap$-stable generator (Theorem 1.9). ∎

The Brownian finite-dimensional laws. Let $I = \mathbb{R}_+$ and

$$p_t(x, y) := \frac{1}{\sqrt{2\pi t}} e^{-\frac{(y-x)^2}{2t}} \quad (t > 0), \qquad p_0(x, y)\, dy := \delta_x(dy). \tag{4.5}$$

For $0 \leq t_1 < \cdots < t_n$ and $A \in \mathcal{B}(\mathbb{R}^n)$ set, with $t_0 := 0$, $x_0 := 0$,

$$\mu_{t_1 \dots t_n}(A) := \int \cdots \int \mathbb{1}_A(x_1, \dots, x_n) \prod_{i=1}^{n} p_{t_i - t_{i-1}}(x_{i-1}, x_i)\, dx_n \cdots dx_1$$

(if $t_1 = 0$, the first factor means $\delta_0(dx_1)$). This is a probability measure: the integrand is a nonnegative Borel density (Theorem 1.47), and integrating out $x_n, x_{n-1}, \dots$ successively gives total mass 1. By Fubini and Theorem 1.9 it is the unique probability measure with

$$\mu_{t_1 \dots t_n}(A_1 \times \cdots \times A_n) = \int_{A_1} \cdots \int_{A_n} \prod_{i=1}^{n} p_{t_i - t_{i-1}}(x_{i-1}, x_i)\, dx_n \cdots dx_1. \tag{4.6}$$

(Replacing $x_0 = 0$ by $x_0 = x$ gives Brownian motion started at x.) For unordered times, sort them by the unique permutation σ with $t_{\sigma(1)} < \cdots < t_{\sigma(n)}$ and permute the coordinates accordingly. The family is consistent: (4.3) holds by construction, and (4.4) amounts to deleting one time,

$$\mu_{t_1 \dots t_{i-1} t_{i+1} \dots t_n}(\cdots \times A_{i-1} \times A_{i+1} \times \dots) = \mu_{t_1 \dots t_n}(\cdots \times A_{i-1} \times \mathbb{R} \times A_{i+1} \times \dots). \tag{4.7}$$

For $i = n$ this is immediate. For $i < n$ it is the **Chapman–Kolmogorov equation**

$$\int_A p_{t+s}(x, z)\, dz = \int_A \int_{\mathbb{R}} p_s(x, y) p_t(y, z)\, dy\, dz, \qquad \text{i.e.} \quad \int p_s(x, y) p_t(y, z)\, dy = p_{s+t}(x, z),$$

which holds because $N(0, s) * N(0, t) = N(0, s + t)$ (Example 3.44(i)). [Supplement]

The difficulty. By Theorem 4.7 there is a unique $\tilde{P}$ on $(\mathbb{R}^{\mathbb{R}_+}, \sigma(\mathcal{C}_{\mathbb{R}_+}))$ with these marginals, and $X_t(\omega) = \omega(t)$ satisfies Definition 4.1(i)–(iii) (see below). But (iv) fails to make sense: $C(\mathbb{R}_+) \notin \sigma(\mathcal{C}_{\mathbb{R}_+})$. Sets in $\sigma(\mathcal{C}_{\mathbb{R}_+})$ depend on only countably many coordinates, and continuity cannot be decided from countably many values. So we first restrict to the dyadic rationals $\mathbb{Q}_2^+$: the restricted family is still consistent, and Theorem 4.7 gives a unique Q on $(\mathbb{R}^{\mathbb{Q}_2^+}, \sigma(\mathcal{C}_{\mathbb{Q}_2^+}))$ with these marginals for $X_t(\omega) = \omega(t)$, $t \in \mathbb{Q}_2^+$.

Theorem 4.8

Let $C(\mathbb{Q}_2^+) := \bigcap_{n\ge1}\{\omega \in \mathbb{R}^{\mathbb{Q}_2^+} \mid \omega|_{[0,n]\cap\mathbb{Q}_2^+} \text{ is uniformly continuous}\}$. Then $C(\mathbb{Q}_2^+) \in \sigma(\mathcal{C}_{\mathbb{Q}_2^+})$ and $Q(C(\mathbb{Q}_2^+)) = 1$; so $(C(\mathbb{Q}_2^+), C(\mathbb{Q}_2^+) \cap \sigma(\mathcal{C}_{\mathbb{Q}_2^+}), Q)$ is a probability space (with the trace σ-algebra).

From Theorem 4.8 to Brownian motion. Let $\Omega := C(\mathbb{R}_+)$ with the metric

$$\rho(\omega,\omega') := \sum_{n\ge1} 2^{-n}\Big(1 \wedge \sup_{t\in[0,n]} |\omega(t)-\omega'(t)|\Big),$$

under which Ω is a complete separable metric space; $\mathcal{F} := \mathcal{B}(\Omega)$ and $B_t(\omega) := \omega(t)$, which is ρ-continuous and hence measurable, so $\sigma(B_t \mid t \ge 0) \subset \mathcal{F}$. Conversely, for a countable dense $D \subset \Omega$ and $\omega_0 \in D$, $\rho(\omega,\omega_0) = \sum_n 2^{-n}(1 \wedge \sup_{t\in[0,n]\cap\mathbb{Q}_2^+} |B_t(\omega)-\omega_0(t)|)$ is $\sigma(B_t \mid t \in \mathbb{Q}_2^+)$-measurable. Every open set is a countable union of balls $B_\rho(\omega_0, r)$, $\omega_0 \in D$, $r \in \mathbb{Q}_2^+$, so $\mathcal{F} = \sigma(B_t \mid t \ge 0)$.

The map $\psi : C(\mathbb{Q}_2^+) \to \Omega$, $\omega \mapsto$ the unique continuous extension of ω to $\mathbb{R}_+$, is measurable: for $q_m \to t$ in $\mathbb{Q}_2^+$,

$$\psi^{-1}(\{B_t < \alpha\}) = \bigcup_k \bigcup_n \bigcap_{m\ge n} \{\omega \mid \omega(q_m) < \alpha - \tfrac{1}{k}\},$$

and the sets $\{B_t < \alpha\}$ generate $\mathcal{F}$. Let $P := Q \circ \psi^{-1}$. For dyadic t_i, $(\psi\omega)(t_i) = \omega(t_i)$, so

$$P((B_{t_1},\dots,B_{t_n}) \in A_1\times\cdots\times A_n) = Q((X_{t_1},\dots,X_{t_n}) \in A_1\times\cdots\times A_n) \qquad (t_i \in \mathbb{Q}_2^+). \tag{4.8}$$

Then $B_t(\omega) = \omega(t)$ is a Brownian motion on $(\Omega,\mathcal{F},P)$, every path being continuous. The key is that for all $0 \le t_1 < \cdots < t_n$ in $\mathbb{R}_+$, with $t_0 = 0$, $x_0 = 0$,

$$P((B_{t_1},\dots,B_{t_n}) \in A_1\times\cdots\times A_n) = \int_{A_1}\cdots\int_{A_n} \prod_{i=1}^{n} p_{t_i-t_{i-1}}(x_{i-1},x_i)\,dx_n\cdots dx_1, \tag{4.9}$$

and that (4.9) is equivalent to Definition 4.1(i)–(iii).

[Supplement] *Proof of (4.9).* Take dyadic $t_i^k \to t_i$ (strictly ordered). For $f \in C_b(\mathbb{R}^n)$, $\mathbb{E}_P[f(B_{t_1^k},\dots,B_{t_n^k})] = \int f\,d\mu_{t_1^k\dots t_n^k}$ by (4.8). As $k\to\infty$ the left side tends to $\mathbb{E}_P[f(B_{t_1},\dots,B_{t_n})]$ by path continuity and the DCT. In the right side substitute $x_i = x_{i-1} + \sqrt{t_i^k - t_{i-1}^k}\,z_i$: it becomes an integral of f evaluated at points depending continuously on k against a fixed Gaussian density, so the DCT gives the limit $\int f\,d\mu_{t_1\dots t_n}$. By Example 2.30 on $\mathbb{R}^n$ the two laws agree.

(4.9) gives (i)–(iii). $n=1$ gives $B_t \sim N(0,t)$ and $B_0 = 0$ a.s. For $t_1 < t_2$, with $\bar{A} = \{(x,y) \mid y-x \le \alpha\}$ and $z = y-x$,

$$P(B_{t_2}-B_{t_1} \le \alpha) = \int_{\mathbb{R}} p_{t_1}(0,x)\int_{-\infty}^{x+\alpha} p_{t_2-t_1}(x,y)\,dy\,dx = \int_{-\infty}^{\alpha} \frac{e^{-z^2/(2(t_2-t_1))}}{\sqrt{2\pi(t_2-t_1)}}dz.$$

More generally, the change of variables $z_i = x_i - x_{i-1}$ turns the density in (4.9) into $\prod_i p_{t_i-t_{i-1}}(0,z_i)$, a product density. So $(B_{t_1}, B_{t_2}-B_{t_1},\dots,B_{t_n}-B_{t_{n-1}})$ has independent $N(0,t_i-t_{i-1})$ coordinates (Theorem 3.42), i.e.

$$P(B_{t_1}\le\alpha_1, B_{t_2}-B_{t_1}\le\alpha_2,\dots,B_{t_n}-B_{t_{n-1}}\le\alpha_n) = P(B_{t_1}\le\alpha_1)\cdots P(B_{t_n}-B_{t_{n-1}}\le\alpha_n). \tag{4.10}$$

Conversely (i)–(iii) determine the joint law of the increments, hence (4.9).

Proof of Theorem 4.8.

Step 1 (measurability). With $\Omega_n := \{\omega \mid \omega|_{[0,n]\cap\mathbb{Q}_2^+} \text{ uniformly continuous}\}$,

$$\Omega_n = \bigcap_{M\ge1}\bigcup_{N\ge1}\bigcap_{t,s\in\mathbb{Q}_2^+\cap[0,n],\ |t-s|\le\frac{1}{N}} \Big\{|X_t - X_s| \le \frac{1}{M}\Big\} \in \sigma(\mathcal{C}_{\mathbb{Q}_2^+}),$$

a countable combination of cylinder sets. It remains to show $Q(\Omega_n) = 1$ for each n.

Step 2 (moments). The marginals of $(X_t)_{t\in\mathbb{Q}_2^+}$ satisfy Definition 4.1(i)–(iii), hence scaling invariance: $X_t - X_s \sim \frac{1}{\sqrt{\lambda}}(X_{\lambda t} - X_{\lambda s}) \sim N(0, t-s)$. For $Y_t^n := \frac{1}{\sqrt{n}} X_{nt}$ and $s < t$ in $\mathbb{Q}_2^+ \cap [0,1]$,

$$\mathbb{E}_Q|Y_t^n - Y_s^n|^{2k} = \mathbb{E}_Q|X_t - X_s|^{2k} = \mathbb{E}_Q|\sqrt{t-s}\, X_1|^{2k} = c_k|t-s|^k, \qquad c_k := \mathbb{E}_Q|X_1|^{2k}. \tag{4.11}$$

Step 3. By Theorem 4.9 and Remark 4.10 there is Ω^* with $Q(\Omega^*) = 1$ on which $t \mapsto Y_t^n(\omega)$ is Hölder continuous of order $\gamma \in (0, \frac{1}{2})$ on $[0,1] \cap \mathbb{Q}_2^+$, i.e. $t \mapsto X_t(\omega)$ is Hölder on $[0,n] \cap \mathbb{Q}_2^+$. So $\Omega^* \subset \Omega_n$. ∎

Theorem 4.9 (Kolmogorov–Čentsov, dyadic version)

Let $(X_t)_{t\in\mathbb{Q}_2^+}$ be a process on $(\mathbb{R}^{\mathbb{Q}_2^+}, \sigma(\mathcal{C}_{\mathbb{Q}_2^+}), Q)$ with

$$\mathbb{E}_Q|X_t - X_s|^\alpha \le K|t-s|^{1+\beta} \qquad \forall t, s \in \mathbb{Q}_2^+ \cap [0,1]$$

for some $\alpha, \beta, K > 0$, and let $\gamma \in (0, \beta/\alpha)$. Then there is $\Omega^* \in \sigma(\mathcal{C}_{\mathbb{Q}_2^+})$ with $Q(\Omega^*) = 1$ such that for every $\omega \in \Omega^*$ there is $n^*(\omega)$ with

$$\sup_{t,s\in\mathbb{Q}_2^+\cap[0,1],\ 0<t-s<2^{-n^*(\omega)}} \frac{|X_t(\omega) - X_s(\omega)|}{|t-s|^\gamma} \le C := \frac{2}{1-2^{-\gamma}}.$$

In particular every $\omega \in \Omega^*$ is Hölder continuous of order γ on $\mathbb{Q}_2^+ \cap [0,1]$: with $n_0 := n^*(\omega)+1$, splitting $[s,t]$ into 2^{n_0} equal pieces gives $|X_t(\omega) - X_s(\omega)| \le 2^{n_0} C\big((t-s)2^{-n_0}\big)^\gamma = 2^{n_0(1-\gamma)} C|t-s|^\gamma$.

Remark 4.10

By (4.11) we may take $\alpha = 2k$, $\beta = k-1$, and $\beta/\alpha = \frac{k-1}{2k} \nearrow \frac{1}{2}$. So Theorem 4.9 applies to every $\gamma \in (0, \frac{1}{2})$: Q-a.e. path is locally Hölder continuous of every order $\gamma < \frac{1}{2}$. The map ψ preserves local Hölder continuity, so P-a.e. Brownian path in $C(\mathbb{R}_+)$ is locally γ-Hölder for every $\gamma < \frac{1}{2}$.

Proof of Theorem 4.9.

Step 1 (Borel–Cantelli on dyadic increments). By Markov's inequality and the hypothesis,

$$\begin{aligned} Q\Big(\max_{1\le k\le 2^n} |X_{k2^{-n}} - X_{(k-1)2^{-n}}| \ge 2^{-\gamma n}\Big) &\le \sum_{k=1}^{2^n} \frac{\mathbb{E}_Q|X_{k2^{-n}} - X_{(k-1)2^{-n}}|^\alpha}{2^{-\gamma n\alpha}} \\ &\le 2^n \frac{K 2^{-n(1+\beta)}}{2^{-\gamma n\alpha}} = K 2^{-n(\beta-\gamma\alpha)}. \end{aligned}$$

This is summable since $\beta > \gamma\alpha$. By Borel–Cantelli, $\Omega^* := \bigcup_l \bigcap_{n\ge l} \{\max_k |X_{k2^{-n}} - X_{(k-1)2^{-n}}| < 2^{-\gamma n}\}$ has $Q(\Omega^*) = 1$, and for $\omega \in \Omega^*$ there is $n^*(\omega)$ with

$$\max_{1\le k\le 2^n} |X_{k2^{-n}}(\omega) - X_{(k-1)2^{-n}}(\omega)| < 2^{-\gamma n} \qquad \forall n \ge n^*(\omega). \tag{4.12}$$

Step 2 (chaining). Fix $\omega \in \Omega^*$, $n \ge n^*(\omega)$, and let $D_m := \{k2^{-m} \mid 0 \le k \le 2^m\}$. We claim, for all $m > n$,

$$|X_t(\omega) - X_s(\omega)| \le 2 \sum_{j=n+1}^{m} 2^{-\gamma j} \qquad \forall t, s \in D_m,\ 0 < t-s < 2^{-n}. \tag{4.13}$$

For $m = n+1$, t, s are neighbours in D_{n+1}, and (4.12) gives $< 2^{-\gamma(n+1)}$. Assume (4.13) for $m = M-1$, and let $s, t \in D_M$, $0 < t - s < 2^{-n}$. Put $s_1 := \min\{r \in D_{M-1} \mid r \geq s\}$ and $t_1 := \max\{r \in D_{M-1} \mid r \leq t\}$. Then $s \leq s_1 \leq t_1 \leq t$ and $s_1 - s, t - t_1 \in \{0, 2^{-M}\}$, so (4.12) bounds $|X_{s_1} - X_s|$ and $|X_t - X_{t_1}|$ by $2^{-\gamma M}$, and the induction hypothesis bounds $|X_{t_1} - X_{s_1}|$. Summing gives (4.13) for $m = M$.

Step 3. For $t, s \in \mathbb{Q}_2^+ \cap [0,1]$ with $0 < t - s < 2^{-n^*(\omega)}$ choose $m > n \geq n^*(\omega)$ with $t, s \in D_m$ and $2^{-(n+1)} \leq t - s < 2^{-n}$. By (4.13),

$$|X_t - X_s| \leq 2 \sum_{j \geq n+1} 2^{-\gamma j} = \frac{2 \cdot 2^{-\gamma(n+1)}}{1 - 2^{-\gamma}} \leq \frac{2}{1 - 2^{-\gamma}} |t - s|^{\gamma}.$$

■

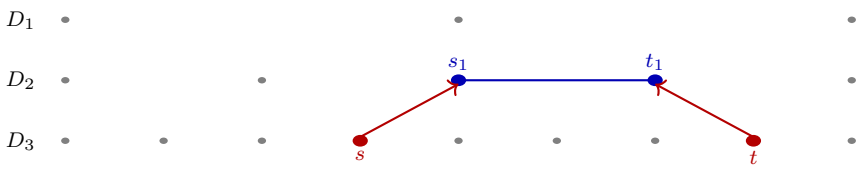


Figure. *Chaining in the proof of Theorem 4.9: $s, t \in D_M$ are moved by at most one step of size 2^{-M} to $s_1, t_1 \in D_{M-1}$, and the induction continues on coarser grids. The increments along the chain are bounded by $2^{-\gamma j}$ at level j, a convergent geometric series.*

Remark 4.11

$((B_t)_{t \geq 0}, \Omega, \mathcal{F}, P)$ as constructed above is the **canonical model** of Brownian motion; $P = P_0$ is the **Wiener measure** (started at 0).

4.3 The Markov property

Motivation. Brownian motion forgets its past: given the present position B_s, the future path $(B_{s+t})_{t \geq 0}$ is a Brownian motion started at B_s, regardless of how it got there. On the canonical space this is expressed with the shift θ_s. Working with the slightly larger filtration $\mathcal{F}_t^{0,+}$ costs nothing and yields Blumenthal's 0–1 law.

Work in the canonical model: $\Omega = C(\mathbb{R}_+)$, $B_t(\omega) = \omega(t)$, $\mathcal{F} = \mathcal{B}(C(\mathbb{R}_+)) = \sigma(B_t \mid t \geq 0)$. For a start in $x \in \mathbb{R}$ there are three equivalent options: (1) (B_t) under P_x, built as P but with $x_0 = x$ in (4.6); (2) $(B_t + x)$ under P_0; (3) (B_t) under $P_0 \circ T_x^{-1}$ with $T_x(\omega) := \omega + x$. All have the same finite-dimensional distributions and continuous paths; in particular $P_x = P_0 \circ T_x^{-1}$. We write $\mathbb{E}_x[Y] := \int Y \, dP_x$.

Let $\mathcal{F}_t^0 := \sigma(B_u \mid u \leq t)$. Then $(\mathcal{F}_t^0)_{t \geq 0}$ is a **filtration** ($\mathcal{F}_s^0 \subset \mathcal{F}_t^0 \subset \mathcal{F}$ for $s \leq t$), the "information up to time t", and (B_t) is **adapted** (B_t is $\mathcal{F}_t^0$-measurable). The **shift** $\theta_s : \Omega \to \Omega$, $\theta_s(\omega)(t) := \omega(s+t)$, is ρ-continuous, hence measurable, and in fact $\hat{\mathcal{F}}_s^0/\mathcal{F}$-measurable with $\hat{\mathcal{F}}_s^0 := \sigma(B_u \mid u \geq s)$, the future after s. For $Y \in \mathcal{F}_b$ (bounded $\mathcal{F}$-measurable), $Y \circ \theta_s$ is "a function of the future after time s". For instance, $Y = f(B_{t_1}, \ldots, B_{t_n})$ gives $Y \circ \theta_s = f(B_{t_1+s}, \ldots, B_{t_n+s})$, and $B_t \circ \theta_s = B_{t+s}$.

The canonical model has the **Markov property** w.r.t. $(\mathcal{F}_t^0)$: for $Y \in \mathcal{F}_b$,

$$\mathbb{E}_x[\underbrace{Y \circ \theta_s}_{\text{future}} \mid \underbrace{\mathcal{F}_s^0}_{\text{past}}] = \mathbb{E}_{\underbrace{B_s}_{\text{present}}}[Y] \qquad P_x\text{-a.s., } \forall s \geq 0,\ x \in \mathbb{R}. \tag{4.14}$$

It will follow from the Markov property (4.15) for a larger filtration (Remark 4.13).

Remark 4.12

$x \mapsto \mathbb{E}_x[Y]$ is Borel measurable. [Supplement] *Reason:* the class of $Y \in \mathcal{F}_b$ for which this holds is a monotone vector space (MCT). It contains the multiplicative class of indicators of cylinder sets, for which $\mathbb{E}_x[\mathbb{1}_A]$ is an integral of Gaussian kernels continuous in x, by (4.6) and the DCT. Theorem 2.29 applies. Hence $\mathbb{E}_{B_s}[Y]$ is $\sigma(B_s)$-measurable, and since $\sigma(B_s) \subset \mathcal{F}_s^0$, P_x-a.s.

$$\begin{aligned}\mathbb{E}_x[Y \circ \theta_s \mid \sigma(B_s)] &= \mathbb{E}_x\big[\mathbb{E}_x[Y \circ \theta_s \mid \mathcal{F}_s^0] \,\big|\, \sigma(B_s)\big] = \mathbb{E}_x[\mathbb{E}_{B_s}[Y] \mid \sigma(B_s)] \\ &= \mathbb{E}_{B_s}[Y] = \mathbb{E}_x[Y \circ \theta_s \mid \mathcal{F}_s^0],\end{aligned}$$

using Theorem 2.35(iv), (4.14) and Example 2.34(i). The best guess for the future given the present equals the best guess given the whole past.

For technical reasons we enlarge $\mathcal{F}_t^0$ to

$$\mathcal{F}_t^{0,+} := \bigcap_{\varepsilon>0} \mathcal{F}_{t+\varepsilon}^0 \ (\supset \mathcal{F}_t^0),$$

which allows an infinitesimal look into the future. It is **right continuous**: $\bigcap_{\varepsilon>0} \mathcal{F}_{t+\varepsilon}^{0,+} = \bigcap_n \bigcap_k \mathcal{F}_{t+\frac{1}{n}+\frac{1}{k}}^0 \subset \mathcal{F}_t^{0,+}$, and trivially $\mathcal{F}_t^{0,+} \subset \bigcap_{\varepsilon>0} \mathcal{F}_{t+\varepsilon}^{0,+}$. (B_t) is adapted to $(\mathcal{F}_t^{0,+})$ and has the Markov property w.r.t. it:

$$\mathbb{E}_x[Y \circ \theta_s \mid \mathcal{F}_s^{0,+}] = \mathbb{E}_{B_s}[Y] \qquad P_x\text{-a.s., } \forall s \geq 0,\ x \in \mathbb{R},\ Y \in \mathcal{F}_b. \tag{4.15}$$

Proof of (4.15).

Step 1 (reduction). Let $\mathcal{C}$ be the cylinder sets $A = \{(B_{t_1}, \dots, B_{t_n}) \in A_1 \times \cdots \times A_n\}$, $0 < t_1 < \cdots < t_n$, $A_i \in \mathcal{B}(\mathbb{R})$; $\mathcal{C}$ is $\cap$-stable and generates $\mathcal{F}$. $\mathcal{H} := \{Y \in \mathcal{F}_b \mid (4.15) \text{ holds}\}$ is an MVS (conditional MCT), so by Theorem 2.29 it suffices to treat $Y = \mathbb{1}_A$, $A \in \mathcal{C}$, i.e. to show

$$\mathbb{E}_x[\mathbb{1}_A \circ \theta_s \, \mathbb{1}_C] = \mathbb{E}_x[\mathbb{E}_{B_s}[\mathbb{1}_A] \, \mathbb{1}_C] \qquad \forall C \in \mathcal{F}_s^{0,+}. \tag{4.16}$$

Step 2 (special C). Fix x, s and $0 < h < t_1$, and take $C = \{(B_{s_1}, \dots, B_{s_m}) \in C_1 \times \cdots \times C_m\}$ with $0 \leq s_1 < \cdots < s_m = s + h$. With $s_0 := 0$, $y_0 := x$, (4.9) gives

$$\mathbb{E}_x[\mathbb{1}_A \circ \theta_s \, \mathbb{1}_C] = \int_{C_1} \cdots \int_{C_m} \prod_{j=1}^{m} p_{s_j - s_{j-1}}(y_{j-1}, y_j) \, F(y_m, h) \, dy_m \cdots dy_1 = \mathbb{E}_x[F(B_{s+h}, h) \mathbb{1}_C], \tag{4.17}$$

where, with $x_0 := y$ and $t_0 := h$,

$$F(y, h) := \int_{A_1} \cdots \int_{A_n} p_{t_1 - h}(y, x_1) \prod_{i=2}^{n} p_{t_i - t_{i-1}}(x_{i-1}, x_i) \, dx_n \cdots dx_1.$$

Step 3 (all $C \in \mathcal{F}_{s+h}^0$). The sets C of Step 2 form a $\cap$-stable generator of $\mathcal{F}_{s+h}^0$, and the class of $\tilde{C} \in \mathcal{F}_{s+h}^0$ satisfying (4.17) is a Dynkin system. By Theorem 1.8,

$$\mathbb{E}_x[\mathbb{1}_A \circ \theta_s \, \mathbb{1}_C] = \mathbb{E}_x[F(B_{s+h}, h) \mathbb{1}_C] \qquad \forall C \in \mathcal{F}_{s+h}^0 \supset \mathcal{F}_s^{0,+}. \tag{4.18}$$

Step 4 ($h \to 0$). By the DCT, F is jointly continuous in $(y, h) \in \mathbb{R} \times (0, t_1)$ and bounded by 1. As $h \to 0$, path continuity gives $F(B_{s+h}(\omega), h) \to \int_{A_1} \cdots \int_{A_n} p_{t_1}(B_s(\omega), x_1) \prod_{i \geq 2} p_{t_i - t_{i-1}}(x_{i-1}, x_i) \, dx = \mathbb{E}_{B_s(\omega)}[\mathbb{1}_A]$. The DCT in (4.18) then yields (4.16) for all $C \in \mathcal{F}_s^{0,+}$. ∎

Remark 4.13

(4.15) implies (4.14): $\mathbb{E}_x[Y\circ\theta_s \mid \mathcal{F}_s^0] = \mathbb{E}_x[\mathbb{E}_x[Y\circ\theta_s \mid \mathcal{F}_s^{0,+}] \mid \mathcal{F}_s^0] = \mathbb{E}_x[\mathbb{E}_{B_s}[Y] \mid \mathcal{F}_s^0] = \mathbb{E}_{B_s}[Y]$ by Theorem 2.35(iv), (4.15) and Example 2.34(i).

Theorem 4.14

For $Z \in \mathcal{F}_b$: $\mathbb{E}_x[Z \mid \mathcal{F}_s^{0,+}] = \mathbb{E}_x[Z \mid \mathcal{F}_s^0]$ P_x-a.s., for all $s \geq 0$, $x \in \mathbb{R}$.

Proof. By the monotone class theorem it suffices to take $Z = \mathbb{1}_A$ with $A = \{(B_{t_1}, \dots, B_{t_n}) \in A_1 \times \cdots \times A_n\}$, $0 \leq t_1 < \cdots < t_n$. Then, P_x-a.s., by Theorem 2.35(iii) and (4.14), (4.15) (both equal $\mathbb{E}_{B_s}[\cdot]$),

$$\begin{aligned}\mathbb{E}_x[\mathbb{1}_A \mid \mathcal{F}_s^{0,+}] &= \prod_{t_i\leq s} \mathbb{1}_{A_i}(B_{t_i})\, \mathbb{E}_x\Big[\prod_{t_i>s} \mathbb{1}_{A_i}(B_{t_i-s})\circ\theta_s \,\Big|\, \mathcal{F}_s^{0,+}\Big] \\ &= \prod_{t_i\leq s} \mathbb{1}_{A_i}(B_{t_i})\, \mathbb{E}_x\Big[\prod_{t_i>s} \mathbb{1}_{A_i}(B_{t_i-s})\circ\theta_s \,\Big|\, \mathcal{F}_s^{0}\Big] = \mathbb{E}_x[\mathbb{1}_A \mid \mathcal{F}_s^0].\end{aligned}$$

■

Remark 4.15

$\mathcal{F}_s^{0,+}$ and $\mathcal{F}_s^0$ differ only by P_x-null sets (for every x). For $Z \in \mathcal{F}_s^{0,+}$ bounded, $Z = \mathbb{E}_x[Z \mid \mathcal{F}_s^{0,+}] = \mathbb{E}_x[Z \mid \mathcal{F}_s^0]$ P_x-a.s., an $\mathcal{F}_s^0$-measurable r.v. With $Z = \mathbb{1}_{B^c}$, $B \in \mathcal{F}_s^{0,+}$, the set $B = \{\mathbb{1}_{B^c} < 1\}$ agrees up to a P_x-null set with $B_0 := \{\mathbb{E}_x[\mathbb{1}_{B^c} \mid \mathcal{F}_s^0] < 1\} \in \mathcal{F}_s^0$.

Theorem 4.16 (Blumenthal's 0–1 law)

If $A \in \mathcal{F}_0^{0,+}$, then $P_x(A) \in \{0,1\}$ for every $x \in \mathbb{R}$.

Proof. $\mathcal{F}_0^0 = \sigma(B_0)$ and $P_x(B_0 = x) = 1$, so every $\mathcal{F}_0^0$-measurable Y is P_x-a.s. constant, $Y = \mathbb{E}_x[Y]$. For $A \in \mathcal{F}_0^{0,+}$, by Example 2.34(i) and Theorem 4.14, P_x-a.s.

$$\mathbb{1}_A = \mathbb{E}_x[\mathbb{1}_A \mid \mathcal{F}_0^{0,+}] = \mathbb{E}_x[\mathbb{1}_A \mid \mathcal{F}_0^0] = \mathbb{E}_x[\mathbb{E}_x[\mathbb{1}_A \mid \mathcal{F}_0^0]] = P_x(A),$$

so $P_x(A) \in \{0,1\}$. ■

Remark 4.17

A general BM as in Definition 4.1 (started at x) also has the Markov property w.r.t. $(\mathcal{F}_t^{0,+})$. Since a shift need not exist, it is formulated as

$$\mathbb{E}_x\Big[\prod_{i=1}^n \mathbb{1}_{A_i}(B_{t_i+s}) \,\Big|\, \mathcal{F}_s^{0,+}\Big] = \mathbb{E}_{B_s}\Big[\prod_{i=1}^n \mathbb{1}_{A_i}(B_{t_i})\Big] \qquad P_x\text{-a.s.}, \tag{4.19}$$

for cylinder sets as above. The proof of (4.16) used only the finite-dimensional distributions (4.9) and a.s. path continuity. A general BM satisfies (4.9) (equivalent to Definition 4.1(i)–(iii), also for start in x), so it has the same finite-dimensional distributions as the canonical one. In particular $\mathbb{E}_x[f(B_{t+s}) \mid \mathcal{F}_s^{0,+}] = \mathbb{E}_{B_s}[f(B_t)]$ for $f \in \mathcal{B}(\mathbb{R})_b$.

4.4 The d-dimensional Brownian motion

Motivation. In $\mathbb{R}^d$, Brownian motion is simply d independent one-dimensional Brownian motions as coordinates. The Gaussian kernel becomes d-dimensional, and the constructions and the Markov property carry over verbatim.

Let $\|\cdot\|$ be the Euclidean norm and

$$p_t(x,y) := \frac{1}{(\sqrt{2\pi t})^d} e^{-\frac{\|y-x\|^2}{2t}} \quad (x,y\in\mathbb{R}^d,\ t>0), \qquad p_0(x,y)\,dy := \delta_x(dy). \tag{4.20}$$

The construction of Section 4.2 goes through with this kernel; Kolmogorov's extension theorem holds for $\mathbb{R}^d$-valued (indeed Polish-space-valued) processes, with the same proof (see also Le Gall, *Brownian Motion, Martingales, and Stochastic Calculus*, Thm. 6.3). Equivalently: if P_0 is the one-dimensional Wiener measure on $(C(\mathbb{R}_+),\mathcal{B}(C(\mathbb{R}_+)))$ and $P^{(i)} = P_0$, $i=1,\dots,d$, then $\mathbf{P}_0 := \bigotimes_{i=1}^d P^{(i)}$ is the **d-dimensional Wiener measure** on $\Omega := C(\mathbb{R}_+)^d = C(\mathbb{R}_+,\mathbb{R}^d)$, $\mathcal{F} := \mathcal{B}(C(\mathbb{R}_+)^d)$. Under $\mathbf{P}_0$, $W_t(\omega) := \omega(t) = (\omega_1(t),\dots,\omega_d(t))$ is the canonical d-dimensional BM from 0, and under $P_x := \mathbf{P}_0\circ\Pi_x^{-1}$, $\Pi_x(\omega)=\omega+x$, it starts at x. With $\mathcal{F}_t^0 := \sigma(W_s \mid s\le t)$ and $\mathcal{F}_t^{0,+} := \bigcap_{\varepsilon>0}\mathcal{F}_{t+\varepsilon}^0$ (again filtrations to which W is adapted),

$$P_x((W_{t_1},\dots,W_{t_n})\in A_1\times\cdots\times A_n) = \int_{A_1}\cdots\int_{A_n}\prod_{i=1}^n p_{t_i-t_{i-1}}(x_{i-1},x_i)\,dx_1\cdots dx_n \tag{4.21}$$

for $0\le t_1<\cdots<t_n$, $A_i\in\mathcal{B}(\mathbb{R}^d)$, $x_0=x$, $t_0=0$, with d-dimensional Lebesgue measure dx_i. The analogues of (4.14), (4.15), Theorem 4.14, Remark 4.15 and Theorem 4.16 hold with essentially the same proofs. The components W^i are independent BMs: by (4.21), which factorises over coordinates, $P_x(\bigcap_{i=1}^d \tilde A_i) = \prod_i P_x(\tilde A_i)$ for cylinder sets $\tilde A_i = \{(W^i_{t_1},\dots,W^i_{t_n})\in B^i_1\times\cdots\times B^i_n\}$, and Theorem 3.10 extends this to the generated σ-algebras.

Definition 4.18 (d-dimensional Brownian motion)

An $\mathbb{R}^d$-valued process $(W_t)_{t\ge0}$ on some $(\Omega,\mathcal{F},P_x)$ is a **d-dimensional BM started at** x if (i) $P_x(W_0=x)=1$; (ii) for $0\le s<t$, $W_t-W_s\sim N(0,(t-s)I_d)$, i.e. $P_x(W_t-W_s\in A)=\int_A p_{t-s}(0,y)\,dy$; (iii) the increments $W_{t_1}-W_0,\dots,W_{t_n}-W_{t_{n-1}}$ are independent for $0\le t_1<\cdots<t_n$; (iv) almost all paths are continuous.

Remark 4.19

A d-dimensional BM satisfies (4.21), which is equivalent to Definition 4.18(i)–(iii); so it has the finite-dimensional distributions of the canonical one. For it only P_x-a.e. path is continuous, whereas for the canonical model every path is. As in Remark 4.17 it has the Markov property

$$\mathbb{E}_x\Big[\prod_{i=1}^n \mathbb{1}_{A_i}(W_{t_i+s})\,\Big|\,\mathcal{F}_s^{0,+}\Big] = \mathbb{E}_{W_s}\Big[\prod_{i=1}^n \mathbb{1}_{A_i}(W_{t_i})\Big] \quad P_x\text{-a.s.}, \tag{4.22}$$

and in particular $\mathbb{E}_x[f(W_{t+s})\mid\mathcal{F}_s^{0,+}] = \mathbb{E}_{W_s}[f(W_t)]$ for $f\in\mathcal{B}(\mathbb{R}^d)_b$.

Chapter 5. The Strong Markov Property

5.1 Stopping times and filtrations

Motivation. The Markov property restarts the process at a fixed time. Many questions (when does B first hit a?) need a restart at a *random* time that is decided by the path itself without looking ahead: a stopping time. This section develops stopping times, the σ-algebra $\mathcal{F}_T$ of information up to T, and the measurability of X_T.

Let $(\Omega, \mathcal{F}, P)$ be a probability space with a **filtration** $(\mathcal{F}_t)_{t\geq 0}$ ($\mathcal{F}_s \subset \mathcal{F}_t \subset \mathcal{F}$ for $s \leq t$); $(\Omega, \mathcal{F}, (\mathcal{F}_t), P)$ is a **filtered probability space**. $(\mathcal{F}_t)$ is **right continuous** if $\mathcal{F}_t = \bigcap_{\varepsilon>0} \mathcal{F}_{t+\varepsilon}$ for all t. It satisfies the **usual conditions** if it is right continuous, $(\Omega, \mathcal{F}, P)$ is complete, and every P-null set of $\mathcal{F}$ lies in $\mathcal{F}_0$; then all $(\Omega, \mathcal{F}_t, P)$ are complete.

Remark 5.1

Every probability space can be completed. With $\mathcal{N}_P := \{A \subset \Omega \mid A \subset B \in \mathcal{F},\ P(B) = 0\}$, the **completion** $\mathcal{F}^P := \{C \cup A \mid C \in \mathcal{F},\ A \in \mathcal{N}_P\} = \sigma(\mathcal{F}, \mathcal{N}_P)$ is a σ-algebra, and $P(C \cup A) := P(C)$ extends P to it. [Supplement] *Reason:* $\mathcal{N}_P$ is closed under countable unions. For complements, $(C \cup A)^c = (C^c \setminus B) \cup (C^c \cap B \setminus A)$ with $A \subset B$ null, which is again of the form $C' \cup A'$. The extension is well defined because two representations differ by a subset of a null set, and it is σ-additive because P is.

A process $X = (X_t)$ with values in a metric space S is **adapted** to $(\mathcal{F}_t)$ if each X_t is $\mathcal{F}_t$-measurable. Its **natural filtration** $\mathcal{F}_t^0 = \mathcal{F}_t^0(X) := \sigma(X_s \mid s \leq t)$ (also $\mathcal{F}_t^X$) is the smallest filtration making X adapted. X is **continuous** if $P(t \mapsto X_t \text{ continuous}) = 1$, and it has **continuous sample paths** if every path is continuous.

Example 5.2

In the canonical d-dimensional BM $\mathbb{M}_W = (C(\mathbb{R}_+, \mathbb{R}^d), \mathcal{B}(C(\mathbb{R}_+, \mathbb{R}^d)), (W_t), P_x)$, the filtration $(\mathcal{F}_t^{0,+})$ is right continuous and W has continuous sample paths. For a general d-dimensional BM $(\Omega, \mathcal{F}, (W_t), P_x)$ as in Definition 4.18, let $\mathcal{F}_t^0 := \sigma(W_s \mid s \leq t)$, $\mathcal{F}_t^{0,+} := \bigcap_{\varepsilon>0} \mathcal{F}_{t+\varepsilon}^0$, $\mathcal{N}_x := \mathcal{N}_{P_x}$, and

$$\mathcal{F}_t^{0,+,x} := \sigma(\mathcal{F}_t^{0,+}, \mathcal{N}_x), \qquad \mathcal{F}_t^{0,x} := \sigma(\mathcal{F}_t^0, \mathcal{N}_x),$$
$$\mathcal{F}_t^{0,x,+} := \bigcap_{\varepsilon>0} \mathcal{F}_{t+\varepsilon}^{0,x}, \qquad \mathcal{F}_\infty^{0,+,x} := \sigma(\mathcal{F}_t^{0,+,x} \mid t \geq 0).$$

Then $\mathcal{F}_t^{0,+,x} = \mathcal{F}_t^{0,x,+}$: completing and making right continuous commute (Rogers–Williams, Vol. 1, II.(67.4)). So $(\Omega, \mathcal{F}_\infty^{0,+,x}, (\mathcal{F}_t^{0,+,x}), P_x)$ satisfies the usual conditions and W is continuous. For the canonical model, as Remark 4.15 suggests (Blumenthal's law holds in d dimensions), even $\mathcal{F}_t^{0,x}$ is right continuous (Karatzas–Shreve, Prop. 2.7.7), so $\mathcal{F}_t^{0,x} = \mathcal{F}_t^{0,x,+} = \mathcal{F}_t^{0,+,x}$. One also uses $\mathcal{F}_t := \bigcap_{x\in\mathbb{R}^d} \mathcal{F}_t^{0,+,x}$ and $\mathcal{F}_\infty := \sigma(\mathcal{F}_t \mid t \geq 0)$. This filtration is right continuous, contains $(\mathcal{F}_t^{0,+})$, and differs from it only by sets that are null for every P_x, so $\mathbb{M}_W$ has the Markov property w.r.t. it as well. Continuity of paths plus right continuity of the filtration (or the usual conditions for merely continuous processes) is what resolves the measurability questions below.

Definition 5.3 (Stopping time)

$T : \Omega \to [0, \infty]$ is an $(\mathcal{F}_t)$**-stopping time** if

$$\{T \le t\} \in \mathcal{F}_t \qquad \forall t \ge 0. \tag{5.1}$$

In particular T is $\mathcal{F}$-measurable.

Lemma 5.4

If $(\mathcal{F}_t)$ is right continuous, (5.1) is equivalent to

$$\{T < t\} \in \mathcal{F}_t \qquad \forall t > 0. \tag{5.2}$$

(A T with (5.2) is an **optional** or **weak stopping time**.)

Proof. For a stopping time, $\{T < t\} = \bigcup_n \{T \le t - \frac{1}{n}\} \in \mathcal{F}_t$. Conversely, under (5.2), $\{T \le t\} = \bigcap_{k \ge n} \{T < t + \frac{1}{k}\} \in \mathcal{F}_{t+\frac{1}{n}}$ for every n, hence $\{T \le t\} \in \bigcap_n \mathcal{F}_{t+\frac{1}{n}} = \mathcal{F}_t$ by right continuity. ∎

Example 5.5

(a) If S, T are stopping times, so are $S \vee T$, $S \wedge T$, $S + T$: $\{S \vee T \le t\} = \{S \le t\} \cap \{T \le t\}$ and $\{S \wedge T \le t\} = \{S \le t\} \cup \{T \le t\}$. For the sum, $\tilde{T} \wedge t$ is $\mathcal{F}_t$-measurable for every stopping time $\tilde{T}$ (Lemma 5.6(ii)), and

$$\{S + T \le t\} = \{S \wedge t + \mathbb{1}_{\{S > t\}} + T \wedge t + \mathbb{1}_{\{T > t\}} \le t\} \in \mathcal{F}_t.$$

(b) For stopping times T_n, $\sup_n T_n$ is a stopping time ($\{\sup_n T_n \le t\} = \bigcap_n \{T_n \le t\}$). $\inf_n T_n$, $\liminf_n T_n$, $\limsup_n T_n$ are in general only weak stopping times ($\{\inf_n T_n < t\} = \bigcup_n \{T_n < t\}$, $\limsup_n T_n = \inf_k \sup_{n \ge k} T_n$, $\liminf_n T_n = \sup_k \inf_{n \ge k} T_n$); they are stopping times if $(\mathcal{F}_t)$ is right continuous.

(c) (*Hitting times*) Let X be adapted with values in a metric space (S, d), and assume either (i) X has continuous sample paths and $(\mathcal{F}_t)$ is right continuous, or (ii) X is continuous and $(\mathcal{F}_t)$ satisfies the usual conditions. Under (ii) we may assume (i), since the null set of discontinuous paths and all its subsets lie in $\mathcal{F}_0$. For $A \subset S$ let

$$\sigma_A := \inf\{t > 0 \mid X_t \in A\} \text{ (first hitting time)},$$
$$D_A := \inf\{t \ge 0 \mid X_t \in A\} \text{ (first entry time)},$$

with $\inf \emptyset := \infty$, so $\{\sigma_A \ge t\} = \{X_s \notin A \ \forall s \in (0, t)\}$ and $\{D_A \ge t\} = \{X_s \notin A \ \forall s \in [0, t)\}$.

A open: by continuity $\sigma_A = D_A$, and since A^c is closed, $\{\sigma_A \ge t\} = \bigcap_{s \in [0,t) \cap \mathbb{Q}} \{X_s \notin A\} \in \mathcal{F}_t$.

A closed: $s \mapsto d(X_s, A)$ is continuous, so it is bounded below by a positive constant on $[0, t]$ whenever it is positive there, and

$$\{D_A > t\} = \{d(X_s, A) > 0 \ \forall s \in [0, t]\} = \bigcup_n \bigcap_{s \in [0,t] \cap \mathbb{Q}} \{d(X_s, A) \ge \tfrac{1}{n}\} \in \mathcal{F}_t.$$

[Supplement] Similarly $\{\sigma_A \ge t\} = \bigcap_m \bigcup_n \bigcap_{s \in \mathbb{Q} \cap [\frac{1}{m}, t - \frac{1}{m}]} \{d(X_s, A) \ge \frac{1}{n}\} \in \mathcal{F}_t$.

In all cases σ_A, D_A are weak stopping times, hence stopping times by Lemma 5.4. (For general Borel A and right continuous paths this remains true under extra conditions on S; the proof is hard, see Blumenthal–Getoor, Ch. I, (10.7).)

(d) (*Dyadic approximation*) For a stopping time T let

$$T_n := \sum_{k\geq 0}(k+1)2^{-n}\mathbb{1}_{[k2^{-n},(k+1)2^{-n})}(T) \text{ on } \{T<\infty\}, \qquad T_n := \infty \text{ on } \{T=\infty\}.$$

Then $T_n \searrow T$ and the T_n are stopping times: for $d \in D_n := \{k2^{-n} \mid k \geq 1\}$, $\{T_n = d\} = \{T < d\} \cap \{T \geq d - 2^{-n}\} \in \mathcal{F}_d$, and $\{T_n \leq t\} = \bigcup_{d\in D_n,\, d\leq t}\{T_n = d\} \in \mathcal{F}_t$.

For a stopping time T, the **σ-algebra of the T-past** is

$$\mathcal{F}_T := \{A \in \mathcal{F} \mid A \cap \{T \leq t\} \in \mathcal{F}_t \ \forall t \geq 0\}.$$

[Supplement] It is a σ-algebra: $\Omega \cap \{T \leq t\} \in \mathcal{F}_t$; $A^c \cap \{T \leq t\} = \{T \leq t\} \setminus (A \cap \{T \leq t\})$; unions are clear. For $T \equiv t$, $\mathcal{F}_T = \mathcal{F}_t$. T is $\mathcal{F}_T$-measurable, since $\{T \leq s\} \cap \{T \leq t\} = \{T \leq s \wedge t\} \in \mathcal{F}_t$. If $(\mathcal{F}_t)$ is right continuous, then $\mathcal{F}_T = \{A \in \mathcal{F} \mid A \cap \{T < t\} \in \mathcal{F}_t \ \forall t\}$, by the argument of Lemma 5.4 applied to $A \cap \{T \leq t\}$.

Lemma 5.6

(i) For stopping times $S \leq T$: $\mathcal{F}_S \subset \mathcal{F}_T$.

(ii) For a stopping time T and $t \geq 0$, $T \wedge t$ is $\mathcal{F}_t$-measurable.

(iii) If T_n are decreasing stopping times w.r.t. a right continuous filtration and $T = \inf_n T_n$, then $\mathcal{F}_T = \bigcap_n \mathcal{F}_{T_n}$.

Proof. (i) For $A \in \mathcal{F}_S$: $\{T \leq t\} \subset \{S \leq t\}$, so $A \cap \{T \leq t\} = (A \cap \{S \leq t\}) \cap \{T \leq t\} \in \mathcal{F}_t$.

(ii) $\{T \wedge t \leq t'\}$ is Ω if $t \leq t'$, and $\{T \leq t'\} \in \mathcal{F}_{t'} \subset \mathcal{F}_t$ if $t' < t$.

(iii) $\mathcal{F}_T \subset \bigcap_n \mathcal{F}_{T_n}$ by (i). Conversely, if $A \in \bigcap_n \mathcal{F}_{T_n}$, then $A \cap \{T_n < t\} \in \mathcal{F}_t$ for all n (right continuity), so $A \cap \{T < t\} = A \cap \bigcup_n \{T_n < t\} \in \mathcal{F}_t$ for all t, i.e. $A \in \mathcal{F}_T$. ■

For $T(\omega) < \infty$ put $X_T(\omega) := X_{T(\omega)}(\omega)$; for a finite stopping time this defines X_T on Ω. For instance, $T \wedge t$ is a finite stopping time.

Lemma 5.7

Let X be adapted with values in a metric space, under the conditions of Example 5.5(c)(i) or (ii), and T a finite stopping time. Then X_T is $\mathcal{F}_T$-measurable. In particular $X_{t\wedge T}$ is $\mathcal{F}_{t\wedge T} \subset \mathcal{F}_t$-measurable, so $(X_{t\wedge T})_{t\geq 0}$ is adapted.

Proof.

Step 1 (dyadic times). Take $T_n \searrow T$ as in Example 5.5(d) and $B \in \mathcal{B}(S)$. $\{X_{T_n} \in B\} = \bigcup_k \{T_n = (k+1)2^{-n}\} \cap \{X_{(k+1)2^{-n}} \in B\} \in \mathcal{F}$, and

$$\{X_{T_n} \in B\} \cap \{T_n \leq t\} = \bigcup_{d\in D_n,\, d\leq t} \{X_d \in B\} \cap \{T_n = d\} \in \mathcal{F}_t,$$

so $X_{T_n} \in \mathcal{F}_{T_n}$.

Step 2 (limit). With continuous paths, $X_{T_n} \to X_T$ pointwise. For each n, $X_T = \lim_{k\geq n} X_{T_k}$ is $\mathcal{F}_{T_n}$-measurable (as $\mathcal{F}_{T_k} \subset \mathcal{F}_{T_n}$ for $k \geq n$), so $X_T \in \bigcap_n \mathcal{F}_{T_n} = \mathcal{F}_T$ by Lemma 5.6(iii). ■

Now let X take values in a Borel set $S \subset \mathbb{R}^d$ with $0 \in S$. For a (not necessarily finite) stopping time T define $\mathbb{1}_{\{T<\infty\}}X_T := X_T$ on $\{T < \infty\}$ and $:= 0$ on $\{T = \infty\}$.

Lemma 5.8

Under the assumptions of Lemma 5.7 (with $S \subset \mathbb{R}^d$ Borel, $0 \in S$), $\mathbb{1}_{\{T<\infty\}} X_T$ is $\mathcal{F}_T$-measurable.

Proof. By Lemmas 5.6(i) and 5.7, $\mathbb{1}_{\{T<\infty\}} X_{T\wedge n} \in \mathcal{F}_T$ for each n, and $\mathbb{1}_{\{T<\infty\}} X_{T\wedge n} \to \mathbb{1}_{\{T<\infty\}} X_T$ pointwise. ∎

Remark 5.9

Example 5.5(c) remains true with "continuous" replaced by "right continuous". Since $T_n \searrow T$ in Example 5.5(d), Lemmas 5.7 and 5.8 hold for right continuous paths as well.

5.2 Markov processes and the strong Markov property

Motivation. A Markov process is a family of processes, one for each starting point, linked by a semigroup of transition kernels. If the paths are right continuous and the semigroup maps bounded continuous functions to bounded continuous functions (the Feller property), the Markov property extends from fixed times to stopping times: the strong Markov property. Brownian motion is the main example.

Throughout, $S \subset \mathbb{R}^d$ is Borel with $0 \in S$, equipped with $\mathcal{B}(S)$.

Definition 5.10 (Markovian semigroup)

Maps $P_t : S \times \mathcal{B}(S) \to [0,1]$, $t \geq 0$, form a **markovian semigroup of transition kernels** if (i) $P_t(x,\cdot)$ is a probability measure for all t, x; (ii) $x \mapsto P_t(x,A)$ is $\mathcal{B}(S)$-measurable; (iii) (Chapman–Kolmogorov) $P_t P_s = P_{t+s}$, i.e. $\int_S P_s(y,A)\, P_t(x,dy) = P_{t+s}(x,A)$; (iv) $P_0(x,\cdot) = \delta_x$. For bounded (or nonnegative) measurable f put

$$P_t f(x) := \int_S f(y)\, P_t(x,dy), \tag{5.3}$$

so that $P_t \mathbb{1}_A(x) = P_t(x,A)$.

Definition 5.11 (Normal Markov process)

$\mathbb{M} = (\Omega, \mathcal{F}, (X_t)_{t\geq 0}, (P_x)_{x\in S})$ with state space $(S, \mathcal{B}(S))$ is a **normal Markov process** w.r.t. a filtration $(\mathcal{F}_t)$ if

(M1) $x \mapsto P_x(F)$ is $\mathcal{B}(S)$-measurable for every $F \in \mathcal{F}$;

(M2) (X_t) is $(\mathcal{F}_t)$-adapted and $P_x(X_{s+t} \in A \mid \mathcal{F}_s) = P_{X_s}(X_t \in A)$ P_x-a.s. for all $s,t \geq 0$, $A \in \mathcal{B}(S)$, $x \in S$ (Markov property), where $P_x(\cdot \mid \mathcal{F}_s) = \mathbb{E}_x[\mathbb{1}_A(X_{s+t}) \mid \mathcal{F}_s]$;

(M3) $P_x(X_0 = x) = 1$ for all x (normality).

Theorem 5.12

(i) Let $\mathbb{M} = (\Omega, \mathcal{F}, (X_t), (P_x)_{x\in S})$ with $\mathcal{F} = \sigma(X_t \mid t \geq 0)$.

(a) If there is a markovian semigroup (P_t) with, for all $0 \leq t_1 < \cdots < t_n$, $A_i \in \mathcal{B}(S)$,

$t_0 := 0$, $x_0 := x$,

$$P_x((X_{t_1}, \dots, X_{t_n}) \in A_1 \times \cdots \times A_n) = \int_{A_1} \cdots \int_{A_n} \prod_{i=1}^{n} P_{t_i - t_{i-1}}(x_{i-1}, dx_i), \tag{5.4}$$

then $\mathbb{M}$ is a normal Markov process w.r.t. $(\mathcal{F}_t^0) = (\sigma(X_s \mid s \le t))$.

(b) If $\mathbb{M}$ is a normal Markov process, then

$$P_t(x, A) := \mathbb{E}_x[\mathbb{1}_A(X_t)] = P_x(X_t \in A) \tag{5.5}$$

is a markovian semigroup and (5.4) holds.

(ii) For every markovian semigroup there is a normal Markov process satisfying (5.5), with $\Omega = S^{[0,\infty)}$.

Proof. (i)(a) (M3) is (5.4) with $n = 1$, $t_1 = 0$. (M1) holds for cylinder sets by (5.4) and Definition 5.10(ii) (iterated kernel integrals of measurable functions are measurable), and extends to $\mathcal{F}$ by a Dynkin argument. (M2) is equivalent to

$$\mathbb{E}_x[\mathbb{1}_A(X_{t+s})\mathbb{1}_C] = \mathbb{E}_x[\mathbb{E}_{X_s}[\mathbb{1}_A(X_t)]\mathbb{1}_C] \qquad \forall C \in \mathcal{F}_s^0.$$

For $s = 0$ this is clear ($\mathcal{F}_0^0 = \sigma(X_0)$ is P_x-trivial). For $s > 0$, by the monotone class theorem it suffices to take $C = \{(X_{s_1}, \dots, X_{s_m}) \in C_1 \times \cdots \times C_m\}$, $0 \le s_1 < \cdots < s_m = s$. With $s_{m+1} := t + s$, $y_0 := x$, (5.4) gives

$$\begin{aligned}\mathbb{E}_x[\mathbb{1}_A(X_{t+s})\mathbb{1}_C] &= \int_{C_1} \cdots \int_{C_m} \int_A \prod_{j=1}^{m+1} P_{s_j - s_{j-1}}(y_{j-1}, dy_j) \\ &= \int_{C_1} \cdots \int_{C_m} \mathbb{E}_{y_m}[\mathbb{1}_A(X_t)] \prod_{j=1}^{m} P_{s_j - s_{j-1}}(y_{j-1}, dy_j),\end{aligned}$$

which by (5.4) again equals $\mathbb{E}_x[\mathbb{E}_{X_s}[\mathbb{1}_A(X_t)]\mathbb{1}_C]$.

(i)(b) Definition 5.10(i), (ii),

(iv) are clear from (M1), (M3). By (5.5) and (5.3), $P_t\mathbb{1}_A(x) = \mathbb{E}_x[\mathbb{1}_A(X_t)]$, and by linearity and monotone convergence

$$P_t f(x) = \mathbb{E}_x[f(X_t)] \qquad \text{for bounded or nonnegative measurable } f. \tag{5.6}$$

Chapman–Kolmogorov: with (M2) and (5.6),

$$P_{t+s}(x, A) = \mathbb{E}_x[\mathbb{E}_x[\mathbb{1}_A(X_{t+s}) \mid \mathcal{F}_s]] = \mathbb{E}_x[P_t(X_s, A)] = P_s(P_t\mathbb{1}_A)(x) = \int P_t(y, A)\, P_s(x, dy).$$

For (5.4), it suffices to prove

$$\mathbb{E}_x[f_1(X_{t_1}) \cdots f_n(X_{t_n})] = \int \cdots \int f_1(x_1) \cdots f_n(x_n) \prod_{i=1}^{n} P_{t_i - t_{i-1}}(x_{i-1}, dx_i) \tag{5.7}$$

for bounded measurable f_i, by induction on n; $n = 1$ is (5.6). Conditioning on $\mathcal{F}_{t_{n-1}}$, Theorem 2.35(iii) and (M2) give

$$\mathbb{E}_x\Big[\prod_{i \le n} f_i(X_{t_i})\Big] = \mathbb{E}_x\Big[\prod_{i \le n-1} f_i(X_{t_i})\, P_{t_n - t_{n-1}} f_n(X_{t_{n-1}})\Big],$$

and the induction hypothesis applied to $f_1, \dots, f_{n-2}$, $f_{n-1} \cdot P_{t_n - t_{n-1}} f_n$ yields (5.7).

(ii) The family (5.4) is consistent by Chapman–Kolmogorov. Kolmogorov's extension theorem, whose proof (Theorem 4.7) works verbatim for $\mathbb{R}^d$-valued coordinates (see also Le Gall, Thm. 6.3; Revuz–Yor, Ch. I, (3.2)), gives P_x on the path space. (M1) follows as in (i)(a), and (i)(a) does the rest. ∎

The canonical model. Let $\Omega \subset S^{[0,\infty)}$ (e.g. $S^{[0,\infty)}$ or $C([0,\infty),S)$), $X_t(\omega) := \omega(t)$, $\mathcal{F} := \sigma(X_t \mid t \geq 0)$, $\mathcal{F}_t^0 := \sigma(X_s \mid s \leq t)$, and let P_x be probability measures with $P_x(X_0 = x) = 1$. Then $\mathbb{M} = (\Omega, \mathcal{F}, (X_t), (P_x))$ is the **canonical model** of (X_t). The shifts $\theta_s(\omega)(t) := \omega(s+t)$ are $\hat{\mathcal{F}}_s^0/\mathcal{F}$-measurable with $\hat{\mathcal{F}}_t^0 := \sigma(X_s \mid s \geq t)$. If $\mathbb{M}$ is a normal Markov process w.r.t. $(\mathcal{F}_t)$ (hence also w.r.t. $(\mathcal{F}_t^0)$), we call $\mathbb{M} = (\Omega, \mathcal{F}, (\mathcal{F}_t), (X_t), (P_x), (\theta_t))$ a **canonical Markov process**.

Lemma 5.13

Let the canonical model satisfy (M1). Then (M2) holds w.r.t. $\mathcal{F}_t := \mathcal{F}_t^{0,+} = \bigcap_{\varepsilon>0} \mathcal{F}_{t+\varepsilon}^0$ if and only if

$$\mathbb{E}_x[Y \circ \theta_s \mid \mathcal{F}_s] = \mathbb{E}_{X_s}[Y] \qquad P_x\text{-a.s., } \forall s \geq 0,\ x \in S,\ Y \in \mathcal{F}_b. \tag{5.8}$$

As in Remark 4.12, (5.8) implies $\mathbb{E}_x[Y \circ \theta_s \mid \mathcal{F}_s] = \mathbb{E}_x[Y \circ \theta_s \mid \sigma(X_s)]$.

Proof. (M3) holds by definition of the canonical model. "⇐": take $Y = \mathbb{1}_A(X_t)$. "⇒" [Supplement]: by Theorem 5.12(i)(b), (5.4) holds with the semigroup (5.5). For $Y = \prod_{i\leq n} f_i(X_{t_i})$, $0 \leq t_1 < \cdots < t_n$, condition successively on $\mathcal{F}_{s+t_{n-1}}, \mathcal{F}_{s+t_{n-2}}, \ldots$ and use (M2) and Theorem 2.35(iii),(iv), exactly as in the induction for (5.7). The result is

$$\mathbb{E}_x[Y \circ \theta_s \mid \mathcal{F}_s] = \Big[P_{t_1}\big(f_1 P_{t_2-t_1}(f_2 \cdots P_{t_n-t_{n-1}} f_n)\big)\Big](X_s),$$

and by (5.7) the bracket evaluated at y is $\mathbb{E}_y[Y]$. The class of $Y \in \mathcal{F}_b$ satisfying (5.8) is an MVS containing these products, so Theorem 2.29 gives (5.8). ∎

For a stopping time T put $(\theta_T\omega)(t) := \omega(T(\omega)+t)$ on $\{T < \infty\}$. (In Markov process theory one sets $\theta_T\omega \equiv \Delta$ on $\{T = \infty\}$, with a cemetery point Δ and $f(\Delta) := 0$.)

Definition 5.14 (Strong Markov property)

A canonical Markov process $\mathbb{M}$ with right continuous paths and right continuous filtration $(\mathcal{F}_t)$ has the **strong Markov property** (SMP) w.r.t. $(\mathcal{F}_t)$ if for every $(\mathcal{F}_t)$-stopping time T,

$$\mathbb{E}_x[\mathbb{1}_{\{T<\infty\}} Y \circ \theta_T \mid \mathcal{F}_T] = \mathbb{1}_{\{T<\infty\}} \mathbb{E}_{X_T}[Y] \qquad P_x\text{-a.s., } \forall Y \in \mathcal{F}_b,\ x \in S.$$

The right side, $\mathbb{1}_{\{T<\infty\}}\mathbb{E}_{\mathbb{1}_{\{T<\infty\}}X_T}[Y]$, is $\mathcal{F}_T$-measurable by Lemma 5.8, (M1) and Remark 5.9. For finite T the SMP reads $\mathbb{E}_x[Y \circ \theta_T \mid \mathcal{F}_T] = \mathbb{E}_{X_T}[Y]$.

Theorem 5.15

Let $\mathbb{M} = (\Omega, \mathcal{F} = \sigma(X_t \mid t \geq 0), (\mathcal{F}_t), (X_t), (P_x), (\theta_t))$ be a canonical Markov process w.r.t. $\mathcal{F}_t := \mathcal{F}_t^{0,+}$ such that (i) $\mathbb{M}$ has right continuous sample paths, and (ii) the semigroup (5.5) has the **Feller property** $P_t(C_b(S)) \subset C_b(S)$. Then $\mathbb{M}$ has the SMP w.r.t. $(\mathcal{F}_t)$.

Proof.

Step 1 (reduction). Fix x. By the monotone class theorem it suffices to take $Y = f_1(X_{t_1}) \cdots f_m(X_{t_m})$ with $f_i \in C_b(S)$, $0 \leq t_1 < \cdots < t_m$. Let $T_n \searrow T$ be the dyadic approximations of Example 5.5(d) and $C \in \mathcal{F}_T$.

Step 2 (discretise). By right continuity of paths, $Y \circ \theta_{T_n} = \prod_i f_i(X_{t_i+T_n}) \to Y \circ \theta_T$ on $\{T < \infty\}$,

so by the DCT

$$\mathbb{E}_x[\mathbb{1}_{\{T<\infty\}} Y\circ\theta_T \mathbb{1}_C] = \lim_n \sum_{k\geq 1} \mathbb{E}_x\big[\mathbb{1}_{C\cap\{(k-1)2^{-n}\leq T<k2^{-n}\}}\, Y\circ\theta_{k2^{-n}}\big].$$

Here $C\cap\{(k-1)2^{-n}\leq T<k2^{-n}\}\in\mathcal{F}_{k2^{-n}}$, since $C\cap\{T<k2^{-n}\}\in\mathcal{F}_{k2^{-n}}$ by right continuity and $\{T\geq(k-1)2^{-n}\}\in\mathcal{F}_{(k-1)2^{-n}}$.

Step 3 (Markov at fixed times). By (5.8) (Lemma 5.13) each summand equals

$$\mathbb{E}_x\big[\mathbb{1}_{C\cap\{(k-1)2^{-n}\leq T<k2^{-n}\}}\,\mathbb{E}_{X_{k2^{-n}}}[Y]\big],$$

so the sum is $\mathbb{E}_x[\mathbb{1}_{\{T<\infty\}}\mathbb{E}_{X_{T_n}}[Y]\mathbb{1}_C]$.

Step 4 (Feller). $x\mapsto\mathbb{E}_x[Y]=P_{t_1}(f_1P_{t_2-t_1}(f_2\cdots P_{t_m-t_{m-1}}f_m))(x)$ is continuous by the Feller property, and $X_{T_n}\to X_T$ on $\{T<\infty\}$. The DCT gives the limit $\mathbb{E}_x[\mathbb{1}_{\{T<\infty\}}\mathbb{E}_{X_T}[Y]\mathbb{1}_C]$. ∎

Time-dependent functionals. Let $Y:\mathbb{R}\times\Omega\to\mathbb{R}$ be bounded and $\mathcal{B}(\mathbb{R})\otimes\mathcal{F}$-measurable, and define $\mathbb{1}_{\{T<\infty\}}\mathbb{E}_{X_T}[Y(T,\cdot)]$ as

$$\omega\mapsto\mathbb{1}_{\{T<\infty\}}(\omega)\,\mathbb{E}_{\mathbb{1}_{\{T<\infty\}}X_T(\omega)}[Y(T(\omega),\cdot)], \tag{5.9}$$

and $(Y(T,\cdot)\circ\theta_T)(\omega):=Y(T(\omega),\theta_T\omega)$ (the shift acts only on the second variable). If Y does not depend on time, these reduce to $\mathbb{1}_{\{T<\infty\}}\mathbb{E}_{X_T}[Y]$ and $Y\circ\theta_T$.

Definition 5.16 (Extended strong Markov property)

$\mathbb{M}$ as in Definition 5.14 has the **extended strong Markov property** (ESMP) w.r.t. $(\mathcal{F}_t)$ if for every stopping time T, $Y\in(\mathcal{B}(\mathbb{R})\otimes\mathcal{F})_b$ and $x\in S$,

$$\mathbb{E}_x[\mathbb{1}_{\{T<\infty\}}Y(T,\cdot)\circ\theta_T\mid\mathcal{F}_T]=\mathbb{1}_{\{T<\infty\}}\mathbb{E}_{X_T}[Y(T,\cdot)]\qquad P_x\text{-a.s.}$$

[Supplement] The right side (5.9) is $\mathcal{F}_T$-measurable. For Y as in (5.10) below it equals $\mathbb{1}_{\{T<\infty\}}\Phi(T,X_T)$ with Φ continuous (Feller), and T, $\mathbb{1}_{\{T<\infty\}}X_T$ are $\mathcal{F}_T$-measurable (Lemma 5.8). The monotone class argument at the end of the next proof extends this to all bounded Y.

Theorem 5.17

Under the assumptions of Theorem 5.15 ($\mathcal{F}_t=\mathcal{F}_t^{0,+}$, right continuous paths, Feller property), $\mathbb{M}$ has the ESMP w.r.t. $(\mathcal{F}_t)$.

Proof.

Step 1 (product functionals). Fix x and first let

$$Y(s,\omega)=g(s)f_1(X_{t_1}(\omega))\cdots f_m(X_{t_m}(\omega)),\qquad g\in C_b(\mathbb{R}),\ f_i\in C_b(S). \tag{5.10}$$

With T_n, $C\in\mathcal{F}_T$ as before, right continuity and the DCT give $\mathbb{E}_x[\mathbb{1}_{\{T<\infty\}}Y(T,\theta_T)\mathbb{1}_C]=\lim_n\mathbb{E}_x[\mathbb{1}_{\{T<\infty\}}Y(T_n,\theta_{T_n})\mathbb{1}_C]$. Splitting according to $\{(k-1)2^{-n}\leq T<k2^{-n}\}\in\mathcal{F}_{k2^{-n}}$, on which $Y(T_n,\cdot)=Y(k2^{-n},\cdot)\in\mathcal{F}_b$, the Markov property at the fixed times $k2^{-n}$ turns this into $\lim_n\mathbb{E}_x[\mathbb{1}_{\{T<\infty\}}\mathbb{E}_{X_{T_n}}[Y(T_n,\cdot)]\mathbb{1}_C]$. Since $(s,y)\mapsto\mathbb{E}_y[Y(s,\cdot)]=g(s)P_{t_1}(f_1P_{t_2-t_1}(\cdots P_{t_m-t_{m-1}}f_m))(y)$ is continuous (Feller), the DCT gives $\mathbb{E}_x[\mathbb{1}_{\{T<\infty\}}\mathbb{E}_{X_T}[Y(T,\cdot)]\mathbb{1}_C]$.

Step 2 (monotone class). The $Y\in(\mathcal{B}(\mathbb{R})\otimes\mathcal{F})_b$ satisfying the ESMP form an MVS. The multiplicative class $\mathcal{M}$ of indicators of $U_0\times\{(X_{t_1},\dots,X_{t_m})\in U_1\times\cdots\times U_m\}$, U_j open, generates $\mathcal{B}(\mathbb{R})\otimes\mathcal{F}$. Each such indicator is an increasing limit of functions of the form (5.10) (use $\phi(m\,d(\cdot,U^c))$-type cut-offs as in Theorem 3.35, (ii)⇒(iii)), so $\mathcal{M}\subset\mathcal{H}$ and Theorem 2.29 concludes. ∎

Example 5.18

The canonical d-dimensional Brownian motion $\mathbb{M}_W$ has the Markov property w.r.t. $(\mathcal{F}_t^{0,+})$ (Section 4.4), continuous paths, and the Feller property: $P_t f(x) = \int f(y) p_t(x,y)\,dy$ is continuous in x for $f \in C_b$ by the DCT. So it has the SMP and the ESMP w.r.t. $(\mathcal{F}_t^{0,+})$, and also w.r.t. $(\mathcal{F}_t)$ of Example 5.2, which differs only by sets that are null for every P_x.

Remark 5.19

Going through the proofs of Theorems 5.15, 5.17 (cf. Remarks 4.17, 4.19), a d-dimensional BM as in Definition 4.18 satisfies the ESMP for fixed x w.r.t. $(\mathcal{F}_t^{0,+,x})$ (needed for measurability, as it satisfies the usual conditions), for Y of the form (5.10) with $Y(T,\cdot)\circ\theta_T$ replaced by $g(T)f_1(W_{t_1+T})\cdots f_m(W_{t_m+T})$. In particular, for $f \in \mathcal{B}(\mathbb{R}^d)_b$ and every $(\mathcal{F}_t^{0,+,x})$-stopping time T,

$$\mathbb{E}_x[\mathbb{1}_{\{T<\infty\}} f(W_{t+T}) \mid \mathcal{F}_T^{0,+,x}] = \mathbb{1}_{\{T<\infty\}} \mathbb{E}_{W_T}[f(W_t)] \qquad P_x\text{-a.s.} \tag{5.11}$$

If T is an $(\mathcal{F}_t^{0,+})$-stopping time (hence an $(\mathcal{F}_t^{0,+,x})$-stopping time for every x), (5.11) holds simultaneously for all x.

5.3 Applications to Brownian motion

Motivation. Restarting Brownian motion at the first time it hits a level, and using its symmetry, gives the reflection principle. From it come the exact laws of hitting times and of the running maximum, and then the arcsine law for the last zero. Time inversion and the 0–1 laws describe the behaviour of paths at 0 and at ∞.

5.3.1 The reflection principle for one-dimensional BM

Let $\mathbb{M}_B = (\Omega, \mathcal{F}, (\mathcal{F}_t^{0,+}), (B_t), P_0, (\theta_t))$ be the canonical one-dimensional BM; it has the ESMP (Example 5.18). For $a, t > 0$ we show

$$P_0(D_{\{a\}} < t,\ B_t > a) = \tfrac{1}{2} P_0(D_{\{a\}} < t). \tag{5.12}$$

Let $Y(s,\omega) := \mathbb{1}_{(-\infty,t)}(s)\mathbb{1}_{(a,\infty)}(B_{(t-s)^+}(\omega))$ (bounded, $\mathcal{B}(\mathbb{R})\otimes\mathcal{F}$-measurable) and $T := \inf\{0 \le s < t \mid B_s = a\}$ (a stopping time, Example 5.5(c)). Then $\{T<\infty\} = \{T<t\} = \{D_{\{a\}} < t\}$ and $\mathbb{1}_{\{T<\infty\}} Y(T,\theta_T) = \mathbb{1}_{\{D_{\{a\}}<t\}}\mathbb{1}_{\{B_t>a\}}$. Taking expectations in the ESMP (Theorem 5.17), and using $B_T = a$ and $\mathbb{E}_a[\mathbb{1}_{\{B_r>a\}}] = \frac{1}{2}$ for $r > 0$ (symmetry),

$$P_0(D_{\{a\}} < t,\ B_t > a) = \mathbb{E}_0[\mathbb{1}_{\{T<t\}}\mathbb{E}_{B_T}[\mathbb{1}_{\{B_{t-T}>a\}}]] = \tfrac{1}{2}P_0(T<t),$$

which is (5.12). Since $B_0 = 0 < a$, path continuity gives $\{B_t > a\} \subset \{D_{\{a\}} < t\}$, hence

$$P_0(D_{\{a\}} < t) = 2P_0(B_t > a), \tag{5.13}$$

and, by continuity from above,

$$P_0(D_{\{a\}} \le t) = \lim_n P_0(D_{\{a\}} < t + \tfrac{1}{n}) = 2\lim_n P_0(B_{t+\frac{1}{n}} > a) = \frac{2}{\sqrt{2\pi t}}\int_a^\infty e^{-\frac{x^2}{2t}}\,dx = 2P_0(B_t \ge a). \tag{5.14}$$

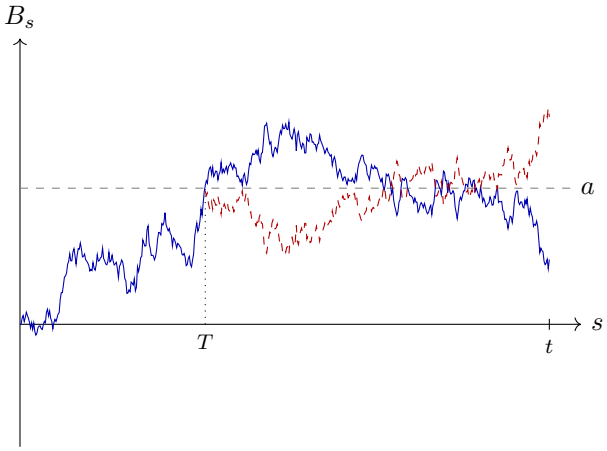


Figure. *The reflection principle. After the first hitting time $T = D_{\{a\}}$ the path (blue) and its mirror image in the level a (red, dashed) are equally likely. So among the paths that reach a before t, exactly half end above a, which gives $P_0(D_{\{a\}} < t) = 2P_0(B_t > a)$.*

Proposition 5.20

(i) For $a \in \mathbb{R}\setminus\{0\}$ and $t \geq 0$: $P_0(D_{\{a\}} \leq t) = \int_0^t \frac{|a|}{\sqrt{2\pi s^3}} e^{-\frac{a^2}{2s}}\, ds$. (ii) For $a, t > 0$: $P_0(\sup_{s\leq t} B_s \leq a) = P_0(|B_t| \leq a) = \frac{2}{\sqrt{2\pi t}} \int_0^a e^{-\frac{x^2}{2t}}\, dx$.

Proof. (i) For $a, t > 0$ substitute $x = t^{1/2} a\, s^{-1/2}$ in (5.14):

$$P_0(D_{\{a\}} \leq t) = \frac{2}{\sqrt{2\pi t}} \int_a^\infty e^{-\frac{x^2}{2t}}\, dx = 2\int_0^t \frac{1}{\sqrt{2\pi t}} \frac{t^{1/2} a}{2s^{3/2}} e^{-\frac{a^2}{2s}}\, ds = \int_0^t \frac{a}{\sqrt{2\pi s^3}} e^{-\frac{a^2}{2s}}\, ds.$$

For $a < 0$ use that $(-B_t)$ is again a canonical BM; $t = 0$ is trivial.

(ii) For $a, t > 0$, $\{D_{\{a\}} > t\} = \{\sup_{s\leq t} B_s < a\}$. By continuity from above and (5.13),

$$P_0\Big(\sup_{s\leq t} B_s \leq a\Big) = \lim_n P_0(D_{\{a+\frac{1}{n}\}} > t) = 1 - 2\lim_n P_0(B_t \geq a + \tfrac{1}{n}) = 1 - 2P_0(B_t > a) = P_0(|B_t| \leq a).$$

■

Remark 5.21

Proposition 5.20 and the arcsine law below hold for every BM $(\tilde{B}_t)$ on $(\tilde{\Omega}, \tilde{\mathcal{F}}, \tilde{P}_0)$ as in Definition 4.1. A shift-free strong Markov property gives this directly (Le Gall, Ch. 2.4); alternatively, the finite-dimensional distributions agree with those of $\mathbb{M}_B$, and the events involved are determined by countably many times:

$$\tilde{P}_0\Big(\sup_{s\leq t} \tilde{B}_s \leq a\Big) = \tilde{P}_0\Big(\sup_{s\in\mathbb{Q}\cap[0,t]} \tilde{B}_s \leq a\Big) = P_0\Big(\sup_{s\in\mathbb{Q}\cap[0,t]} B_s \leq a\Big) = P_0\Big(\sup_{s\leq t} B_s \leq a\Big).$$

Similarly (with a filtration satisfying the usual conditions) $\tilde{P}_0(\tilde{D}_{\{a\}} \leq t) = P_0(D_{\{a\}} \leq t)$, because $\{D_{\{a\}} > t\} = \bigcup_n \bigcap_{s\in[0,t]\cap\mathbb{Q}} \{|B_s - a| \geq \frac{1}{n}\}$, and likewise for $\tilde{B}$.

Example 5.22 (Arcsine law)

Let $L := \sup\{t \le 1 \mid B_t = 0\}$, the last zero before time 1. L is not a stopping time, but its law can be computed. For $s \in [0,1)$, $\{L \le s\} = \{B_t \neq 0 \ \forall t \in (s,1]\}$, and $P_0(L \le 1) = 1$. By the Markov property,

$$P_0(L \le s) = \mathbb{E}_0\big[\mathbb{E}_0[\mathbb{1}_{\{B_{t-s}\neq 0 \ \forall t\in(s,1]\}}\circ\theta_s \mid \mathcal{F}_s^{0,+}]\big] = \mathbb{E}_0[f(B_s)] = \int p_s(0,y)\, P_y(\sigma_{\{0\}} > 1-s)\, dy,$$

with $f(y) := P_y(B_t \neq 0 \ \forall t \in (0, 1-s])$. Since $P_y(\sigma_{\{0\}} > 1-s) = P_y(D_{\{0\}} > 1-s) = P_0(D_{\{-y\}} > 1-s)$, Proposition 5.20(i) and symmetry give

$$\begin{aligned} P_0(L \le s) &= 2\int_0^\infty \frac{e^{-y^2/(2s)}}{\sqrt{2\pi s}}\Big(\int_{1-s}^\infty \frac{y}{\sqrt{2\pi r^3}} e^{-\frac{y^2}{2r}}\, dr\Big) dy = \frac{1}{\pi}\int_{1-s}^\infty \frac{1}{\sqrt{sr^3}} \int_0^\infty y e^{-\frac{(r+s)y^2}{2rs}}\, dy\, dr \\ &= \frac{1}{\pi}\int_{1-s}^\infty \frac{1}{\sqrt{sr^3}} \frac{rs}{r+s}\, dr = \frac{1}{\pi}\int_{1-s}^\infty \sqrt{\frac{r}{s}}\, \frac{s}{(r+s)^2}\, dr. \end{aligned}$$

The substitution $t = \frac{s}{r+s}$, $dt = -\frac{s}{(r+s)^2} dr$, gives

$$P_0(L \le s) = \frac{1}{\pi}\int_0^s \frac{dt}{\sqrt{t(1-t)}} = \frac{2}{\pi}\arcsin\sqrt{s}, \qquad s \in [0,1].$$

The density $\frac{1}{\pi\sqrt{t(1-t)}}$ is largest near 0 and 1: the last zero before time 1 is most likely to occur either very early or very late.

5.3.2 Further path properties of BM

Theorem 5.23 (Time inversion)

If (B_t) is a standard BM, so is $X_t := tB_{1/t}$ for $t > 0$, $X_0 := 0$.

Proof.

Step 1 (Definition 4.1(i), (ii)). (i) is clear. For $0 < s < t$, $0 < \frac{1}{t} < \frac{1}{s}$ and

$$X_t - X_s = (t-s)\underbrace{B_{1/t}}_{=:Y_1} - s\underbrace{(B_{1/s} - B_{1/t})}_{=:Y_2},$$

with independent $Y_1 \sim N(0, \frac{1}{t})$, $Y_2 \sim N(0, \frac{1}{s} - \frac{1}{t})$. By Example 3.44(i), $X_t - X_s \sim N\big(0, (t-s)^2\frac{1}{t} + s^2(\frac{1}{s} - \frac{1}{t})\big) = N(0, t-s)$. The case $s = 0$ is similar.

Step 2 (Definition 4.1(iii)). For $s,t > 0$, $\mathbb{E}[X_sX_t] = st(\frac{1}{s} \wedge \frac{1}{t}) = s \wedge t$, and this also holds if s or t is 0. [Supplement] This covariance and Gaussianity give independent increments. For $0 \le t_1 < \cdots < t_n$ the vector of increments $Z = (X_{t_1}, X_{t_2} - X_{t_1}, \dots)$ is a linear image MG of a vector G of independent standard normals (the normalised increments of B at the times $1/t_i$). So, by Theorem 3.52, $\varphi_Z(u) = \mathbb{E}[e^{i\langle M^\top u, G\rangle}] = e^{-\frac{1}{2}\|M^\top u\|^2} = e^{-\frac{1}{2}u^\top \Sigma u}$ with $\Sigma = MM^\top$ the covariance matrix of Z. From $\mathbb{E}[X_sX_t] = s \wedge t$, Σ is diagonal with entries $t_i - t_{i-1}$. So $\varphi_Z(u) = \prod_i e^{-\frac{1}{2}(t_i - t_{i-1})u_i^2} = \prod_i \varphi_{Z_i}(u_i)$, and Remark 3.50 gives independence.

Step 3 (continuity). By Steps 1–2, X satisfies (4.9), so X and B have the same finite-dimensional distributions. Paths of X are continuous on $(0,\infty)$ a.s. (everywhere if B is canonical). At 0: the laws of X and B on $\mathbb{Q} \cap (0,\infty)$ agree, and $\{\lim_{t\downarrow 0,\, t\in\mathbb{Q}} X_t = 0\}$ is determined by countably many coordinates, so $P(\lim_{t\downarrow 0,\, t\in\mathbb{Q}} X_t = 0) = P(\lim_{t\downarrow 0,\, t\in\mathbb{Q}} B_t = 0) = 1$. With continuity on $(0,\infty)$ this gives $\lim_{t\downarrow 0} X_t = 0$ a.s. ∎

Remark 5.24

Time inversion links the behaviour of B_t as $t \to \infty$ with that of X_t as $t \to 0$. For example $B_t/t = X_{1/t} \to 0$, so paths grow slower than linearly; Theorem 5.26 says more. From now on $\mathbb{M}_B = (\Omega, \mathcal{F}, (B_t), (P_x), (\theta_t))$ is the canonical one-dimensional BM, $\hat{\mathcal{F}}_t^0 := \sigma(B_s \mid s \geq t)$ (decreasing in t) and

$$\mathcal{T} := \bigcap_{t \geq 0} \hat{\mathcal{F}}_t^0$$

is the **tail σ-algebra** of B: the events of the remote future. For example, $\limsup_n A_n \in \mathcal{T}$ for $A_m \in \hat{\mathcal{F}}_m^0$, since $\bigcup_{m \geq n} A_m \in \hat{\mathcal{F}}_n^0$ and $(\hat{\mathcal{F}}_n^0)$ is decreasing.

Theorem 5.25

P_0 is deterministic on $\mathcal{T}$: $P_0(A) \in \{0, 1\}$ for all $A \in \mathcal{T}$.

Proof.

Step 1 (tail of B = germ of X). With X from Theorem 5.23, using $X_0 \equiv 0$ (so adding it changes no σ-algebra),

$$\begin{aligned}\mathcal{T} &= \bigcap_{t>0} \sigma(B_s \mid s \geq t) = \bigcap_{t>0} \sigma(B_{1/s} \mid 0 < s \leq \tfrac{1}{t}) = \bigcap_{t>0} \sigma(X_s \mid 0 < s \leq t) \\ &= \bigcap_{t>0} \sigma(X_s \mid 0 \leq s \leq t) = \mathcal{F}_0^{0,+}(X).\end{aligned}$$

Step 2 (Blumenthal for X) [Supplement]. The lectures cite Blumenthal's law for general BM (Le Gall, Thm. 2.13); here is a transfer from Theorem 4.16. Modify X on the null set where it is discontinuous at 0 (set it to 0 there); then $X : \Omega \to C(\mathbb{R}_+)$ is measurable, $P_0 \circ X^{-1} = P_0^W$ (Wiener measure, same finite-dimensional laws), and $\sigma(X_s \mid s \leq t) = X^{-1}(\mathcal{F}_t^0)$. For $A \in \mathcal{F}_0^{0,+}(X)$ choose $\Gamma_n \in \mathcal{F}_{1/n}^0$ with $A = X^{-1}(\Gamma_n)$. Then $P_0^W(\Gamma_n \triangle \Gamma_m) = P_0(X^{-1}(\Gamma_n \triangle \Gamma_m)) = 0$, so $\Gamma := \limsup_n \Gamma_n \in \mathcal{F}_0^{0,+}$ differs from each Γ_n by a P_0^W-null set. Hence $P_0(A) = P_0^W(\Gamma) \in \{0, 1\}$ by Theorem 4.16. ∎

Theorem 5.26

For the canonical one-dimensional BM, P_0-a.s.

$$\limsup_{t \to \infty} \frac{B_t}{\sqrt{t}} = +\infty, \qquad \liminf_{t \to \infty} \frac{B_t}{\sqrt{t}} = -\infty.$$

Proof. Let $K > 0$. By Fatou applied to indicators and scaling invariance (Proposition 4.2),

$$P_0\Big(\limsup_n \Big\{\frac{B_n}{\sqrt{n}} \geq K\Big\}\Big) \geq \limsup_n P_0\Big(\frac{B_n}{\sqrt{n}} \geq K\Big) = P_0(B_1 \geq K) > 0.$$

The event on the left lies in $\mathcal{T}$, so by Theorem 5.25 it has probability 1. As K is arbitrary, $\limsup_t B_t/\sqrt{t} = \infty$ a.s.; the second claim follows by symmetry ($-B$ is a BM). ∎

The next corollary holds for every BM, but we prove it for the canonical one, as a consequence of the SMP w.r.t. $(\mathcal{F}_t^{0,+})$.

Corollary 5.27

Let $N(\omega) := \{s \in [0, 1] \mid B_s(\omega) = 0\}$. For P_0-a.e. ω: (i) $N(\omega)$ is closed;
(ii) $N(\omega)$ has Lebesgue measure zero;
(iii) $N(\omega)$ has no isolated points. So $N(\omega)$ is a Cantor-type set: closed, null, and every point

is a cluster point.

Proof. (i) ω is continuous and $\{0\}$ is closed.

(ii) By Fubini, $\mathbb{E}_0\big[\int_0^1 \mathbb{1}_{N(\cdot)}(s)\,ds\big] = \int_0^1 P_0(B_s = 0)\,ds = 0$, so $\lambda(N(\omega)) = 0$ for P_0-a.e. ω.

(iii)

Step 1. Let $A_{r,s} := \{N(\omega) \cap (r,s) \text{ is exactly one point}\}$; then

$$J := \{N(\omega) \text{ has isolated points}\} = \bigcup_{r<s\in\mathbb{Q}_+} A_{r,s}.$$

Let $T_r := r + \sigma_{\{0\}} \circ \theta_r = \inf\{t > r \mid B_t = 0\}$.

Step 2 (T_r is a stopping time) [Supplement]. $\{T_r < t\} = \bigcup_n \{\exists s \in [r + \frac{1}{n}, t - \frac{1}{n}] : B_s = 0\} = \bigcup_n \bigcap_m \bigcup_{q\in\mathbb{Q}\cap[r+\frac{1}{n},t-\frac{1}{n}]} \{|B_q| < \frac{1}{m}\} \in \mathcal{F}_t^0$ (continuity and compactness), so T_r is a weak stopping time, hence an $(\mathcal{F}_t^{0,+})$-stopping time (Lemma 5.4).

Step 3 ($P_0(\sigma_{\{0\}} = 0) = 1$) [Supplement]. $A_+ := \bigcap_n \{\sup_{s\le 1/n} B_s > 0\} \in \mathcal{F}_0^{0,+}$ and $P_0(\sup_{s\le 1/n} B_s > 0) \ge P_0(B_{1/n} > 0) = \frac{1}{2}$, so $P_0(A_+) \ge \frac{1}{2}$ and, by Blumenthal (Theorem 4.16), $P_0(A_+) = 1$. Likewise $A_- := \bigcap_n \{\inf_{s\le 1/n} B_s < 0\}$ has probability 1. On $A_+ \cap A_-$ the path takes both signs in every $(0, \frac{1}{n})$, hence vanishes there (intermediate value theorem), so $\sigma_{\{0\}} = 0$.

Step 4 (SMP). Since $B_{T_r} = 0$ on $\{T_r < \infty\}$, by Theorem 2.35(iii) and the SMP,

$$P_0(A_{r,s}) \le P_0(T_r < s,\ \sigma_{\{0\}} \circ \theta_{T_r} > 0) = \mathbb{E}_0\big[\mathbb{1}_{\{T_r<s\}} \mathbb{E}_{B_{T_r}}[\mathbb{1}_{\{\sigma_{\{0\}}>0\}}]\big] = P_0(\sigma_{\{0\}} > 0)\, P_0(T_r < s) = 0.$$

Hence $P_0(J) = 0$. ∎

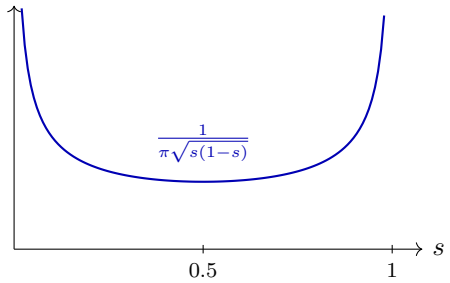


Figure. *The arcsine density of the last zero L before time 1 (Example 5.22). It is U-shaped, so the last zero is most likely near the ends of the interval.*